

SmartSignDeck

Developed By

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Roll No: BC065

ID:23BCUOS076



Group No : M5

Developed At

Technostacks Infotech

As Per

Partial Fulfillment of VI Semester of

B.C.A[Bachelor of Computer Application]

Under the Guidance of

Internal Project Guide

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Assistant Professor

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Ahemdabad

Submitted To:

BCA DEPARTMENT,

Dharmsinh Desai University, NADIAD

DARMSINH DESAI UNIVERSITY

Faculty of Management and Information Science



CERTIFICATE

This is to clarify that Mr. Dhimant.N.Pandya of Faculty of Management and Information Science , Dharmsinh Desai University as part of their Bachelor of Computer Application (BCA) VIth Semester, have done that Project Work on “SmartSignDeck” at Technostacks Infotech Under my guidance.

This Project was carried out from 1st December 2025 to 21st March 2026 as partial Fulfillment for the award of degree of Bachelor of Computer Application.

Prof. Minaz Shaikh

(Internal Guide)

Prof. C P Patel

(Head, BCA Dept.)



0250

Date: 23/03/2026

Certificate

This is to certify that, below mentioned candidate has been working with our organization, **Technostacks Infotech Pvt. Ltd.** for Industrial Training during his 6th Semester of Bachelor of Computer Application.

Student Name: Dhimant Nidhishkumar Pandya

Project Title: Smart Sign Deck

Technology: MERN Stack

Enrollment No: 23BCUOS076

Internship starting date: 1st December, 2025

Internship completion date: 23rd March, 2026

He has completed Industrial training in **MERN Stack** Technology. He was very sincere during his training, his dedication and hard work has been commendable. He performed very well according to our expectation. We are confident that the expertise he have gained will significantly contribute to his professional growth and performance.

Yours truly,

For Technostacks Infotech Pvt. Ltd.

AUTHORIZED SIGNATORY

Challenge Creates Change

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+91 99097 12616 | +91 97129 55934 | info@technostacks.com | CIN: U72900GJ2014PTC078355

www.technostacks.com

ACKNOWLEDGEMENT

The success and final outcome of this project required a lot of guidance and assistance from many people and I am extremely fortunate to have got this all along the completion of my project work. All I have done is only due to such guidance and assistance and I would not forget to thank them.

I feel myself very fortunate to have got an opportunity to work in Bachelor of Computer Application (BCA), **Dharmsinh Desai University** while undertaking the project named "**SmartSignDeck**". This opportunity has proved to be great and immense important to me as it has gifted me with manifold opportunity and benefits. At the end of my successful completion of my B.C.A final semester project training in Dharmsinh Desai University, I am greatly thankful of those personalities, who have made it all possible for me.

I am sincerely thankful to our **H.O.D Prof. C. P. Patel** and all the staff members of B.C.A Department for their constructive thoughts and vision towards successful completion of my project.

I am highly thankful to **Prof. Minaz Shaikh** (BCA department) for her kind support as an internal guide and guided all along till the completion of project work by providing all the necessary information for developing a good system.

I am thankful and fortunate enough to get constant encouragement, support and guidance from Parents, all Teaching staffs of BCA Department which helped in successfully completion of my project. Also, I would like to express my sincere esteem to all staff in laboratory for their timely support.

Thank You All,

Dhimant Pandya [23BCUOS076]

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ABOUT THE SYSTEM

- This website named "**SmartSignDeck**" is built using a frontend library called **React**. **React** makes the site look awesome and work really well. It's like the secret that helps the website be fast and fun to use. With **React**, **Tailwind CSS** for modern styling and **React Router** for easy navigation has also been used. This ensures that everything works well together, making it easy to explore the site and manage your digital advertisements.
- **SmartSignDeck** is a platform that helps businesses manage and display their advertisements effectively across digital screens. It focuses on creating dynamic and effective marketing displays using centralized scheduling methods. With a focus on flexibility, strategy, and results-driven playback, **SmartSignDeck** strives to elevate brands and connect them with their target audiences in meaningful ways.
- At **SmartSignDeck**, we are all about being clear, creative, and making sure our clients see results. Whether a business is launching a new product, refreshing its brand, or trying to reach more customers across multiple physical locations, we are here to help. We are passionate about coming up with unique ideas and making sure our clients stand out in the busy world of advertising.
- **SmartSignDeck** is a **Web Based Application** that can be used over the internet. In this system, Administrators can build custom display templates (using a Canvas-based interface), assign media content (images and videos), and schedule playback across multiple screens.
- Admin can Add, Update (Edit), and Delete the organizations, screens, and templates that have been registered to the site.
- The backend (Api) of this website is built in **Node.js (Express)**.
- In this system following of user is available:
 - **Super Admin**
 - **Organization Admin**
 - **User**
 - **Advertiser**

ORGANIZATION OVERVIEW

TecnoStackS Pvt. Ltd. is a leading IT solutions and consulting company based in Ahmedabad, Gujarat, India. Established in 2014, the organization focuses on delivering innovative and scalable digital solutions to startups and enterprises across various industries.

The company specializes in solving complex business challenges by providing customized digital and technological solutions. TecnoStackS adopts a client-centric approach, deeply understanding customer requirements and delivering solutions that optimize operations and drive business growth.

TecnoStackS offers a wide range of services including Digital Product Development, SaaS Solutions, Mobile and Web Application Development, Artificial Intelligence (AI), Internet of Things (IoT), Cloud & DevOps, and Product Engineering. These services help organizations enhance efficiency, improve decision-making, and stay competitive in the digital era.

The organization is supported by a team of skilled professionals who are dedicated to delivering high-quality, secure, and reliable solutions. The company follows modern development practices and agile methodologies to ensure innovation, flexibility, and timely delivery of projects.

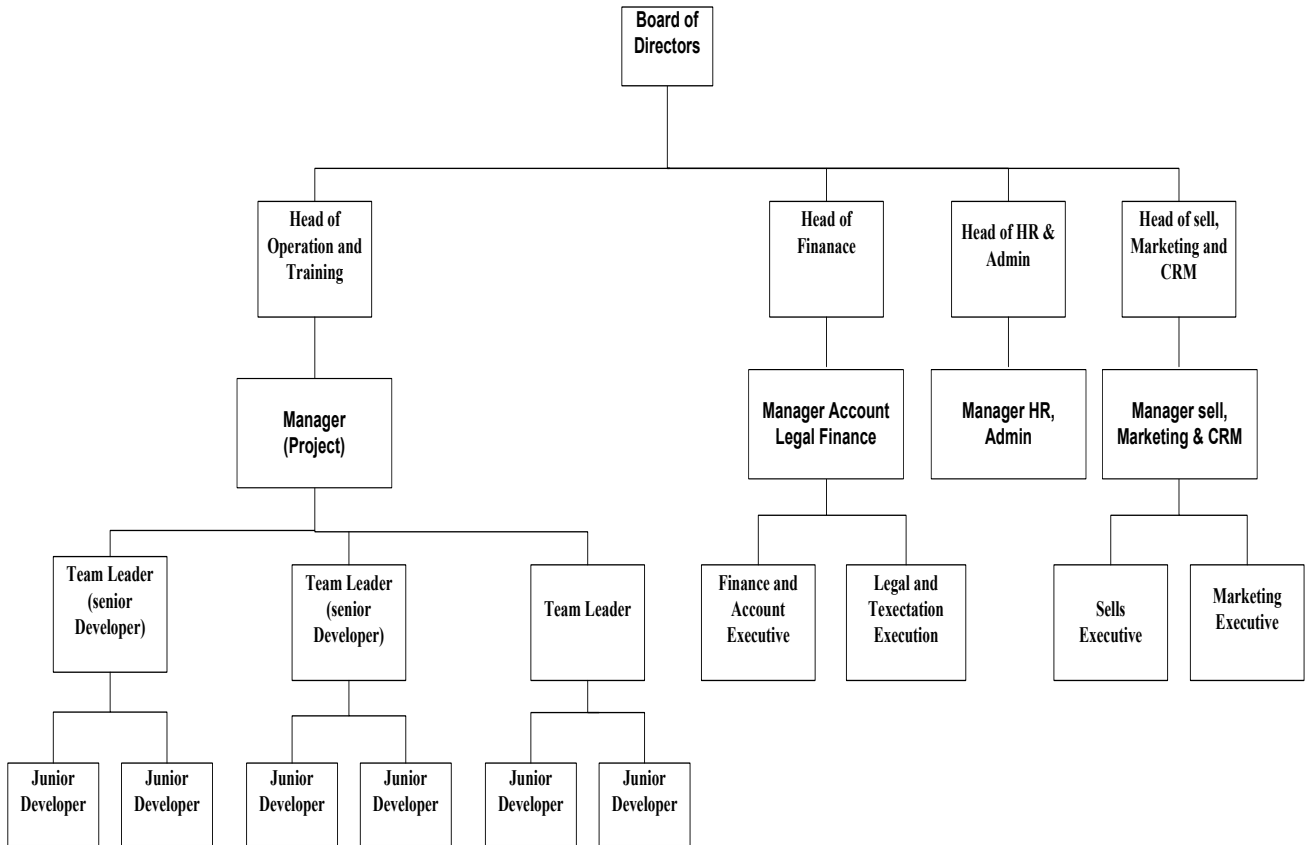
TecnoStackS has collaborated with global brands and serves clients across multiple domains such as healthcare, logistics, finance, and manufacturing. With its strong focus on innovation, security, and customer satisfaction, the company has built a reputation as a trusted technology partner.

The core values of TecnoStackS include customer focus, innovation, reliability, and continuous improvement. The company believes in leveraging advanced technologies and creative problem-solving to deliver impactful solutions and long-term value to its clients.

References:

- [Company Overview – TecnoStackS](#)
- <https://technostacks.com/>
- <https://www.linkedin.com/company/technostacks/>

ORGANIZATION CHART



HARDWARE & SOFTWARE CONFIGURATION

Development Server Side

➤ Hardware specification

- Any machine which has internet facility.

Hardware Configuration	Description
Processor	Intel Pentium 4 CPU, 2.66GHZ
RAM	512 MB
Hard Disk	200 GB

➤ Software specification

Software Configuration	Description
Operating System	Windows 7
Back End	My SQL
Front End	PhpMagento 1.9.2
	All major browser including internet explorer, mozilla Firefox

Development Client Side

➔ Hardware Specification

- ➔ Any machine which has internet facility.

Hardware Configuration	Description
Processor	Intel Pentium 4 CPU, 2.66GHZ
RAM	512 MB
Hard Disk	200 GB

➔ Software Specification

- ➔ Any browser which support following
- ➔ **HTML5**
- ➔ **CSS3** (with Animation support)
- ➔ **JavaScript** (ES6+)
- ➔ **WebSockets** (for Real-time Sync)

FEATURE OF TOOLS USED

➤ HTML :-

HTML, or HyperText Markup Language, is the standard markup language used to create the structure and present content on the World Wide Web. HTML is a fundamental technology for web development, providing the basis for building interactive and visually appealing websites. It works in conjunction with CSS (Cascading Style Sheets) for styling and JavaScript for dynamic behavior to create a complete web experience.

➤ Features:

1. **User Friendly & Simple:** Easy to learn and implement structure.
2. **Semantic Structure:** Uses meaningful tags like `<header>`, `<footer>`, and `<article>`.
3. **SEO – Search Engine Optimisation:** Proper structure helps search engines index content.
4. **LocalStorage & IndexedDB:** Enables client-side data storage for better performance.
5. **Canvas for Rich Graphics:** Supports the **Fabric.js** engine for our template designer.
6. **Platform Independent:** Works on any device with a modern web browser.
7. **Media Support:** Built-in support for images, audio, and video tags.

8. Attributes for Metadata: Uses attributes to provide additional information about elements.

➤ **CSS :-**

CSS (Cascading Style Sheets) is a style sheet language used for describing the presentation of a document written in **HTML**. **CSS** describes how elements should be rendered on screen, on paper, in speech, or on other media.

➤ **Features:**

1.Selector Mechanism: **CSS** uses a powerful selector mechanism that allows you to target **HTML** elements based on their tag names, classes, IDs, attributes, and more.

2.Box Model: The box model describes the layout of elements on a web page, including content, padding, border, and margin.

3.Layout Models: **CSS** provides various layout models like Flexbox and Grid, which make it easier to create complex and responsive layouts.

4.Responsive Designs: Media queries enable developers to create responsive designs that adapt to different screen sizes and devices.

5.Transitions and Animations: CSS allows for smooth transitions between different states and supports animations, enhancing user experience without **JavaScript**.

➤ **JavaScript :-**

JavaScript, often abbreviated as **JS**, is a high-level, versatile programming language primarily used for creating dynamic and interactive content on the web. It is an integral part of the modern web ecosystem, allowing developers to create responsive and interactive web pages.

➤ **Features:**

1. **Client-Side Scripting:** **JavaScript** is primarily used for client-side scripting, meaning it runs on the user's browser rather than on the server.
2. **Event Driven:** **JavaScript** is event-driven, meaning it can respond to events such as user actions (clicks, keypresses) or changes in the environment.
3. **Asynchronous Programming:** **JavaScript** supports asynchronous programming through mechanisms like callbacks, Promises, and Async/Await, enabling non-blocking code execution.
4. **Cross Platform:** **JavaScript** is a cross-platform language and can be run on various devices and operating systems.
5. **DOM Manipulation:** Commonly used to manipulate the Document Object Model (DOM) to dynamically change content, structure, and style.
6. **Single-threaded Event Loop:** Uses an event loop to handle asynchronous operations efficiently while remaining responsive.

7. **Modules (ES6 and beyond):** Native support for modules, promoting better code structure and maintainability.
8. **Extensive Ecosystem:** A rich ecosystem of libraries and frameworks like **React**, which assist in building scalable web applications.

➤ **React :-**

React is a **JavaScript** library used for building user interfaces, particularly for creating dynamic and interactive web applications.

Component-Based Architecture: Build reusable and modular UI components.

1. **Virtual DOM:** Optimizes updates by using a virtual representation in memory.
2. **Reusable Code:** Promotes code reusability across different parts of the application.
3. **Efficient Rendering:** Selective rendering via reconciliation algorithm for improved performance.

➤ **Node.js (Backend) :-**

Node.js is a cross-platform, open-source **JavaScript** runtime environment that executes **JavaScript** code outside a web browser. In our system, it handles all API requests and real-time socket communications.

➤ **Features:**

1. **Fast Request Handling:** Support for handling thousands of concurrent requests and responses.

2. **JSON Support:** Built-in support for **JSON** data formats for API communication.
3. **Database Connectivity:** Robust connectivity with **MongoDB** using Mongoose.
4. **Input Validation:** Integrated security features via Joi for strict input validation.
5. **Scalability:** High performance with caching and real-time load balancing.
6. **RESTful API:** Handling various HTTP methods (GET, POST, PUT, DELETE).
7. **Security Headers:** Implementation of CORS and Helmet for secure headers and monitoring.
8. **Error Handling:** Effective error handling and logging for system monitoring.
9. **Middleware Support:** Robust support for custom processing logic (Authentication, Rate Limiting).

➤ **MongoDB (Database) :-**

MongoDB is a source-available cross-platform document-oriented database program. Classified as a NoSQL database program, **MongoDB** uses **JSON**-like documents with optional schemas.

➤ **Features:**

1. **Document-Oriented Storage:** Data is stored in flexible, **JSON**-like documents, meaning fields can vary from document to document.
2. **Indexing:** High-performance indexing for faster query execution.
3. **Scalability:** Supports horizontal scaling through sharding, making it suitable for large-scale ad playback logs.
4. **Schema-less:** Allows for rapid development as the data structure can evolve over time without complex migrations.

➤ **4.7 Socket.io (Real-time Communication) :-**

Socket.io is a library that enables real-time, bi-directional, and event-based communication between the browser and the server.

➤ **Features:**

1. **Low Latency:** Provides near-instant updates, essential for our "Live Sync" and heartbeat systems.
2. **Auto-reconnection:** Automatically handles dropped connections, ensuring screens stay updated.
3. **Binary Support:** Efficiently handles diverse data types for collaborative design updates.
4. **Room Support:** Allows grouping of connections by Organization or Screen for targeted event broadcasting.

➤ **Cloudinary (Media Management) :-**

Cloudinary is a cloud-based service that provides an end-to-end image and video management solution.

➤ **Features:**

1. **Automated Optimization:** Automatically adjusts quality and format (WebP/MP4) for the fastest delivery to the player.
2. **Secure Uploads:** Provides a robust widget for secure, direct-to-cloud media uploads.
3. **On-the-fly Transformation:** Allows resizing and cropping of ad assets via URL parameters.
4. **Global CDN:** Ensures media assets are cached and delivered from the closest server to the physical screen.

➤ **Tailwind CSS (Styling) :-**

Tailwind CSS is a utility-first CSS framework for rapidly building custom user interfaces without leaving your **HTML/JSEX**.

➤ **Features:**

1. **Utility-First:** Provides low-level utility classes that let you build completely custom designs.
2. **Responsive Utilities:** Built-in breakpoints make it easy to design our dashboard for both desktop and mobile views.
3. **Optimized Bundle:** Automatically removes unused CSS, resulting in a tiny production bundle for faster load times.

4. **Consistent Design System:** Encourages the use of a predefined theme, ensuring UI consistency across the entire platform.

DETAIL DESCRIPTION OF THE SYSTEM

Existing System:

- The current system relies on traditional and digital marketing methods to capture attention and influence purchasing decisions. For an average person not directly involved in the advertising industry, the current system is all about seeing ads on TV, social media, and billboards. These ads try to grab attention and influence what people buy. However, this can sometimes feel overwhelming because there are so many ads everywhere. Main challenging point is the potential for misleading advertising practices, which can create confusion among consumers. Additionally, some individuals may find it challenging to filter and identify products or services that genuinely align with their needs and preferences in a manual environment where content is updated via physical media (USB drives).

Proposed System:

- To solve the problem or drawback of the existing system, I have created a managed advertisement system online called SmartSignDeck. In this system, all screens, templates, and schedules are available in one place.

1. User Registration and Login:

- Implements secure user and organization registration and login.
- Defines roles and permissions for different user types (Super Admin, Org Admin, Collaborator).

- Users can login by their E-mail Id and password or via Google OAuth.

2. Product & Media Management:

- A user-friendly interface for administrators to add, edit, and delete media assets.
- Includes features like Cloudinary-powered image/video uploads and descriptions.
- Admins can add categories (Playlists), designs (Templates), and manage them centrally.

3. Template Designing:

- Real-time collaborative canvas (Fabric.js) where users can create multi-zone layouts.
- Includes "Lock & Sync" mechanisms to prevent overlapping edits.

4. Ad Scheduling:

- Users can assign playlists to specific time slots on specific screens.
- Total control over start/end times and priority levels.

5. Smart Playback Process:

- Implements a multi-step execution process where the Player fetches content via live Socket events.
- Intelligent layout expansion if content is missing from specific zones.

6. Screen Health and Tracking:

- A user dashboard where administrators can view screen status (Online/Offline) and heartbeat history.
- Real-time monitoring of what is currently playing on each screen.

7. User Feedback and Collaboration:

- Allows collaborators to exchange feedback within the template editor.
- System for organization-level collaboration requests and approvals.

8. Playback Notifications:

- Provides facility for notifying administrators if a screen goes offline or if a schedule fails to load.

9. Analytics Generator:

- Detailed playback logs are generated for every ad rendered, which can be viewed by administrators to track campaign performance.

TIMELINE CHART

TIMELINE CHART

ID	Task Name	Start	Finish	Duration	Dec 2025				Jan 2026				Feb 2026				Mar 2026			
					1-7	8-14	15-21	22-28	29-4	5-11	12-18	19-25	26-1	2-8	9-15	16-22	23-1	2-8	9-15	16-23
1	Requirement Gathering	12-01	12-12	12d																
2	Analysis	12-13	12-28	16d																
3	Design	12-29	01-17	20d																
4	Coding	01-18	03-03	45d																
5	Testing	03-04	03-14	11d																
6	Implementation	03-15	03-23	9d																
7	Documentation	12-01	03-23	113d																

FEASIBILITY STUDY

A feasibility study for the **SmartSignDeck** system is carried out for determining whether the proposed system is possible to develop with available resources and what should be the cost consideration. The main objective is to test the Technical, Economical, and Operational aspects for adding new modules and debugging the running system.

1. Technical Feasibility:

- The technical feasibility for **SmartSignDeck** is done to evaluate whether the required technology infrastructure (**Node.js**, **React**, **MongoDB**), systems, and resources are available and capable of supporting the website's functionality and performance.
- The system is developed using the frontend library **ReactJS** and backend framework **Node.js**.
- **SmartSignDeck** requires active internet connectivity for real-time synchronization.
- The proposed system considers the adoption of digital advertising platforms and analytics tools (**Clouinary**, **Socket.io**) to enhance service offerings.
- Assesses the feasibility of integrating with third-party APIs to enhance website functionality and streamline business processes.

2. Economic Feasibility:

- Economic feasibility looks at the financial aspects of the project and return on investment (ROI).
- It determines whether it is worthwhile to invest money in the proposed **SmartSignDeck** system.
- It estimates the initial investment required for establishing the system, including expenses for cloud hosting (**MongoDB Atlas**, **Clouinary**),

development tools, and personnel.

- By addressing economic feasibility, we ensure that resources are allocated efficiently and investments are made with favorable long-term growth prospects.

3. Operational Feasibility:

- Operational feasibility involves assessing the practicality and effectiveness of the system's day-to-day operations.
- It evaluates the operational processes (Ad scheduling, Template creation) required to deliver advertising services effectively.
- **SmartSignDeck** establishes efficient, reliable, and scalable operational processes that support its business objectives and deliver a positive customer experience.
- This lays the foundation for long-term success in the competitive digital advertising market.

4. Legal Feasibility:

- Legal feasibility for **SmartSignDeck** involves ensuring compliance with relevant laws, regulations, and industry standards.
- It understands legal and regulatory requirements for operating an advertising agency or platform, including copyright laws for ad content.
- Ensures compliance with intellectual property laws when creating or hosting advertising materials (trademarks, copyrights).
- Understands consumer protection laws that govern advertising practices, ensuring that claims are substantiated and comply with fair practices.

5. Marketing Feasibility:

- Marketing feasibility involves assessing the viability and effectiveness of marketing strategies to promote the service and attract clients.

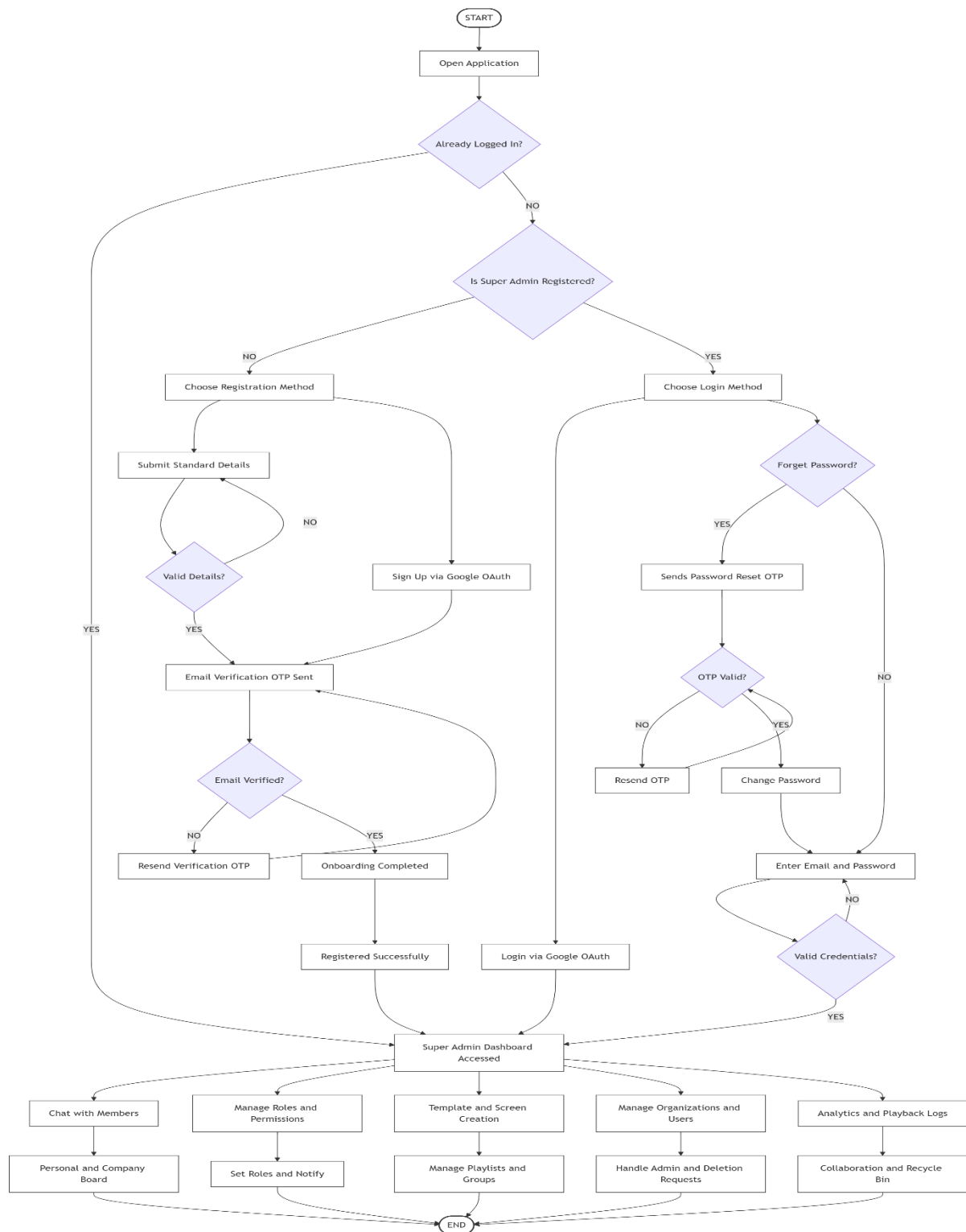
- It conducts market research to identify and understand the needs and preferences of target audiences for digital advertising.
- Analyzes the competitive landscape to identify competitors' strengths and weaknesses, helping **SmartSignDeck** highlight its unique value proposition.
- Identifies potential opportunities for differentiation through real-time collaboration and intelligent layout adaptation.

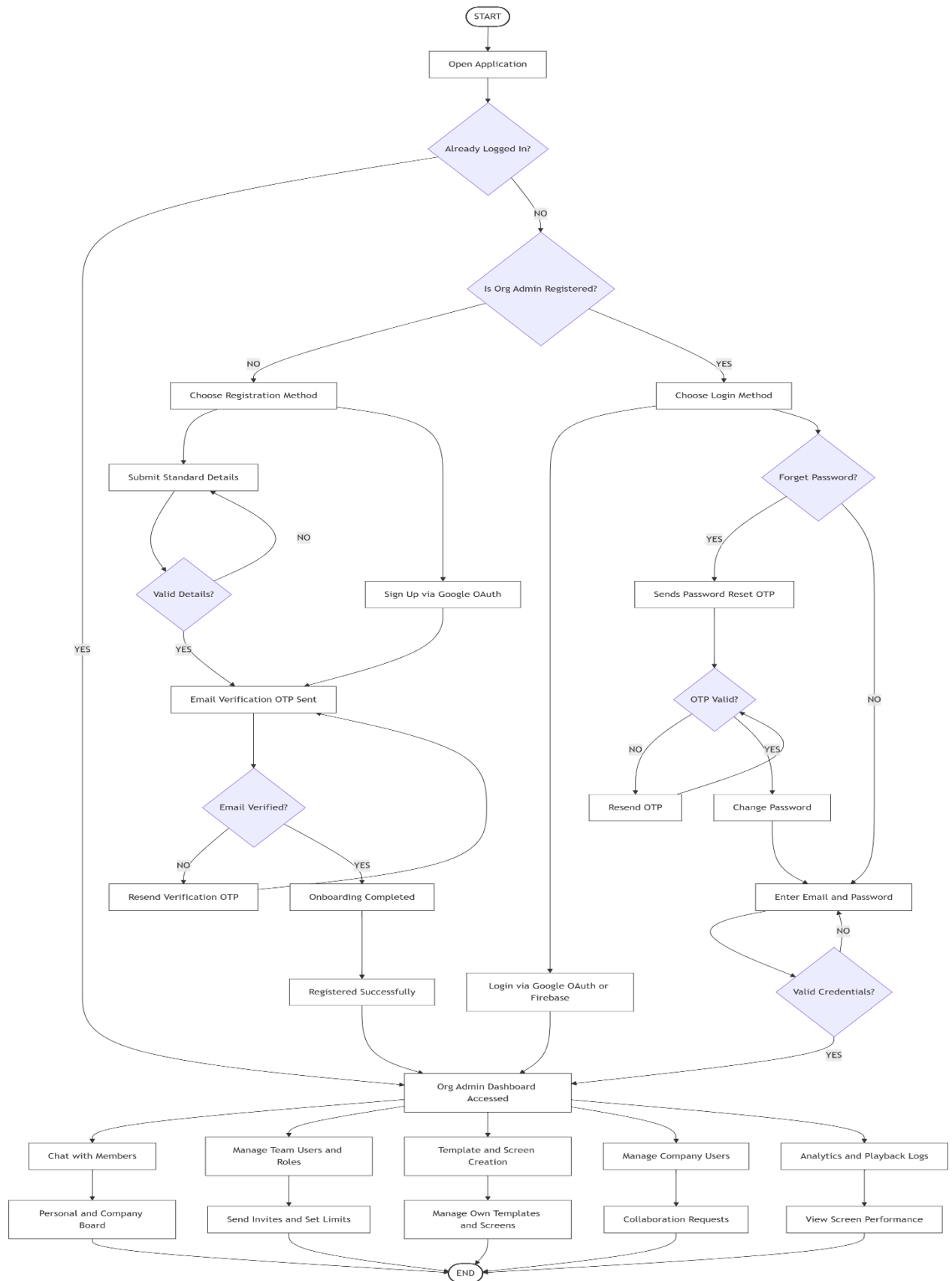
6. Social Feasibility:

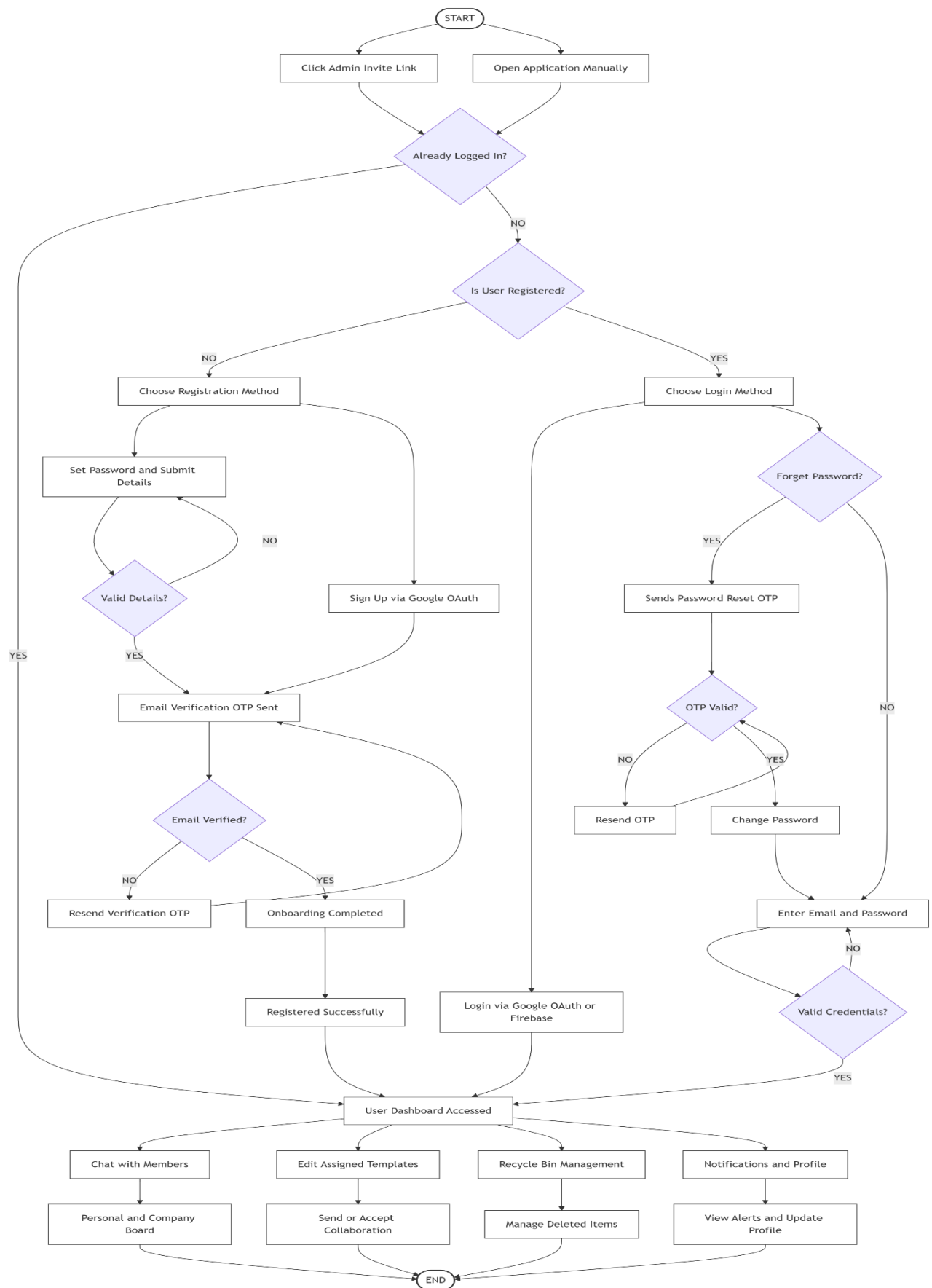
- Social feasibility assesses the acceptance, impact, and alignment of the platform with societal values and expectations.
- It ensures that the **SmartSignDeck** player interface is accessible to people of all abilities and backgrounds.
- Encourages feedback from customers and stakeholders to understand their needs, concerns, and expectations.
- Enhances trust and positive relationships with customers by maintaining a legally and ethically sound advertising environment.

SYSTEM FLOW CHART

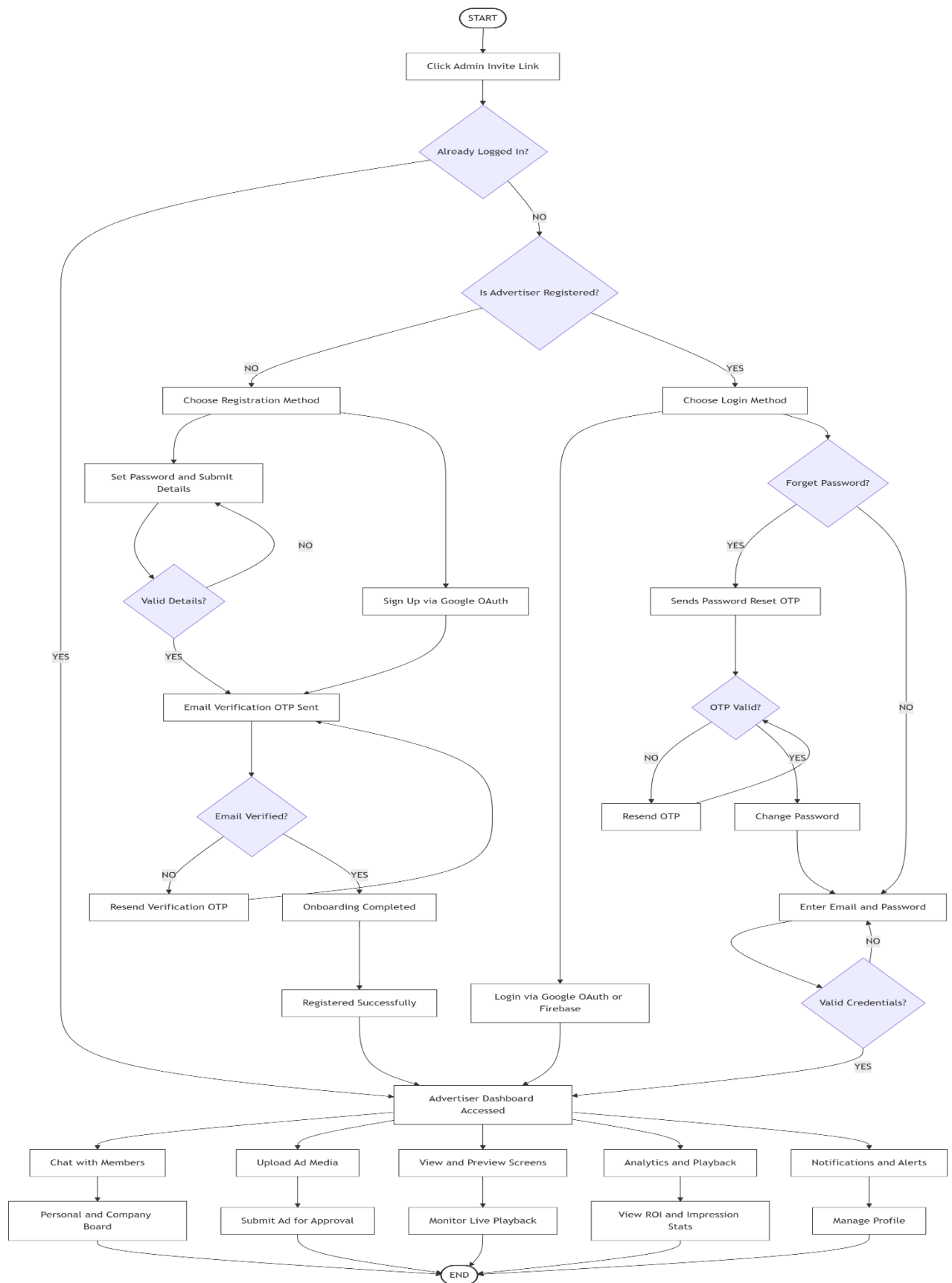
Super Admin :

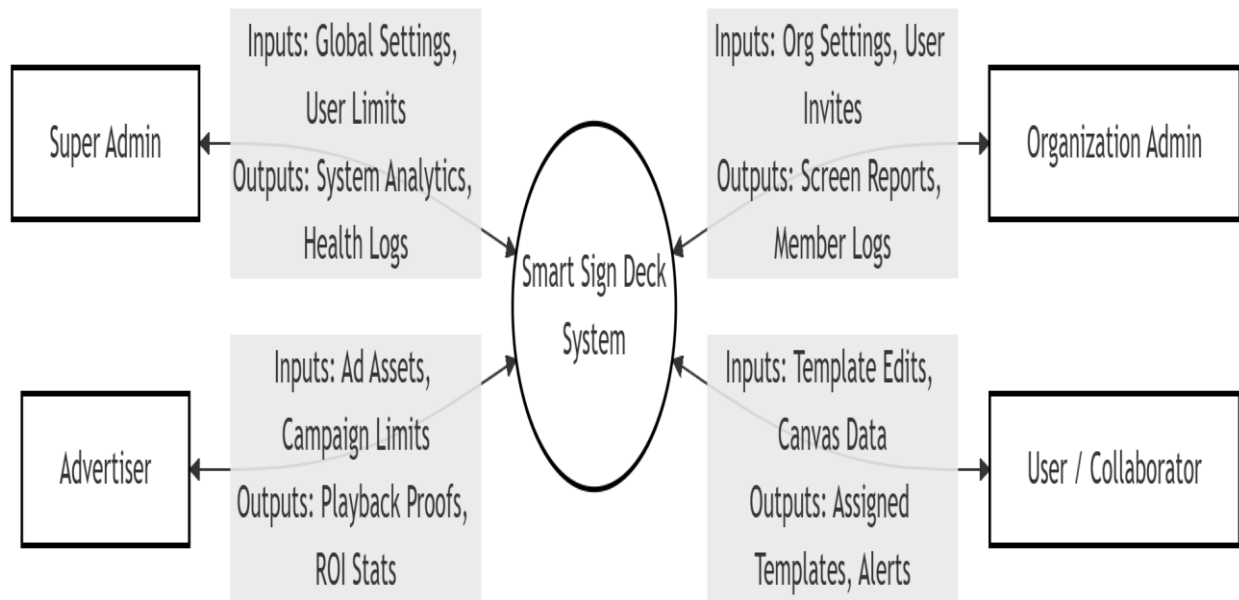


Admin :

User :

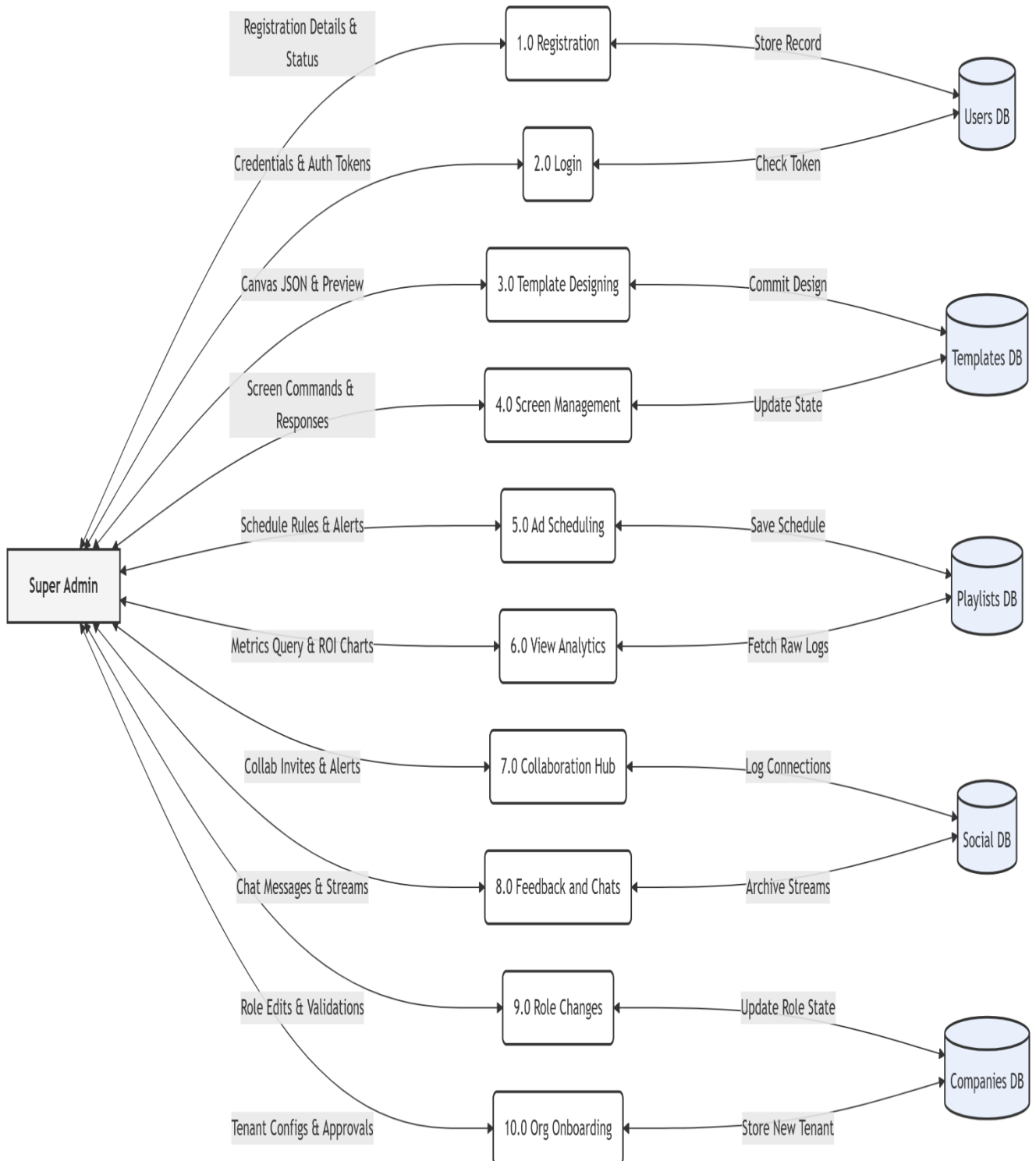
Advertiser :



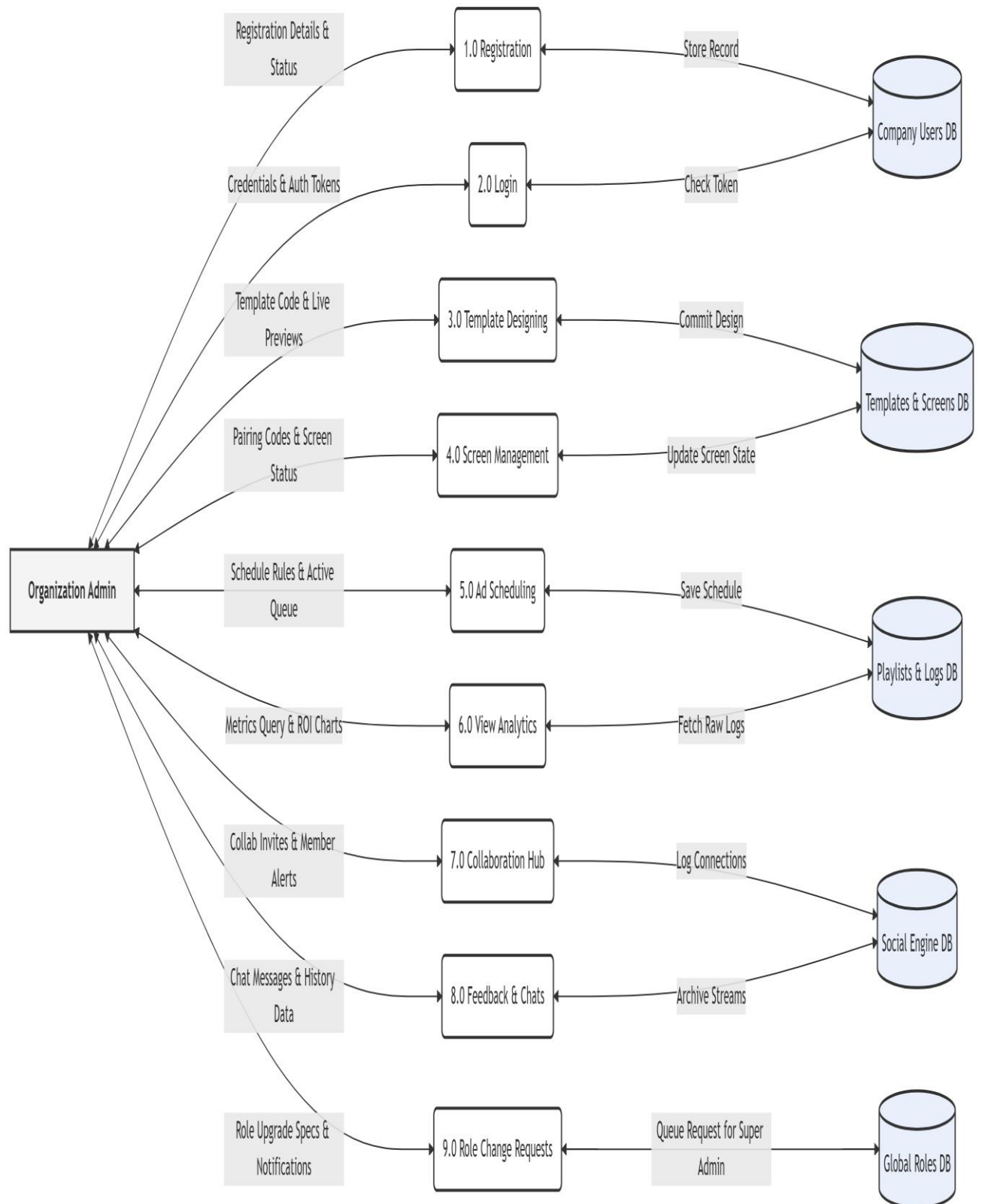
CONTEXT DIAGRAM

DATA FLOW DIAGRAM

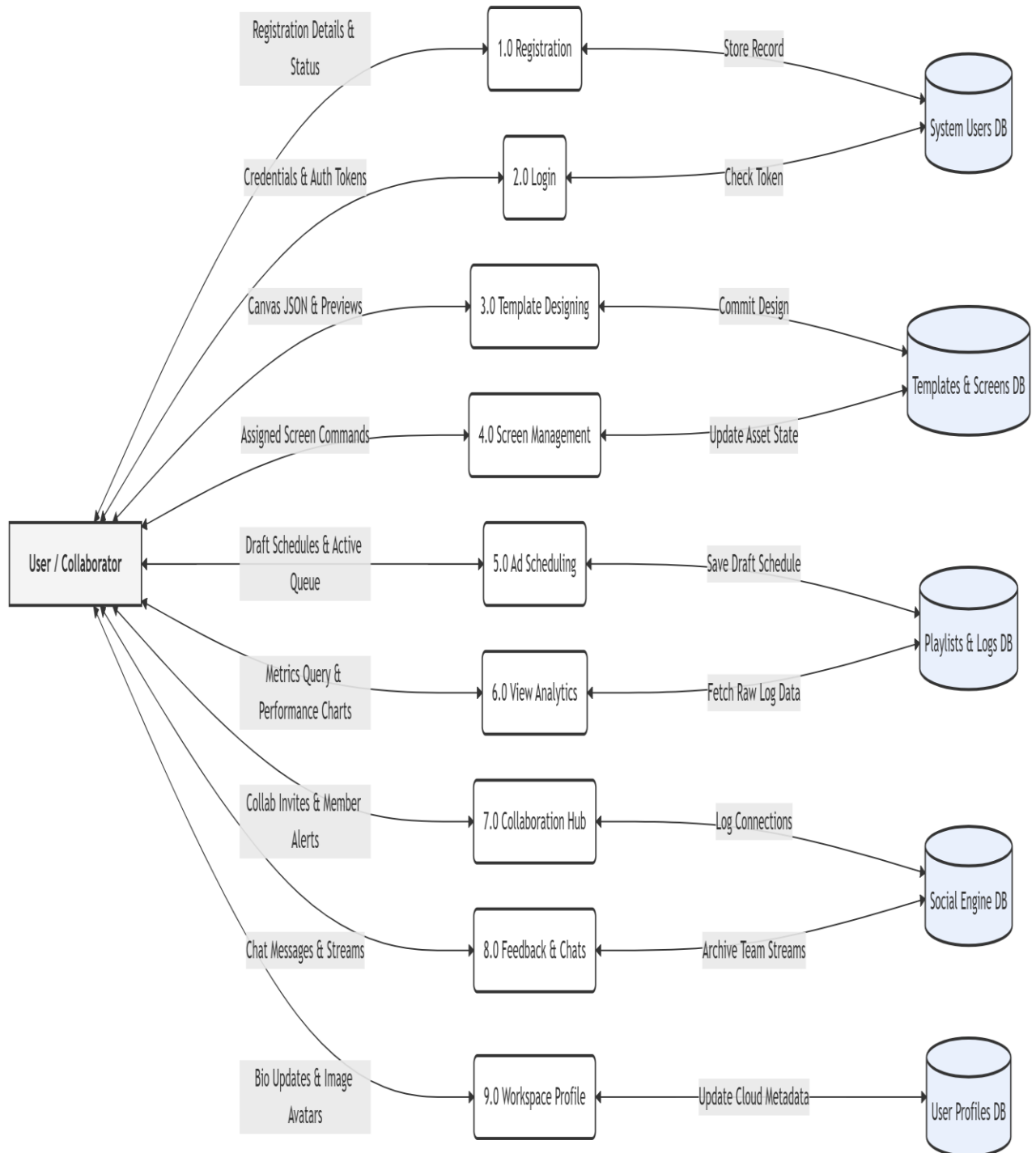
➤ First Level DFD SuperAdmin



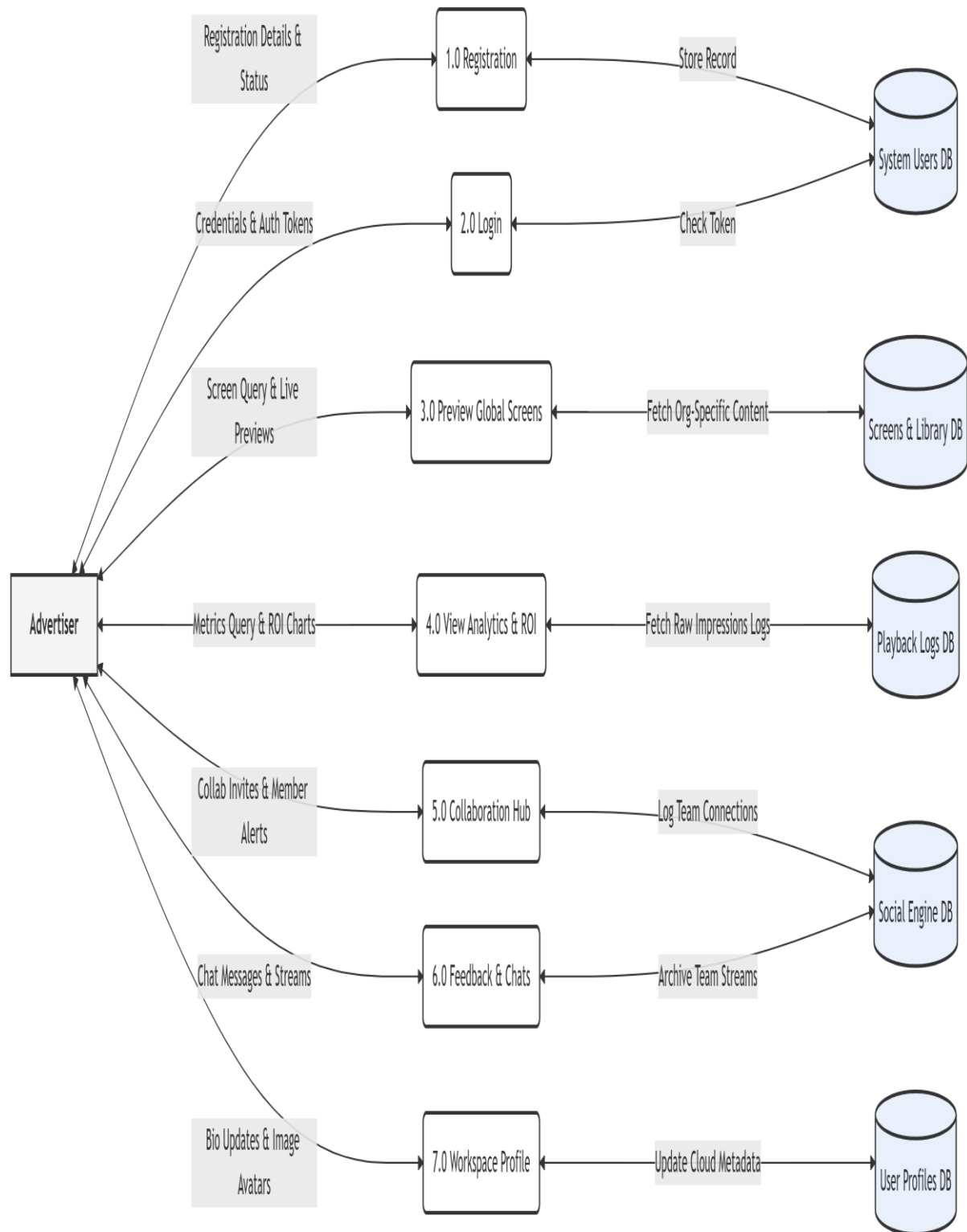
➤ First Level DFD Admin



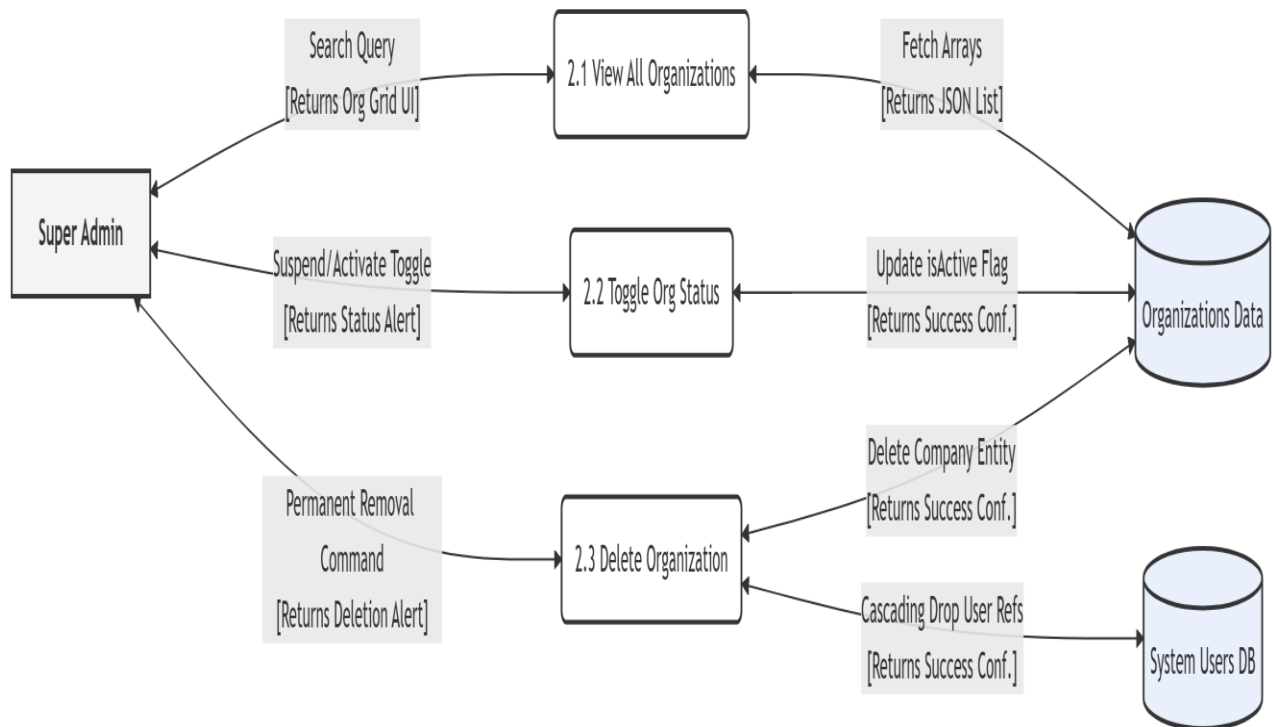
➤ First Level DFD User



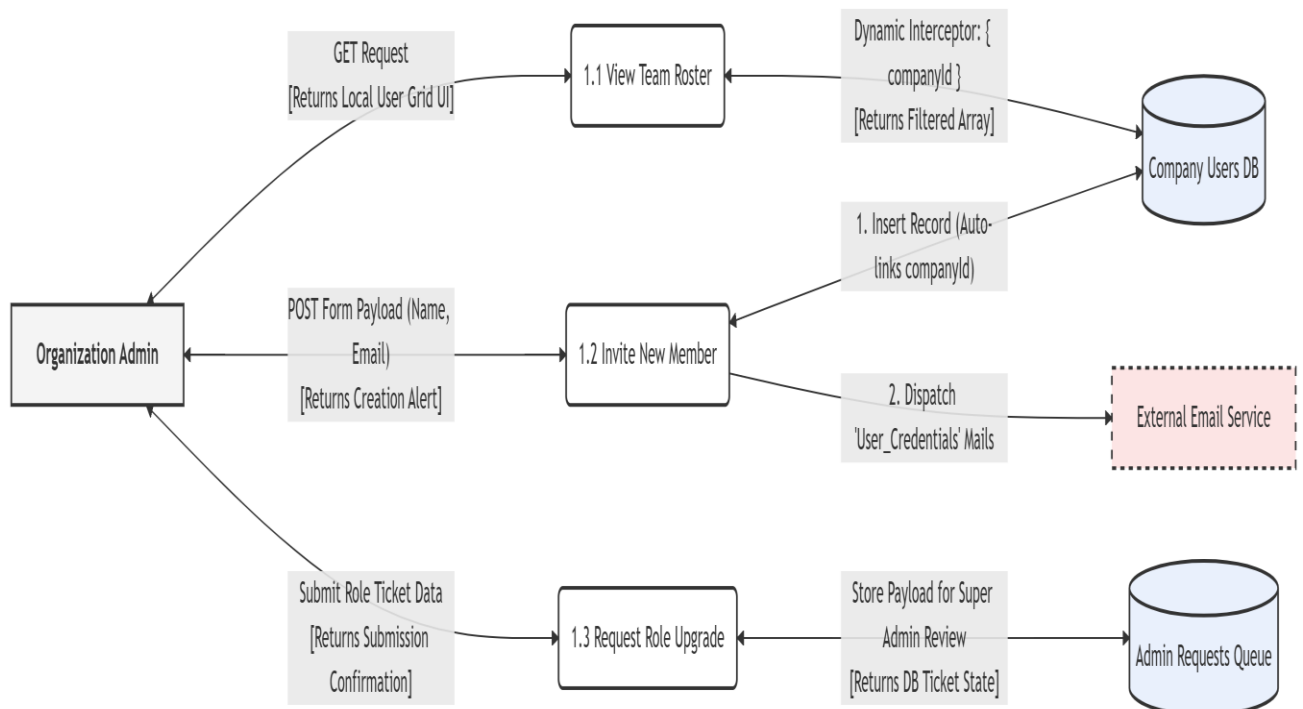
➤ First Level DFD Advertiser



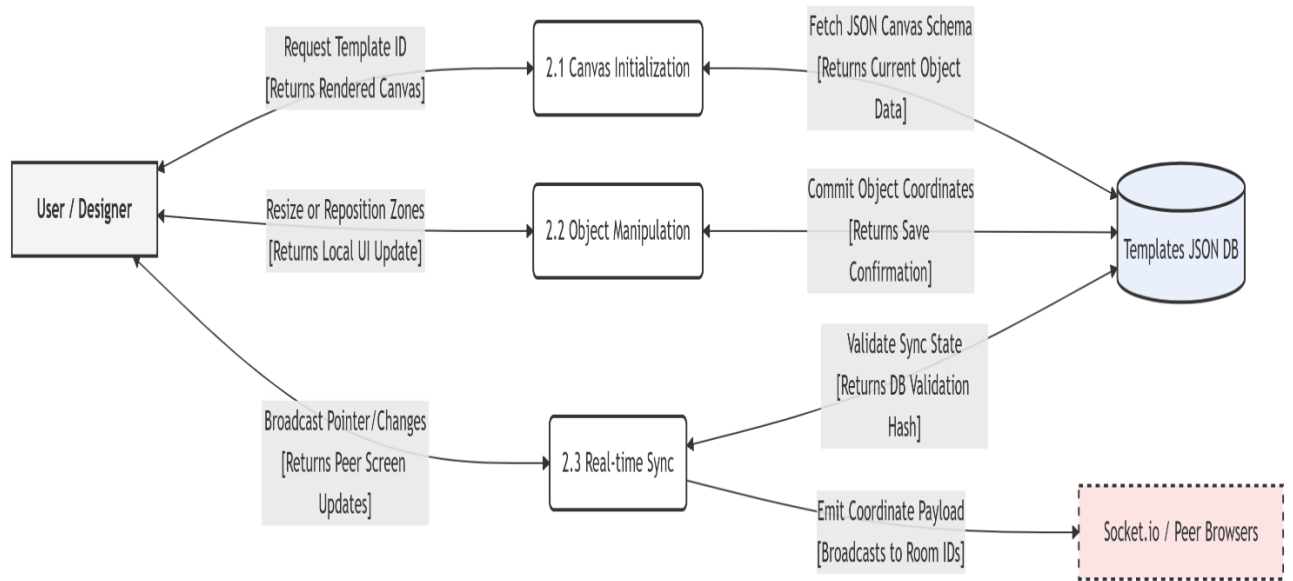
➤ Second Level DFD Super Admin



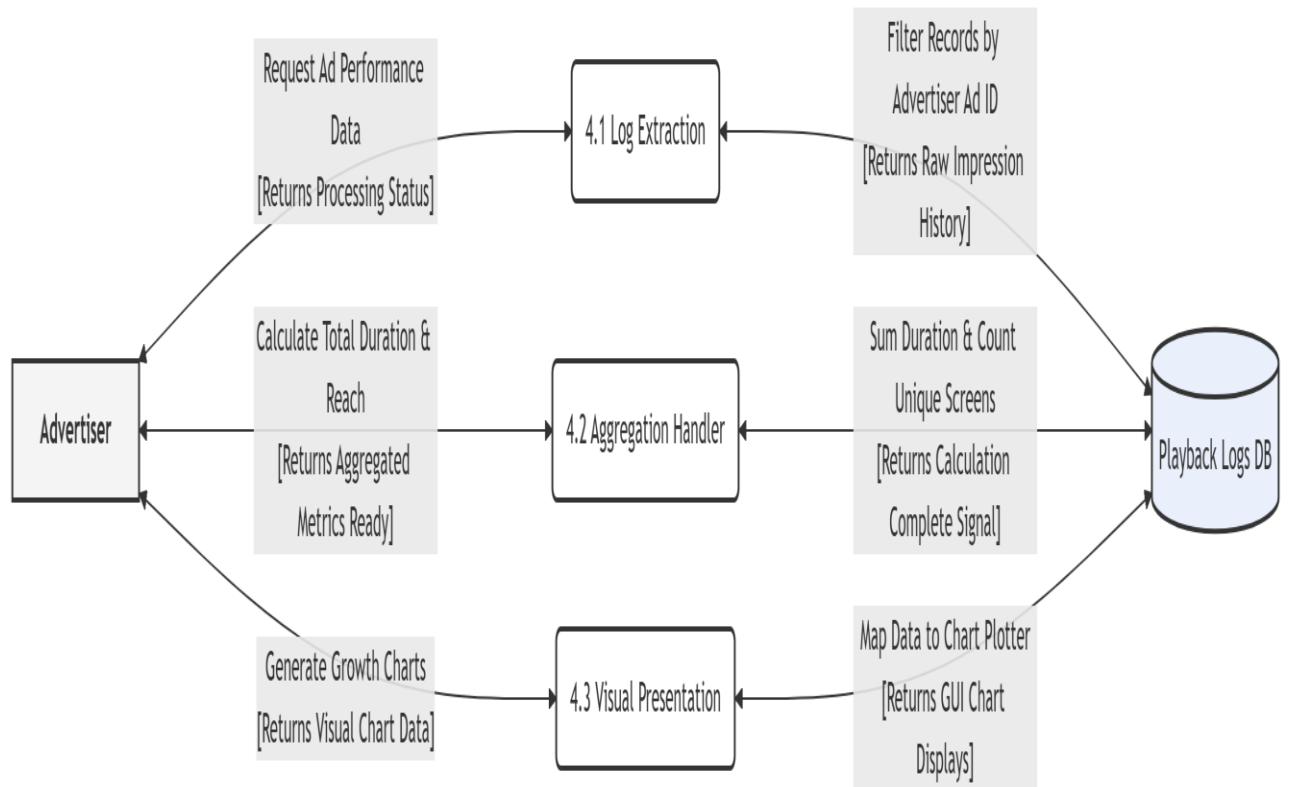
➤ Second Level DFD Admin



➤ Second Level DFD User

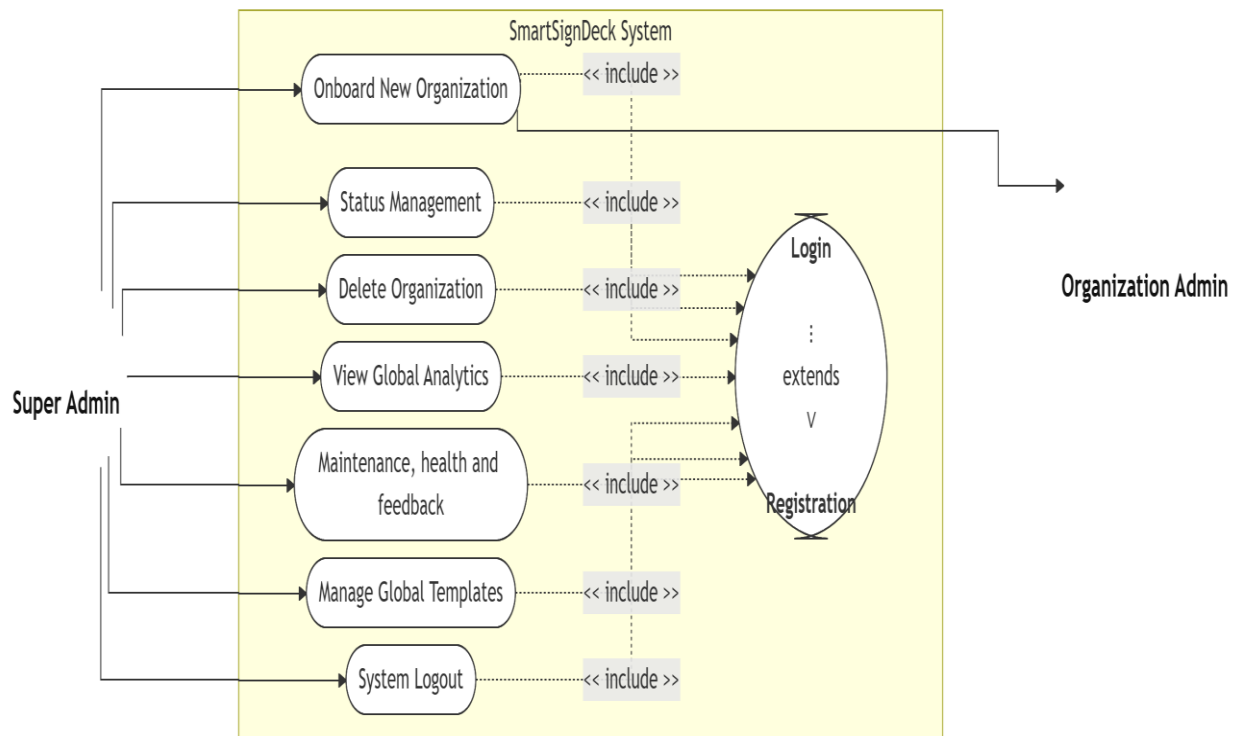


➤ Second Level DFD Advertiser

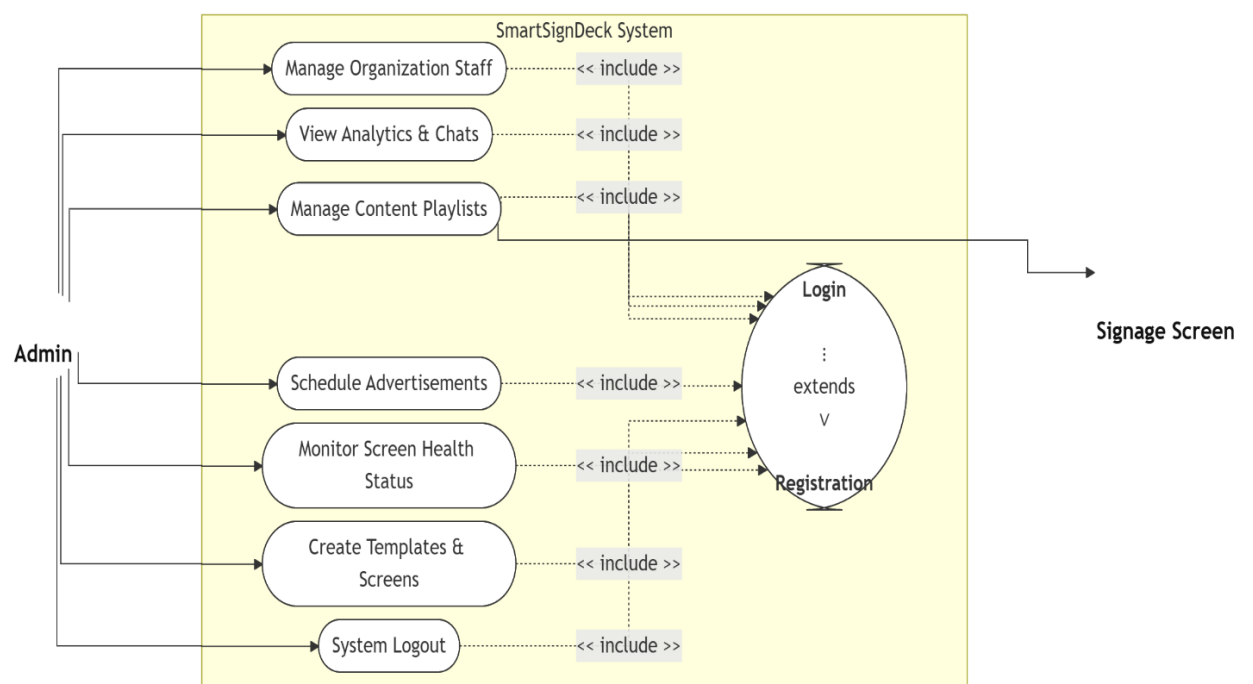


USECASES

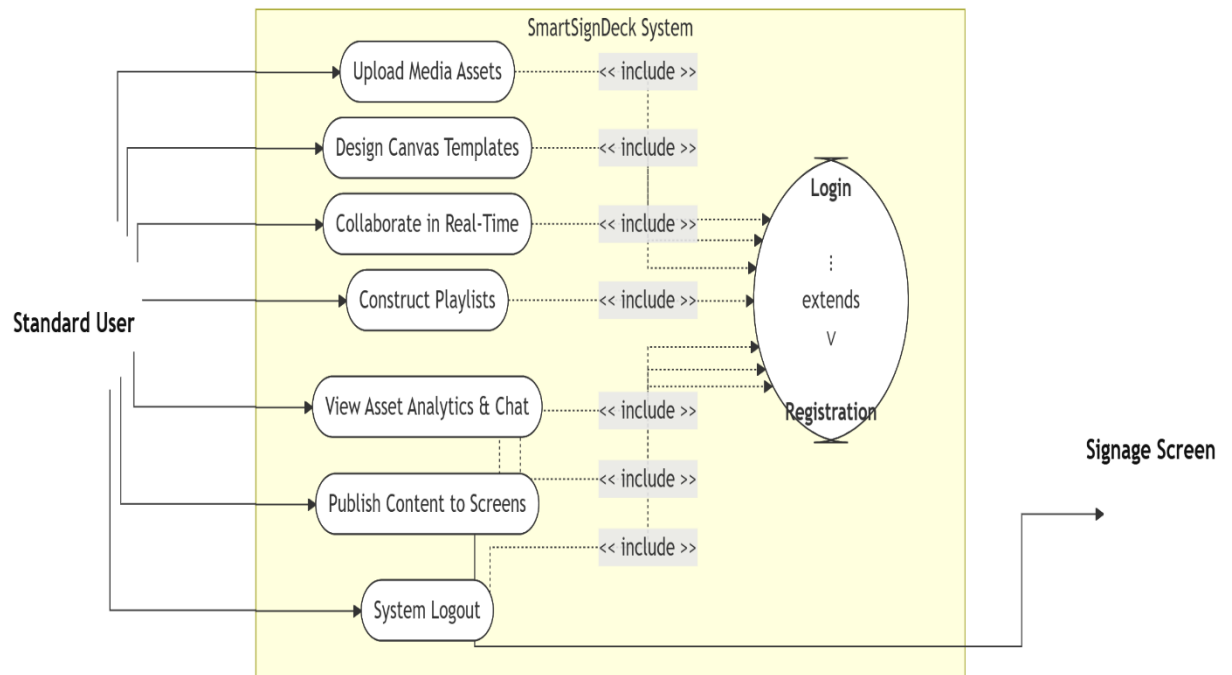
➤ Use CASE Super Admin



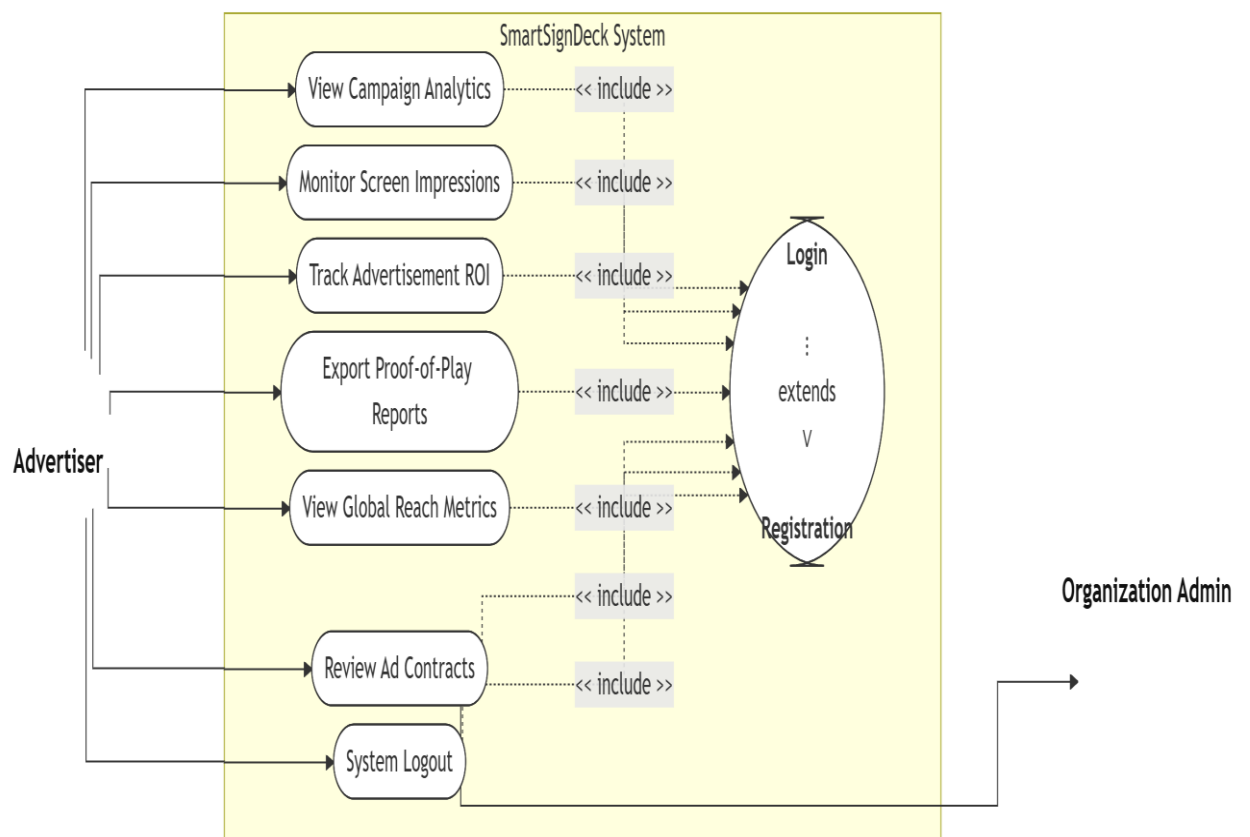
➤ Use CASE Admin



➤ Use CASE User

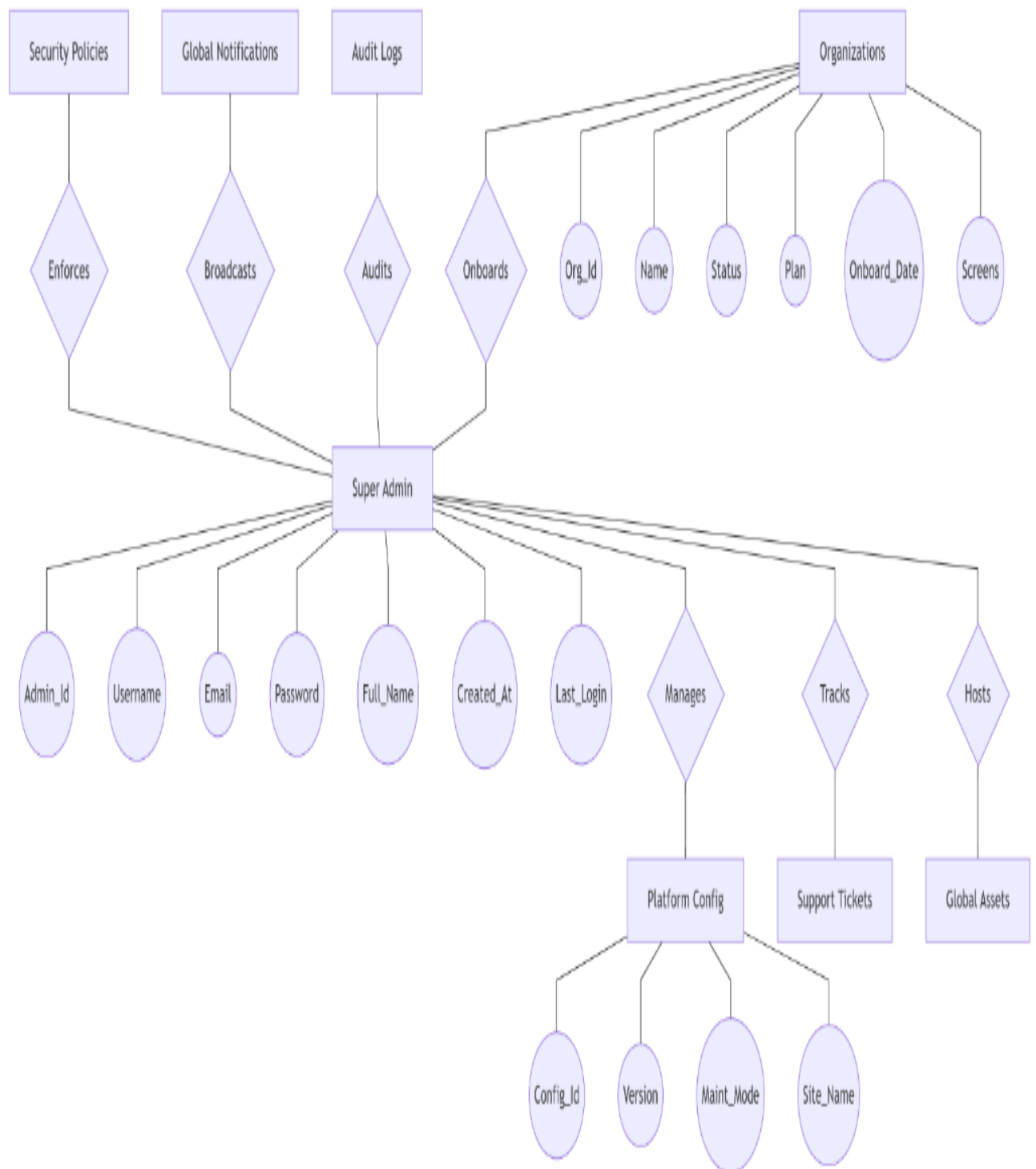


➤ Use CASE Advertiser

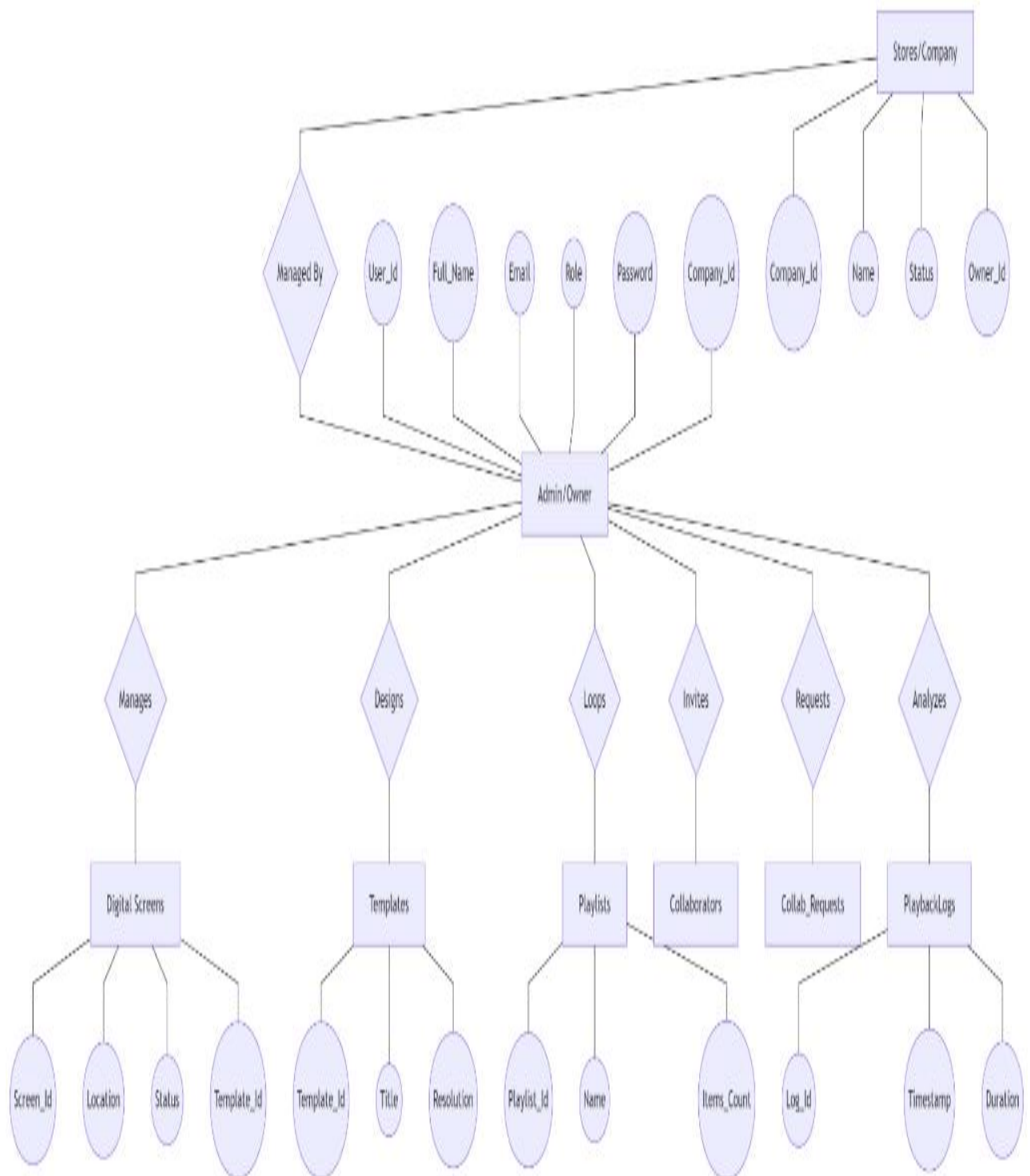


ER DIAGRAMS

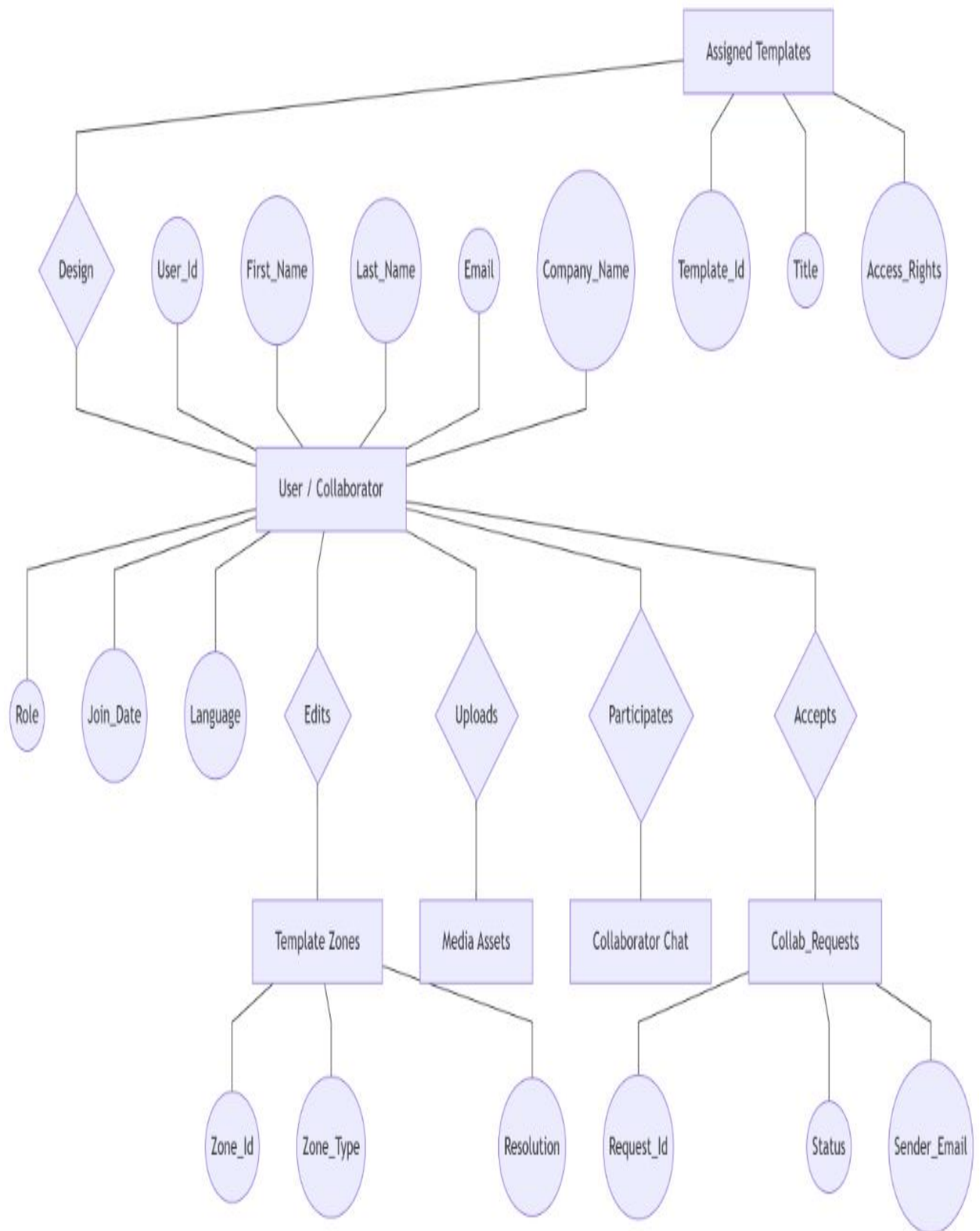
➤ ER Diagram Super Admin



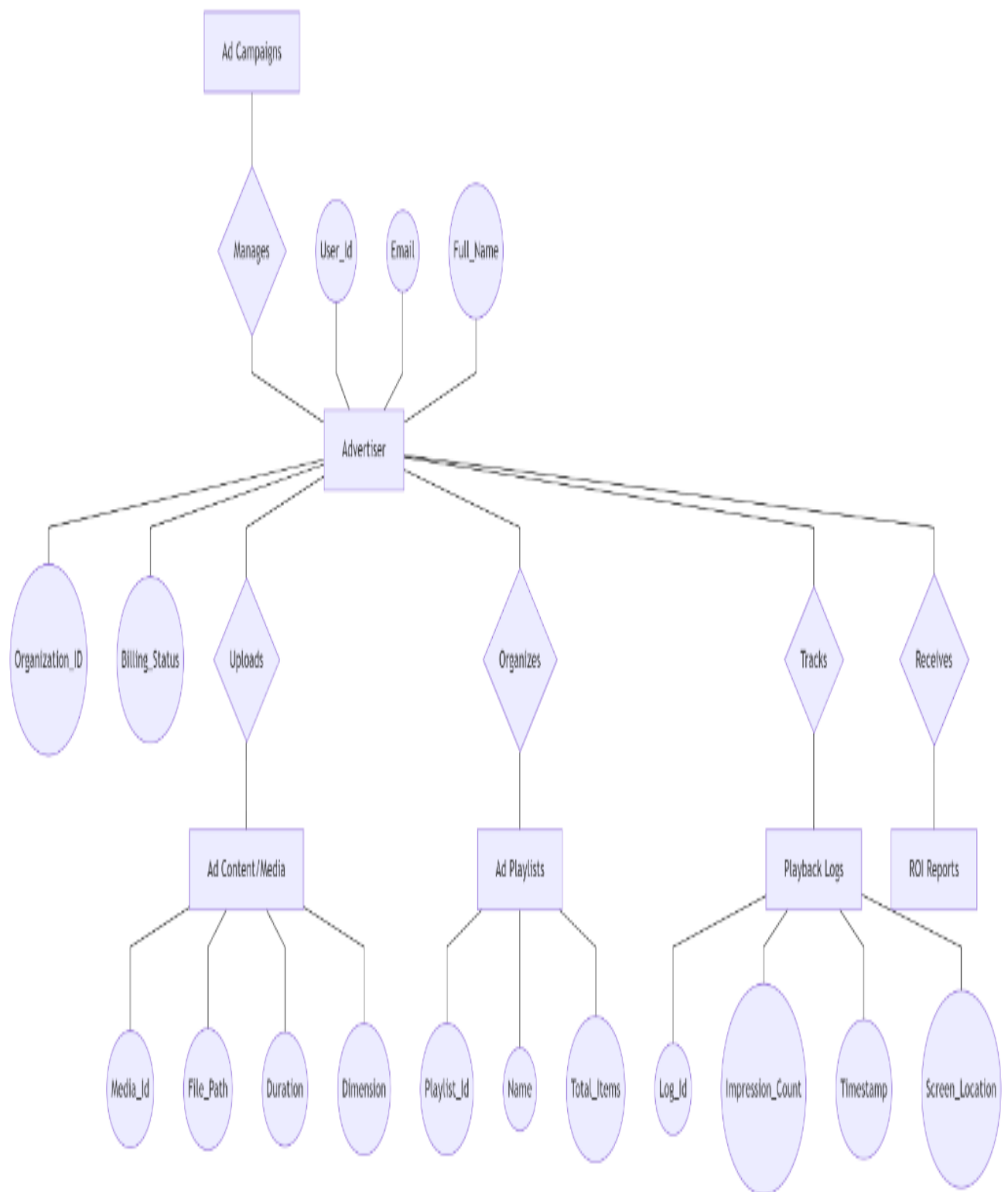
➤ ER Diagram Admin



➤ ER Diagram User



➤ ER Diagram Advertiser



DATA DICTIONARY

- **Data Dictionary** The Data Dictionary provides a comprehensive technical breakdown of the system's global database schema, detailing the constraints and operational purposes of every collection in the **SmartSignDeck** ecosystem.

a. User Table: System-wide user registry and authentication

Sr no.	Field name	Datatype	Size	Description	Constraint	Example
1.	_id	ObjectId	24	Unique identifier for the user record	Primary	60f...
2.	first_name	Varchar	50	User's first name	Required	Dhimant
3.	last_name	Varchar	50	User's last name	Required	Pandya
4.	email	Varchar	100	Unique system-wide identity	Unique, Req	dp@smart.com
5.	authProvider	Enum	10	local / google	Default: local	google
6.	password	Varchar	255	Hashed Bcrypt string	Required (local)	*****
7.	role	Enum	15	admin / user / advertiser / super_admin	Required	admin
8.	is_email_verified	Boolean	1	Flag for OTP confirmation	Default: False	true
9.	companyId	ObjectId	24	Link to owner organization	Ref: Company	60a...

b. Company Table: Organization and Multi-Tenant definitions

Sr no.	Field name	Datatype	Size	Description	Constraint	Example
1.	_id	ObjectId	24	Unique organization identifier	Primary	60a...
2.	name	Varchar	100	Digital storefront name	Unique, Req	Grace Ads
3.	ownerId	ObjectId	24	Account owner/creator reference	Ref: User	60f...
4.	isActive	Boolean	1	Flag for tenant suspension	Default: true	true

c. Screen Table: Physical hardware registration and status

Sr no.	Field name	Datatype	Size	Description	Constraint	Example
1.	_id	ObjectId	24	Internal screen identity	Primary	60s...
2.	name	Varchar	50	Human-readable screen name	Required	Lobby-01
3.	templateId	ObjectId	24	Currently rendering layout	Ref: Template	60t...
4.	visibility	Enum	10	private / company / public	Required	company
5.	secretKey	Varchar	32	Hardware pairing token	Unique/Index	ssh-key...
6.	status	Enum	10	online / offline / syncing	Default: off	online
7.	lastPing	Date	-	Heartbeat timestamp	Index	2024-03-18
8.	schedules	Array	-	Embedded Playback timing logic	Required	[...]

d. Template Table: Digital Canvas logic and zone layout

Sr no.	Field name	Datatype	Size	Description	Constraint	Example
1.	_id	ObjectId	24	Internal template ID	Primary	60t...

Sr no.	Field name	Datatype	Size	Description	Constraint	Example
2.	name	Varchar	50	Design layout title	Required	Hero Grid
3.	resolution	Varchar	20	Target Aspect/Resolution	Required	1920x1080
4.	zones	Mixed	-	JSON Definition of Canvas objects	Required	[...]

e. Template Group Table

Sr no.	Field name	Datatype	Size	Description	Constraint	Example
1.	_id	ObjectId	24	Unique Group Identity	Primary	60tg...
2.	name	Varchar	100	Display name of the group	Required	Holiday Layouts
3.	description	Text	-	Optional details about the group	Optional	Winter Screens
4.	companyId	ObjectId	24	Target Organization	Ref: Company	60c...
5.	createdBy	ObjectId	24	Original creator	Ref: User	60u...
6.	templates	Array	-	List of templates in this group	Ref: Template	[...]

f. Collaboration Request Table: Canvas asset sharing tracking

Sr no.	Field name	Datatype	Size	Description	Constraint	Example
1.	_id	ObjectId	24	Request unique ID	Primary	60cr...
2.	sender	ObjectId	24	User sending the invite	Ref: User	60u...
3.	recipient	ObjectId	24	User receiving the invite	Ref: User	60v...
4.	templateId	ObjectId	24	Subject template being shared	Ref: Template	60t...
5.	status	Enum	15	pending / accepted / declined	Default: pending	pending
6.	message	Varchar	255	Optional context from sender	Optional	Please review

g. Role Table: Access Control and Permission Levels

Sr no.	Field name	Datatype	Size	Description	Constraint	Example
1.	_id	ObjectId	24	Unique Role ID	Primary	60r...
2.	name	Varchar	20	Display name of the role	Required	Org Admin
3.	permissions	Array	-	List of allowed API actions	Required	['read','write']

h. Permission Table: Granular access control logic

Sr no.	Field name	Datatype	Size	Description	Constraint	Example
1.	_id	ObjectId	24	Unique Permission ID	Primary	60pr...
2.	name	Varchar	50	Immutable identifier	Unique, Req	create_user
3.	description	Text	-	Human readable explanation	Required	Allows creation
4.	resource	Varchar	50	Target API or Entity	Required	/users

Sr no.	Field name	Datatype	Size	Description	Constraint	Example
5.	action	Enum	20	create / read / update / delete	Required	create
6.	status	Enum	10	active / inactive	Default: active	active

i. Pending Signup Table : Registration flow staging

Sr no.	Field name	Datatype	Size	Description	Constraint	Example
1.	email	Varchar	100	Staged email for verification	Primary/Unique	new@user.com
2.	otp	Varchar	6	6-digit verification code	Required	123456
3.	otpExpires	Date	-	Expiration for TTL removal	TTL Index	2024-03-18

j. Playlist Table: Logical grouping of Ad files

Sr no.	Field name	Datatype	Size	Description	Constraint	Example
1.	_id	ObjectId	24	Unique Playlist identity	Primary	60p...

Sr no.	Field name	Datatype	Size	Description	Constraint	Example
2.	name	Varchar	50	Campaign or Group name	Required	Summer Ads
3.	companyId	ObjectId	24	Owning Organization	Ref: Company	60c...
4.	items	Array	-	Embedded media asset definitions	Required	[...]

k. Admin Request Table: High-Privilege Operation Queue

Sr no.	Field name	Datatype	Size	Description	Constraint	Example
1.	_id	ObjectId	24	Request Identity	Primary	60ar...
2.	requesterId	ObjectId	24	Admin seeking approval	Ref: User	60u...
3.	targetUserId	ObjectId	24	Subject of the request	Ref: User	60v...
4.	companyId	ObjectId	24	Context organization	Ref: Company	60c...
5.	type	Enum	20	DELETE / ROLE_UPDATE	Required	DELETE
6.	status	Enum	10	PENDING / APPROVED	Default: PENDING	PENDING
7.	adminComment	Text	-	Note from Super Admin	Optional	"Valid request"

I. PlaybackLog Table: Behavioral Analytics Source

Sr no.	Field name	Datatype	Size	Description	Constraint	Example
1.	_id	ObjectId	24	Log identity	Primary	601...
2.	contentUrl	Varchar	255	Media URL being displayed	Required	res...
3.	mediaId	ObjectId	24	Ad asset tracked	Ref: Media	60m...
4.	timestamp	Date	-	Execution time	Required	2024-03-26

m. Message Table: Collaboration Chat Logs

Sr no.	Field name	Datatype	Size	Description	Constraint	Example
1.	_id	ObjectId	24	message ID	Primary	60c...
2.	senderId	ObjectId	24	User source	Ref: User	60u...
3.	receiverId	ObjectId	24	User target	Ref: User	60v...
4.	text	Text	-	Raw message content	Required	Hello team
5.	createdBy	ObjectId	24	Original designer	Ref: User	60f...
6.	deletedAt	Date	-	Soft-delete timestamp for recycle bin	Optional	2024-03-18

n. Token Table: JWT and Session management

Sr no.	Field name	Datatype	Size	Description	Constraint	Example
1.	jti	Varchar	50	Unique JWT ID	Unique, Req	uuid-123
2.	user	ObjectId	24	Bearer of the token	Ref: User	60f...
3.	type	Enum	15	refresh / reset_password / access	Required	refresh
4.	expires	Date	-	Token expiration date	Required	2024-03-25

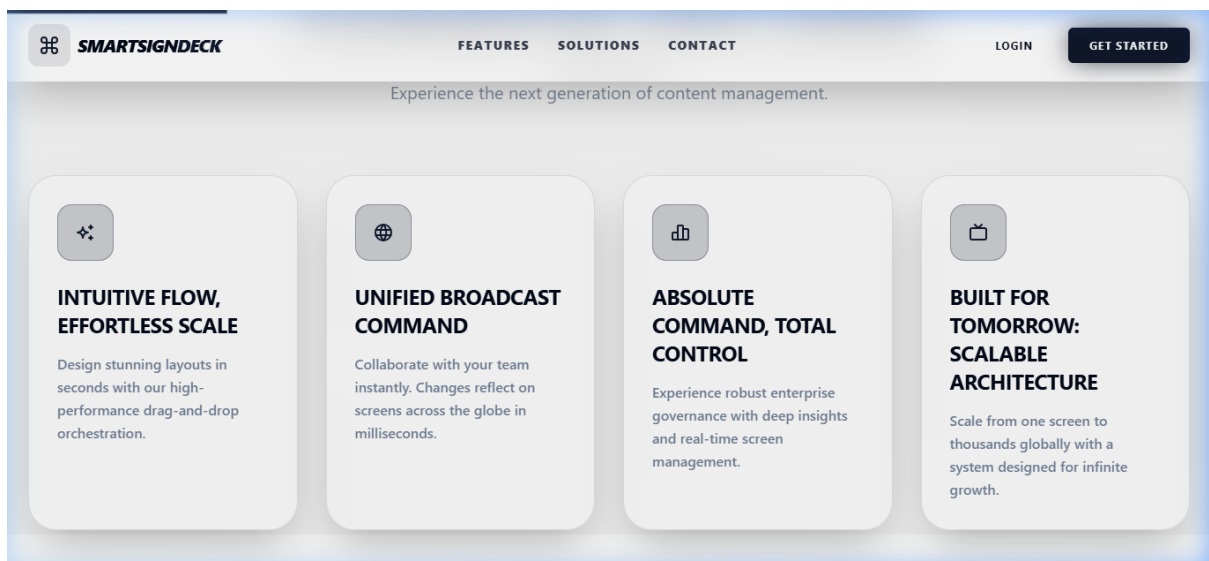
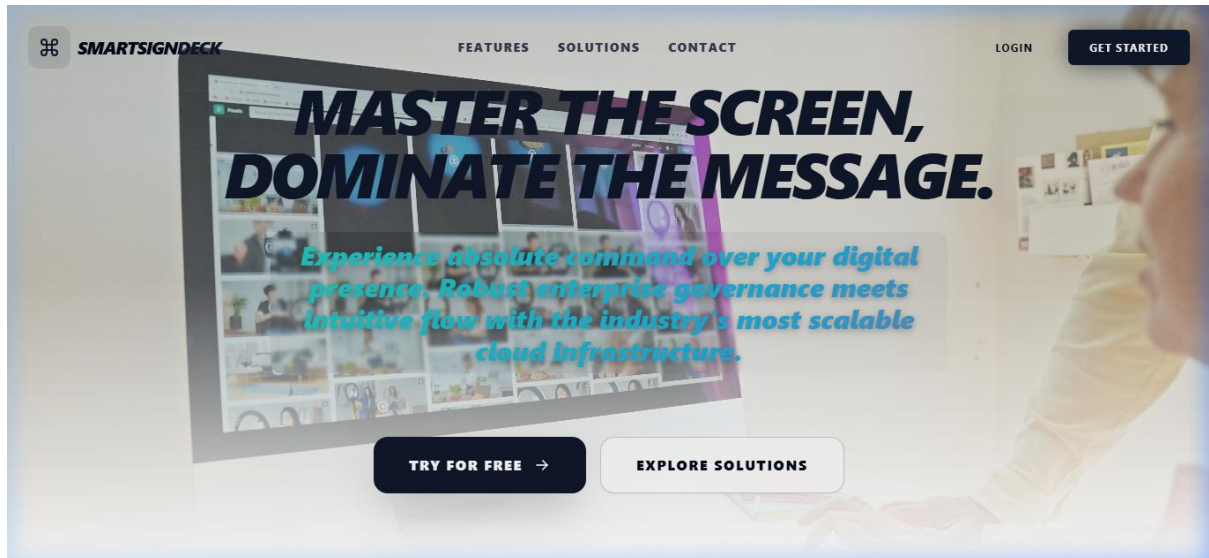
o. Notification Table: Global application alerts and messages

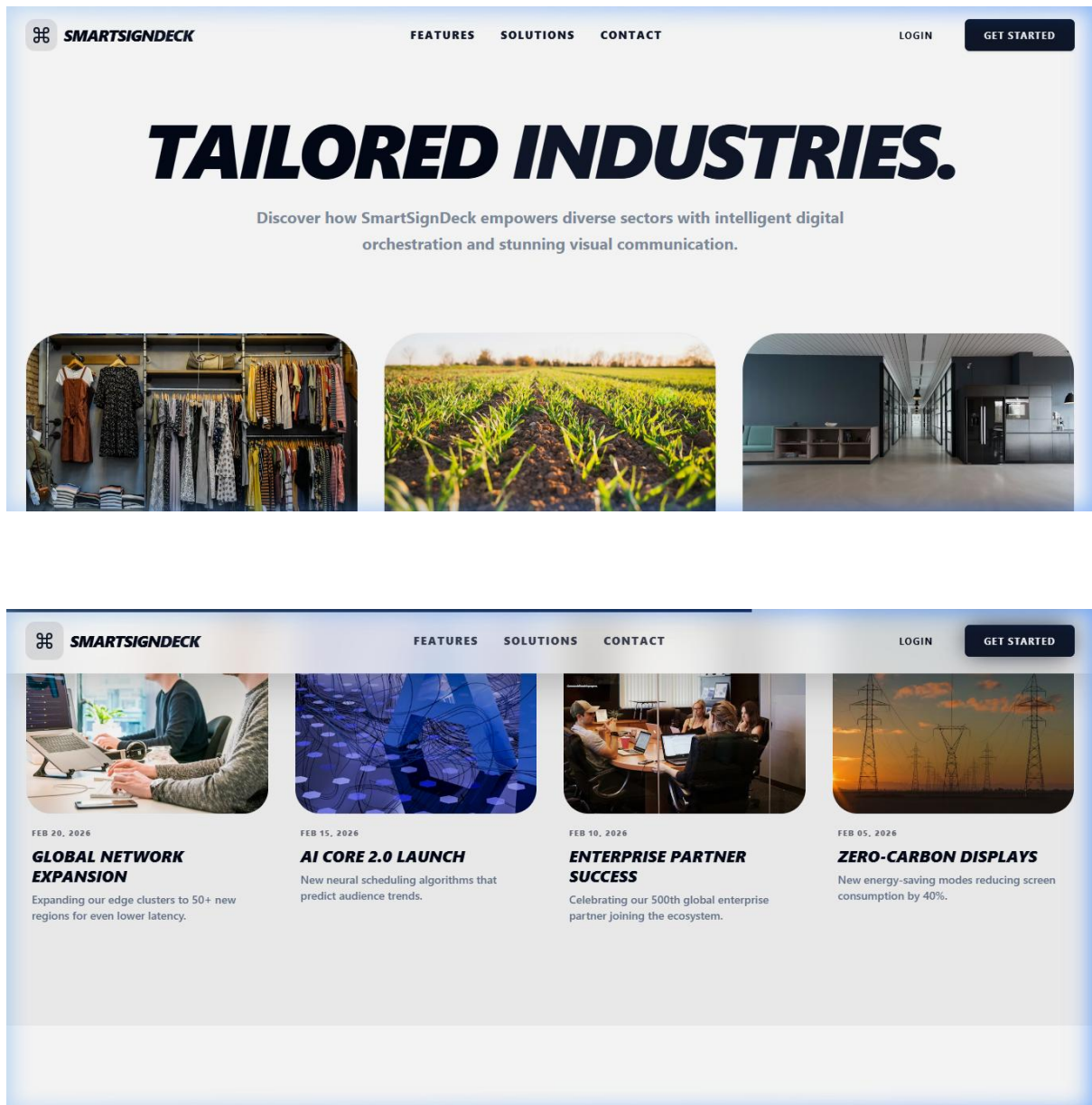
Sr no.	Field name	Datatype	Size	Description	Constraint	Example
1.	_id	ObjectId	24	Notification Identity	Primary	60n...

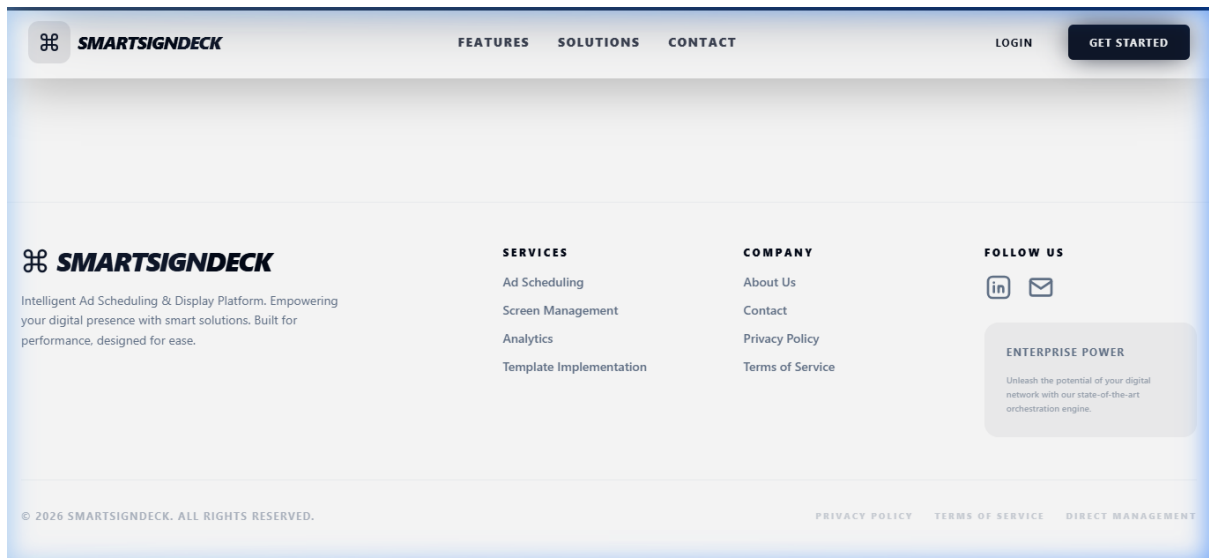
Sr no.	Field name	Datatype	Size	Description	Constraint	Example
2.	recipientId	ObjectId	24	Target User ID	Ref: User	60u...
3.	senderId	ObjectId	24	Originating User (If any)	Ref: User	60v...
4.	type	Enum	25	new_chat / system_alert	Required	system_alert
5.	title	Varchar	100	Subject of the notification	Required	Maintenance at 1
6.	message	Text	-	Raw text body of the alert	Required	Server reboot
7.	isRead	Boolean	1	Tracking whether user saw it	Default: false	false
8.	status	Enum	10	active / archived	Default: active	active

SCREEN LAYOUTS WITH DESCRIPTION

Landing Pages

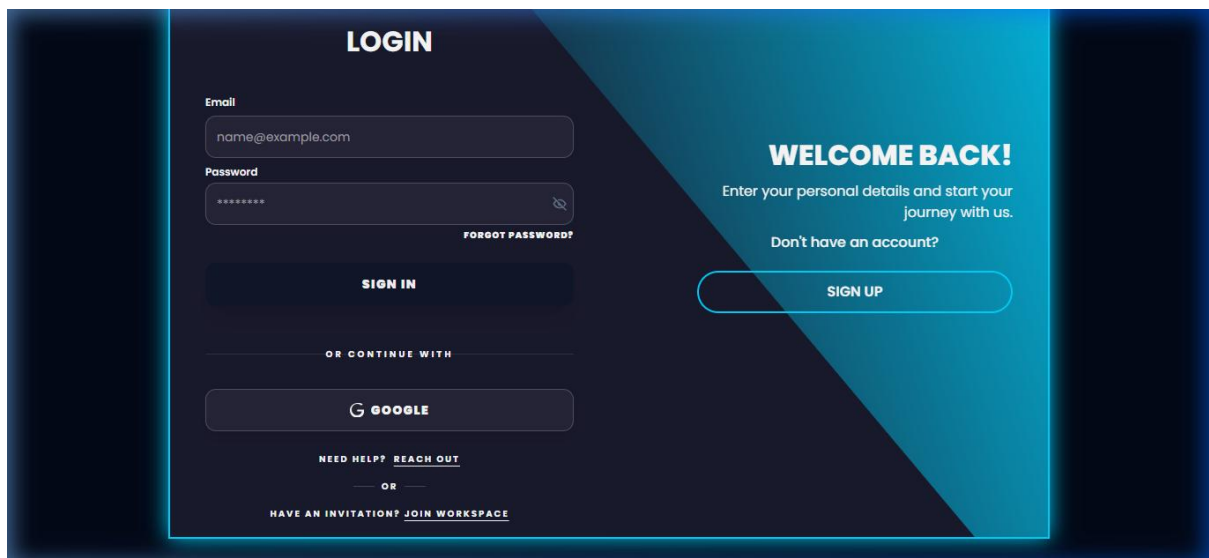






Authentication Pages

Login Page



LOGIN

Email

home@example.com

Please enter your email

Password

Please enter your password

[FORGOT PASSWORD?](#)

SIGN IN

OR CONTINUE WITH

GOOGLE

WELCOME BACK!

Enter your personal details and start your journey with us.

Don't have an account?

SIGN UP

LOGIN

Email

test@example.com

Please enter your email

Password

Please enter your password

[FORGOT PASSWORD?](#)

SIGN IN

OR CONTINUE WITH

GOOGLE

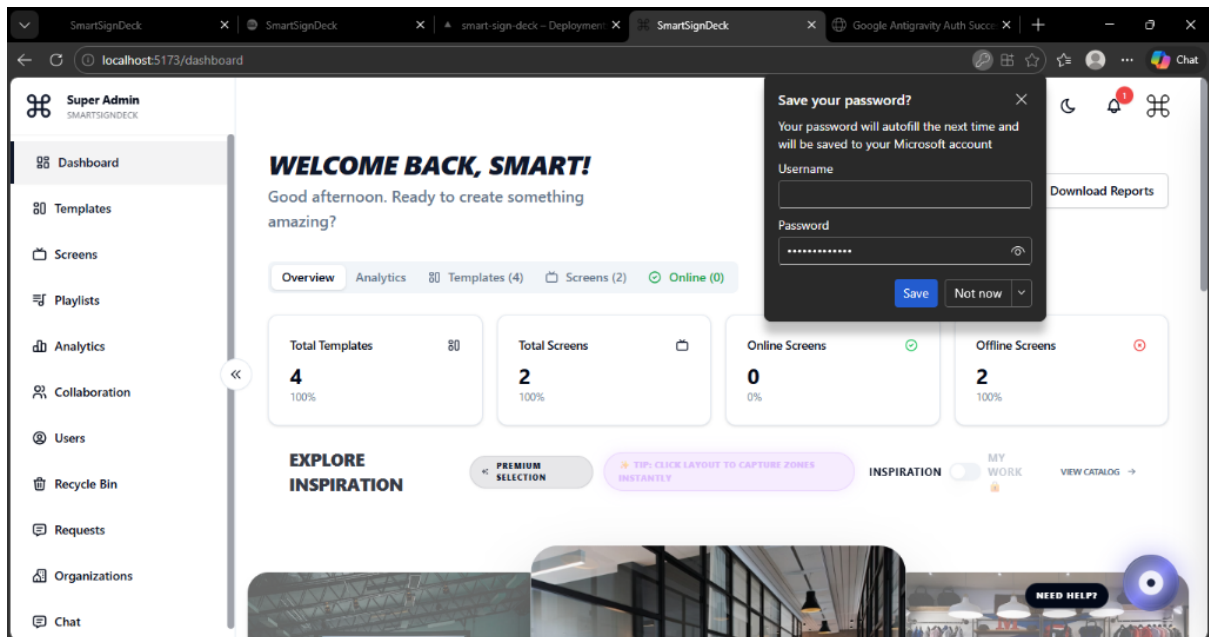
WELCOME BACK!

Enter your personal details and start your journey with us.

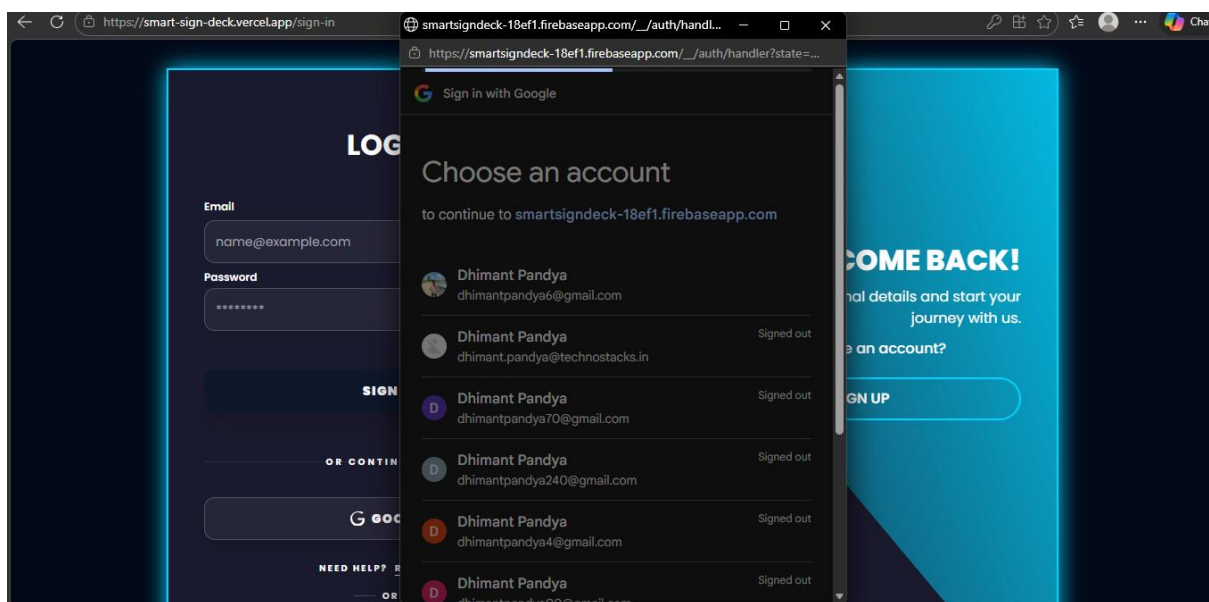
Don't have an account?

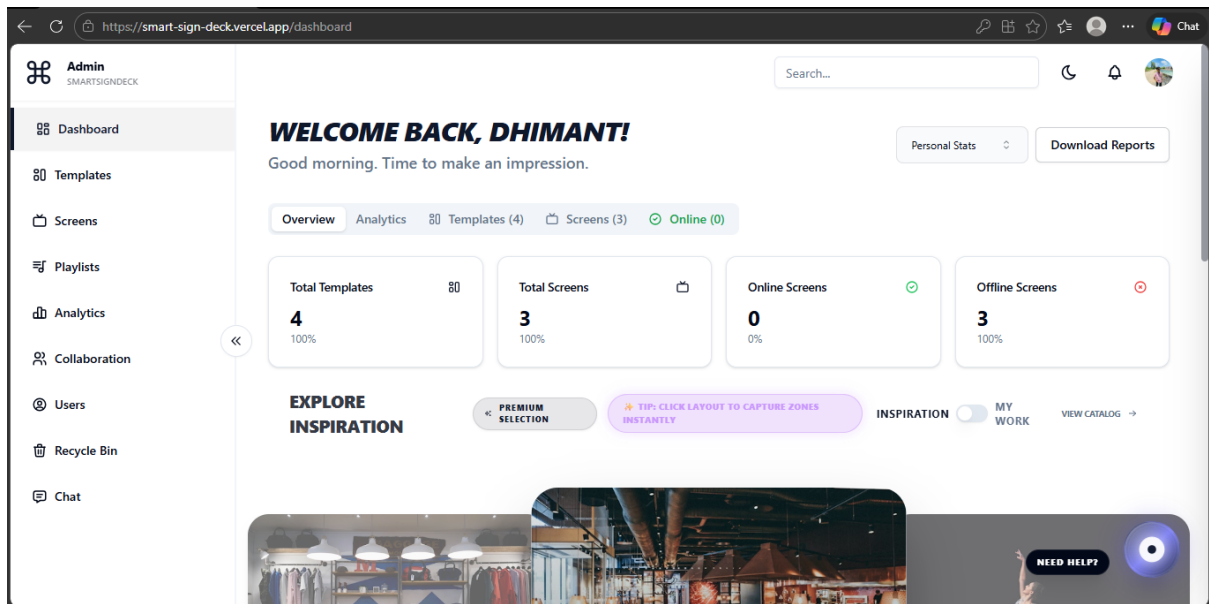
SIGN UP

[NEED HELP? REACH OUT](#)

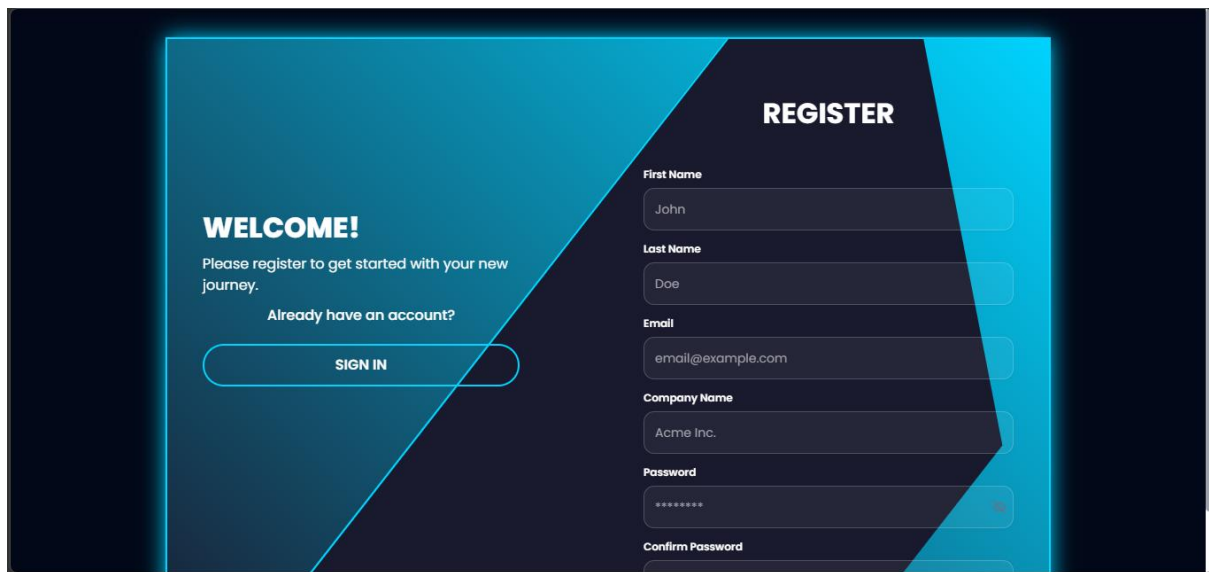


Login Using Google





Registration



SmartSignDeck Thesis Preview

https://smart-sign-deck.vercel.app/sign-up

WELCOME!
Please register to get started with your new journey.
Already have an account?
[SIGN IN](#)

CREATE ACCOUNT

OR CONTINUE WITH

[GOOGLE](#)

NEED HELP? [REACH OUT](#)

OR

HAVE AN INVITATION? [JOIN WORKSPACE](#)

https://smart-sign-deck.vercel.app/sign-up

WELCOME!
Please register to get started with your new journey.
Already have an account?
[SIGN IN](#)

REGISTER

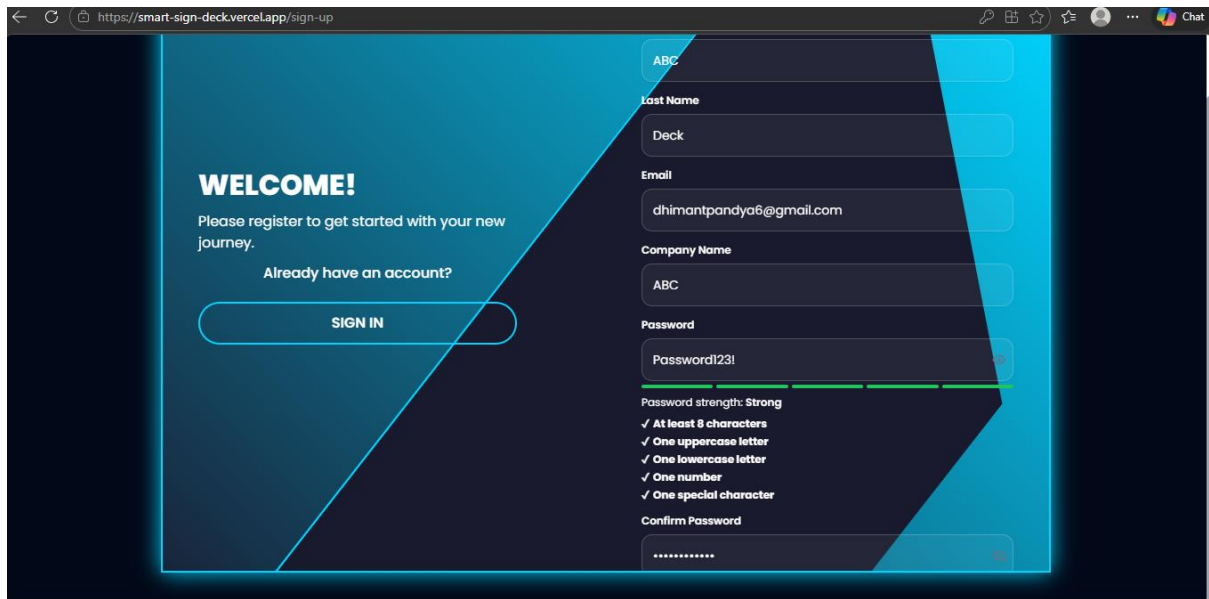
First Name
John
First name is required

Last Name
Doe
Last name is required

Email
email@example.com
Invalid email

Company Name
Acme Inc.
Company Name is required

Password

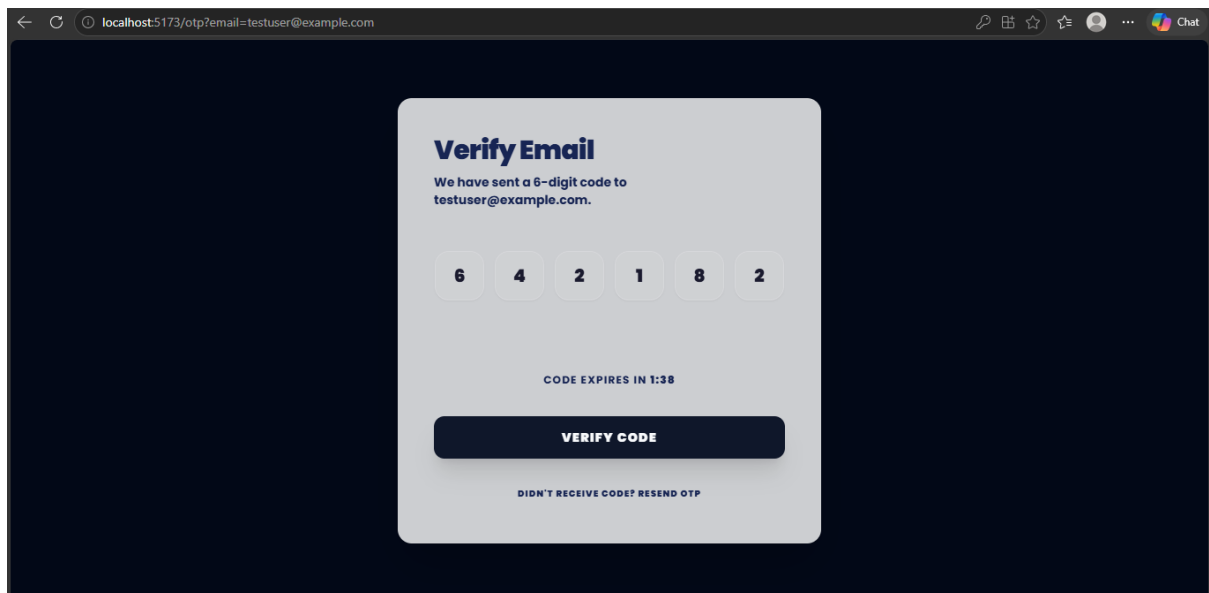


A screenshot of a web browser showing the sign-up page for SmartSignDeck. The page has a dark blue background with a teal diagonal stripe. On the left, a 'WELCOME!' message encourages registration, with a 'SIGN IN' button for existing users. On the right, a form with white input fields collects user details: Last Name (ABC), Deck (Deck), Email (dhimantpandya6@gmail.com), Company Name (ABC), Password (Password123!), and Confirm Password (masked). A password strength indicator shows 'Strong' with five green bars and a list of criteria: at least 8 characters, one uppercase letter, one lowercase letter, one number, and one special character.

https://smart-sign-deck.vercel.app/sign-up

WELCOME!
Please register to get started with your new journey.
Already have an account?
SIGN IN

ABC
Last Name
Deck
Email
dhimantpandya6@gmail.com
Company Name
ABC
Password
Password123!
Password strength: Strong
✓ At least 8 characters
✓ One uppercase letter
✓ One lowercase letter
✓ One number
✓ One special character
Confirm Password



A screenshot of a web browser showing the 'Verify Email' page. The page is dark blue with a central white card. The card contains the title 'Verify Email', a message stating a 6-digit code was sent to testuser@example.com, and a sequence of six circular buttons displaying the code '642182'. Below the code, a timer indicates 'CODE EXPIRES IN 1:38'. A 'VERIFY CODE' button is prominently displayed, and a link for 'DIDN'T RECEIVE CODE? RESEND OTP' is at the bottom.

localhost:5173/otp?email=testuser@example.com

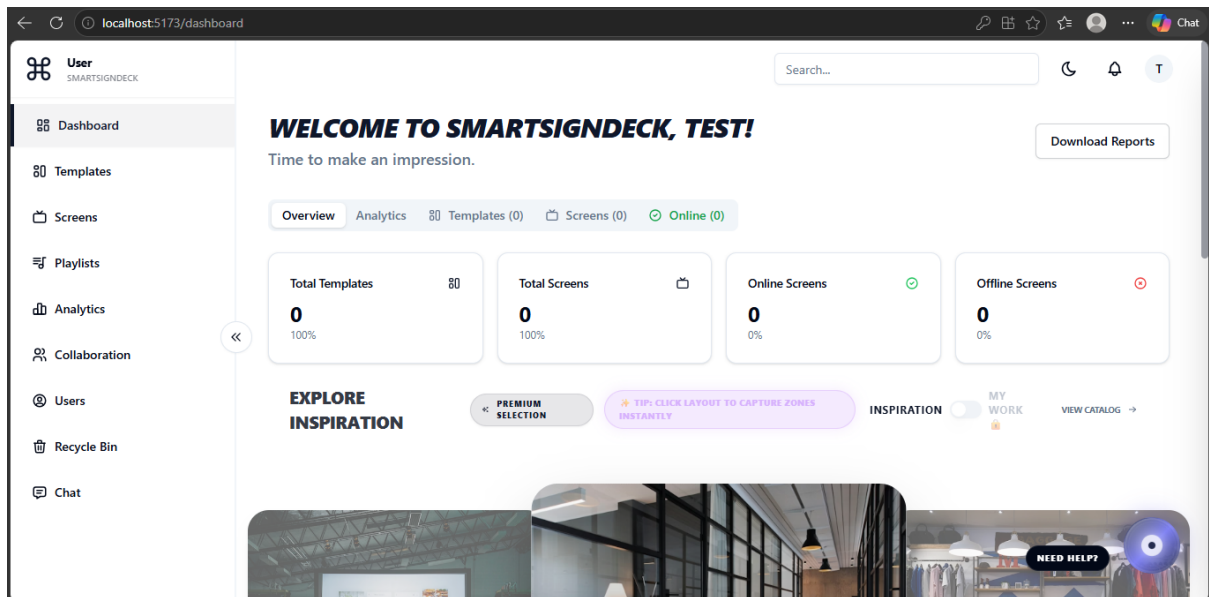
Verify Email
We have sent a 6-digit code to testuser@example.com.

6 4 2 1 8 2

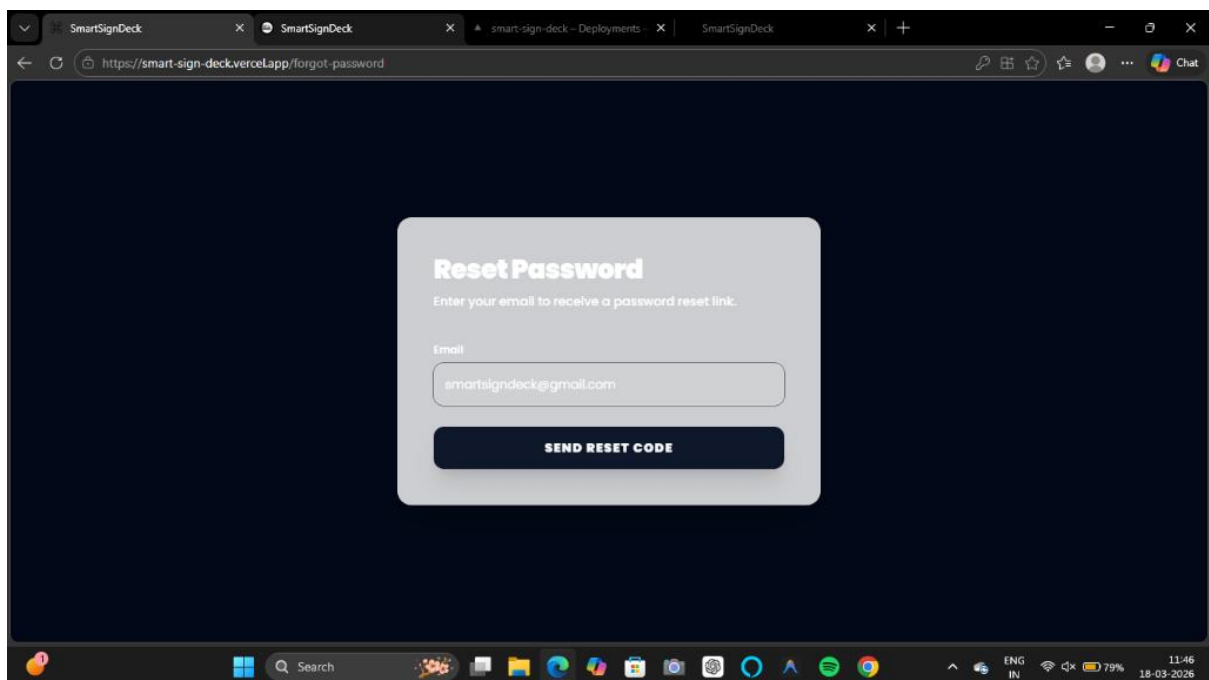
CODE EXPIRES IN 1:38

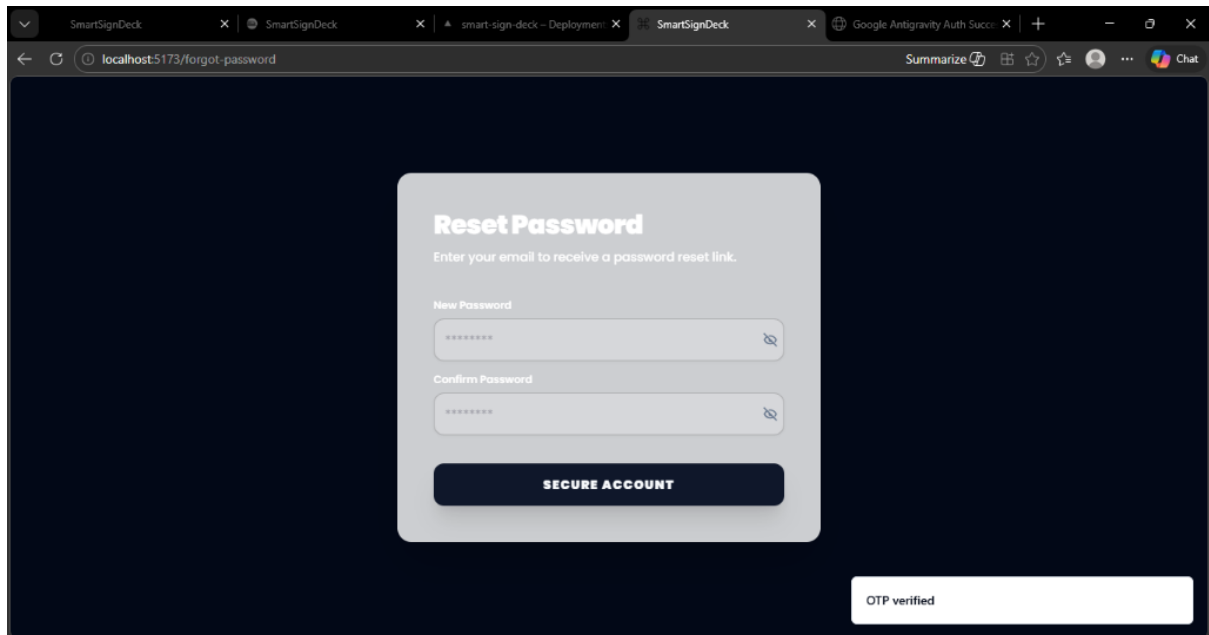
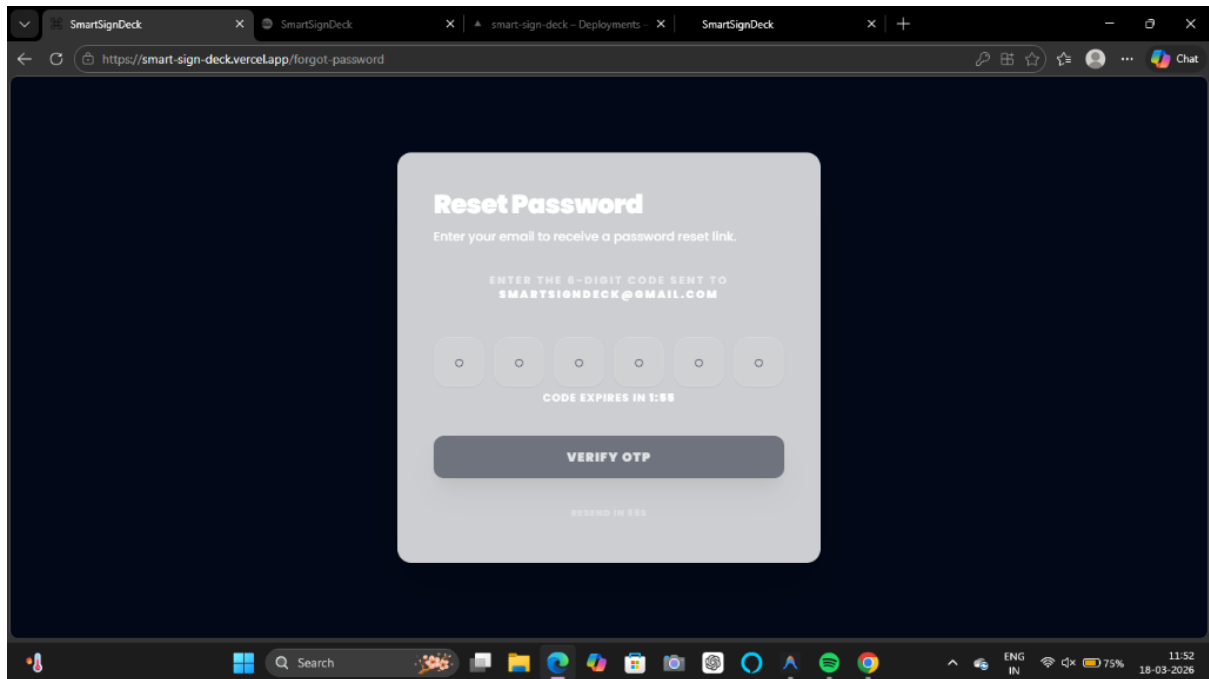
VERIFY CODE

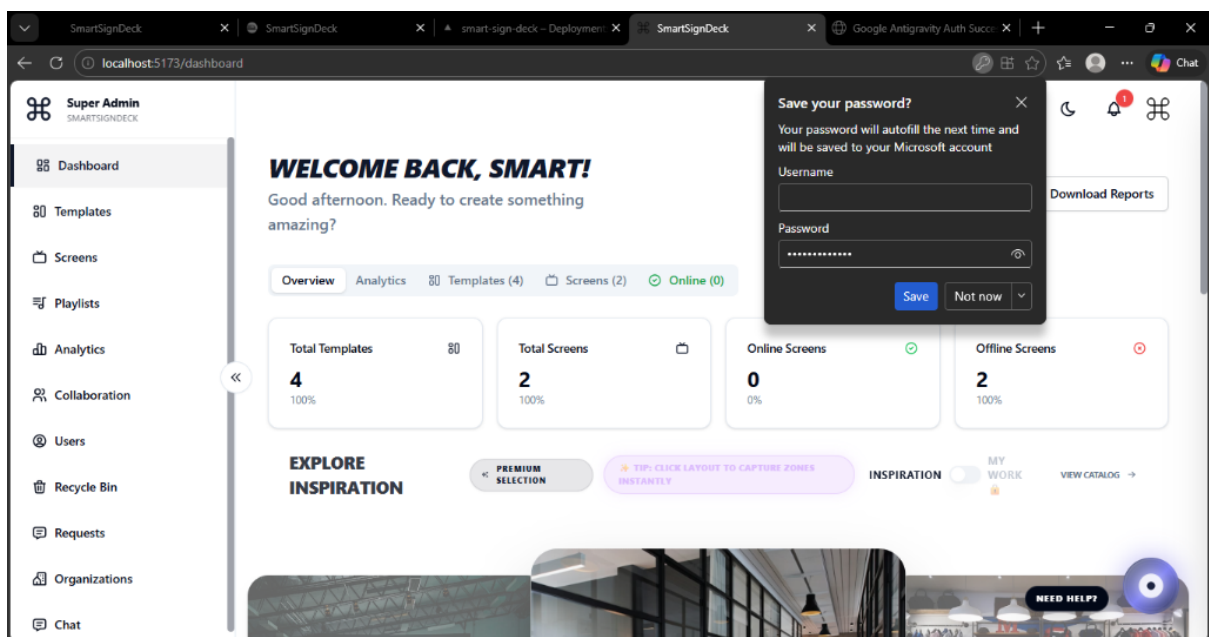
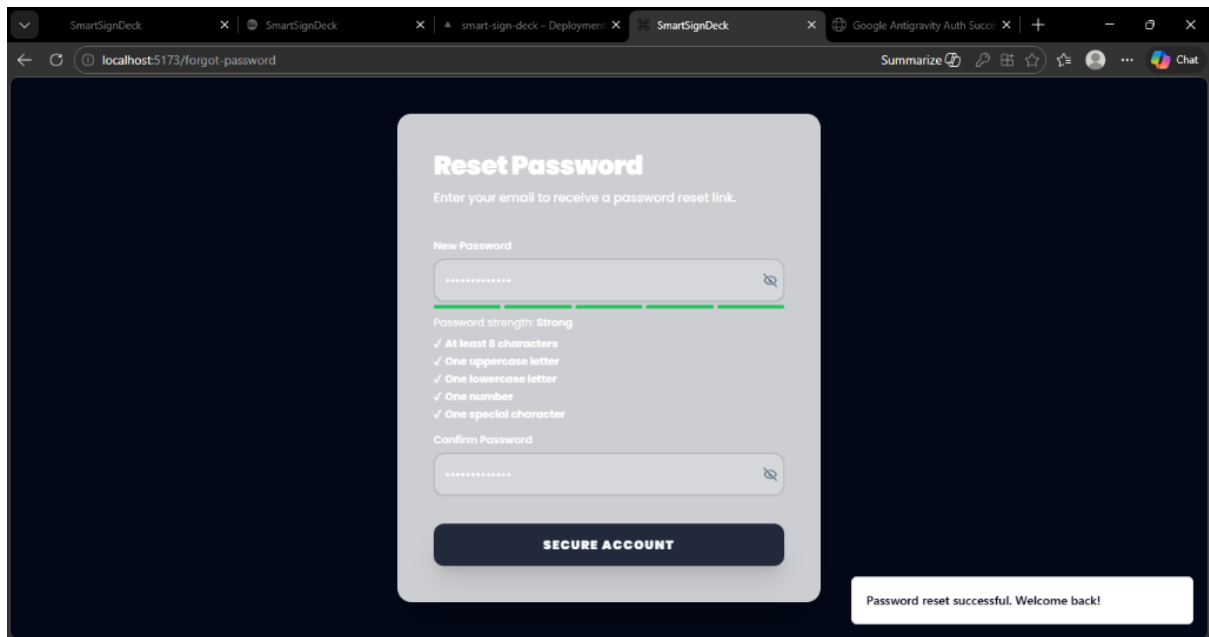
DIDN'T RECEIVE CODE? RESEND OTP



Reset Password







Profile

The screenshot shows the 'General Information' section of the user profile. The left sidebar contains navigation links: Dashboard, Templates, Screens, Playlists, Analytics, Collaboration, Users, Recycle Bin, and Chat. The main content area has a profile picture placeholder with an 'Upload picture' button. The 'General Information' form includes fields for First name* (Test), Last name* (User), Email* (testuser@example.com), Gender* (Male, Female, Other), Date of birth* (dd-mm-yyyy), and Language* (Select language). An 'Update Profile' button is at the bottom of this section. Below it is the 'Password Information' section, which is partially visible.

SmartSignDeck User

Dashboard
Templates
Screens
Playlists
Analytics
Collaboration
Users
Recycle Bin
Chat

Profile picture
Upload picture

General Information

First name*
Test

Last name*
User

Email*
testuser@example.com

Gender*
☐ Male ☐ Female ☐ Other

Date of birth*
dd-mm-yyyy

Language*
Select language

Update Profile

Password Information

Old password*

The screenshot shows the 'Update Password' and 'Danger Zone' sections of the user profile. The 'Update Password' section has fields for Old password*, New password*, and Confirm password*, each with a toggle to show or hide the password. There are 'Update Password' and 'Forgot password?' buttons. The 'Danger Zone' section has a warning message: 'Delete your account and all associated data. This action is permanent and cannot be undone.' and a red 'Delete my account' button.

SmartSignDeck User

Dashboard
Templates
Screens
Playlists
Analytics
Collaboration
Users
Recycle Bin
Chat

Old password*
Enter old password

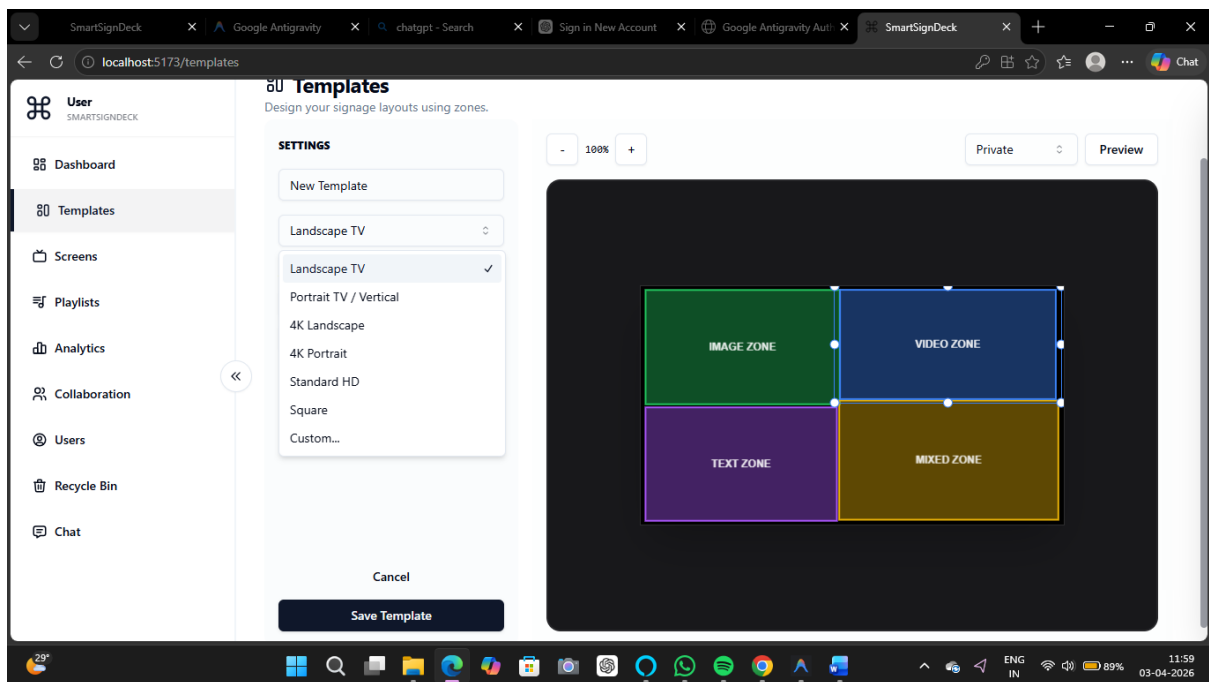
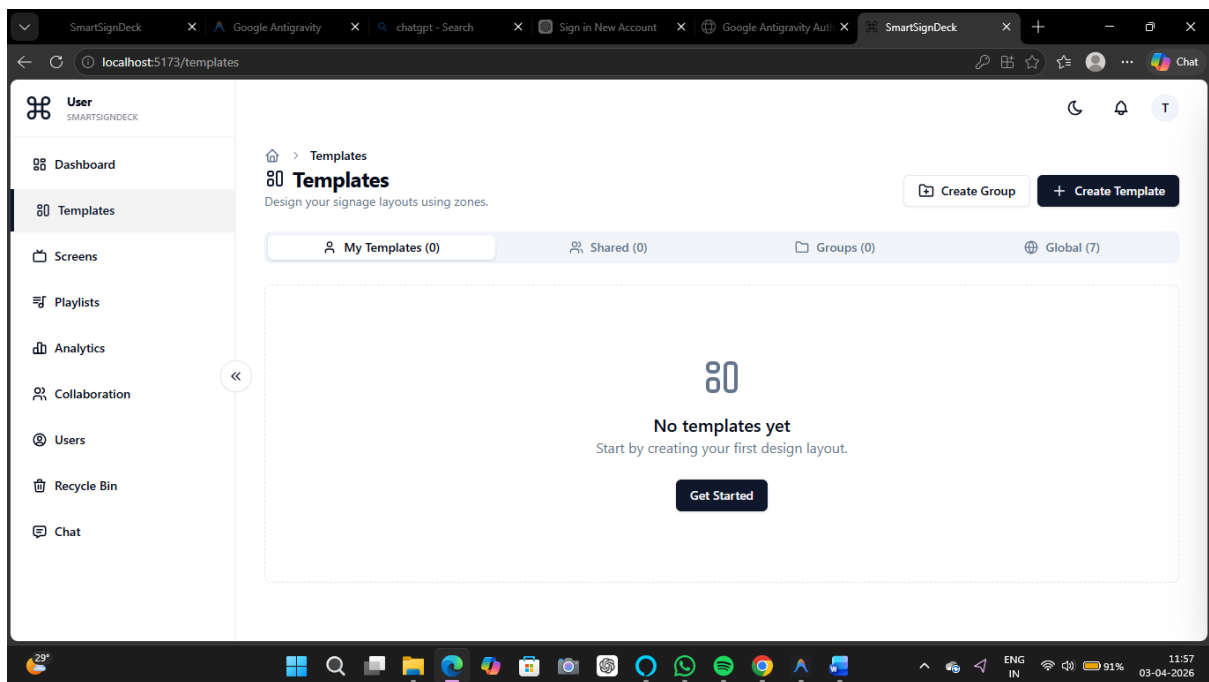
New password*
Enter new password

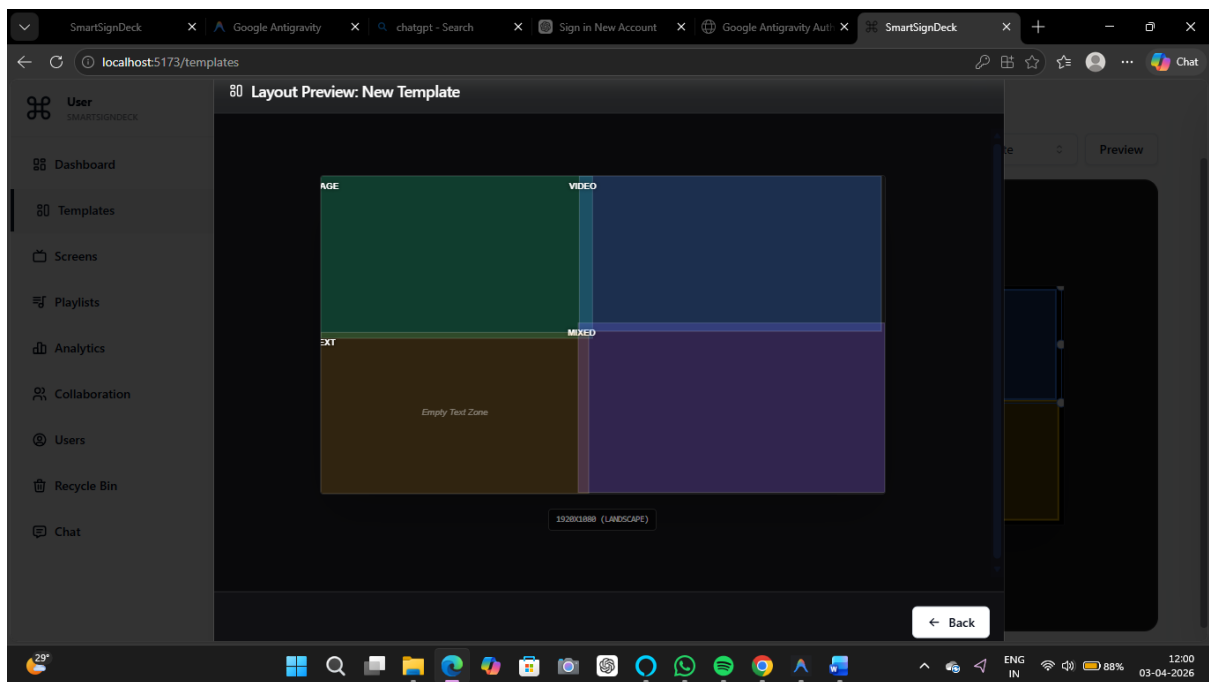
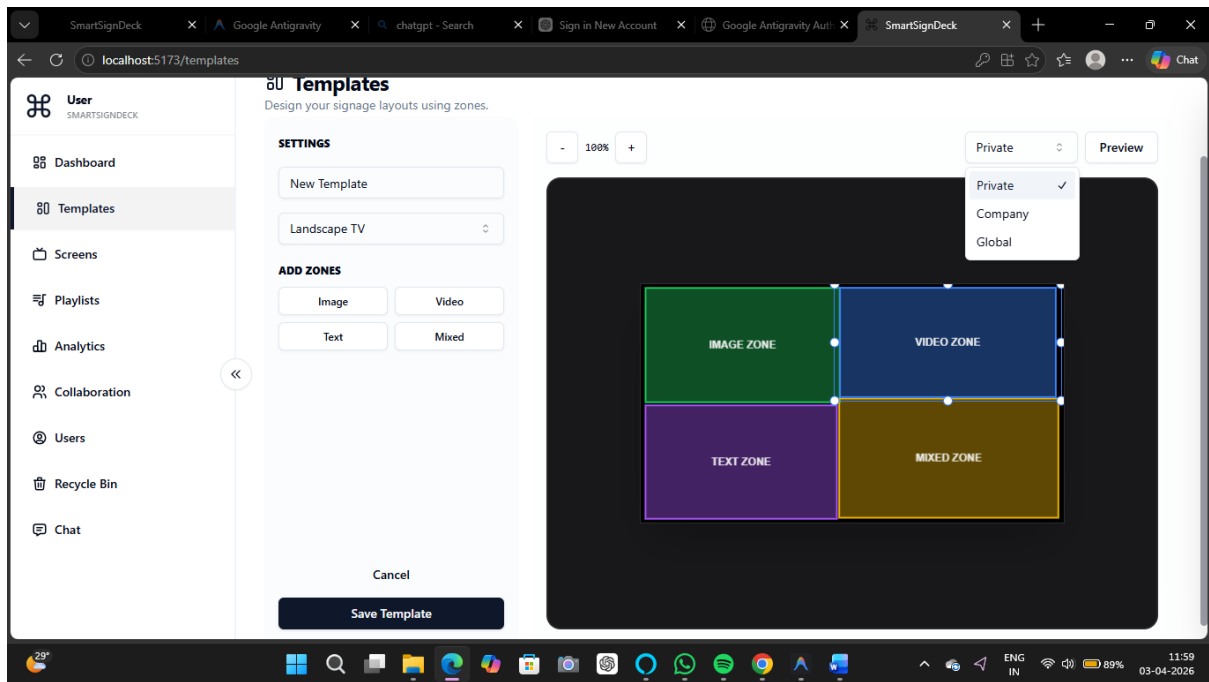
Confirm password*
Enter confirm password

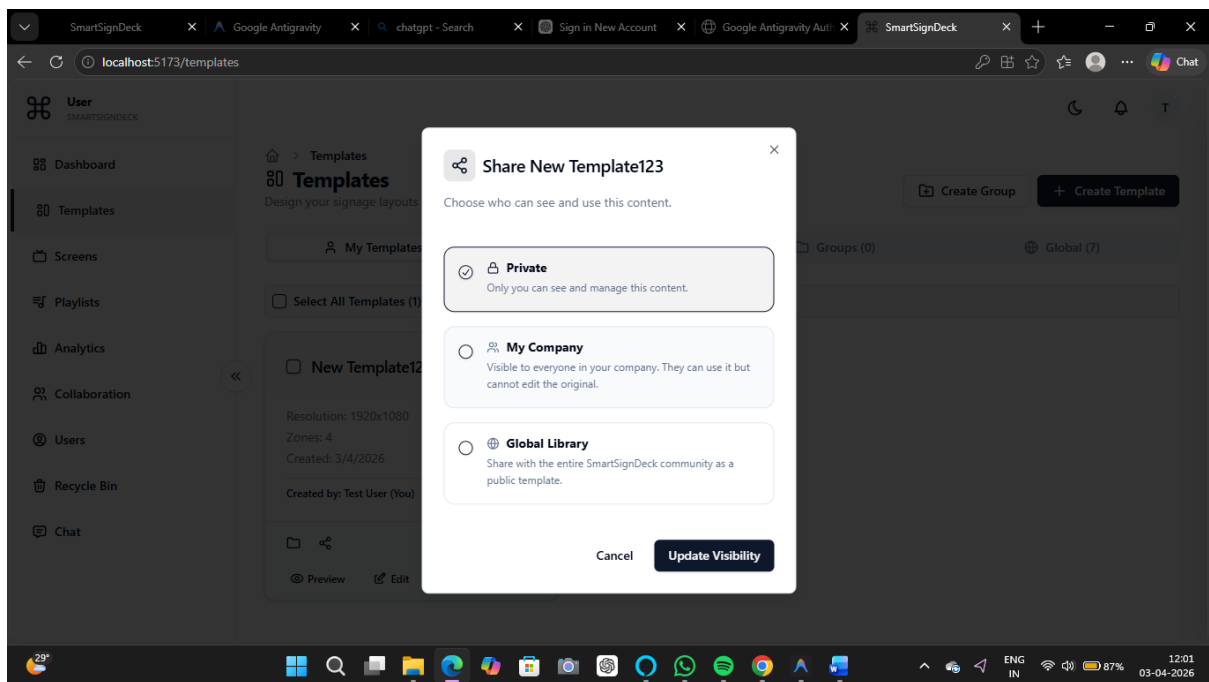
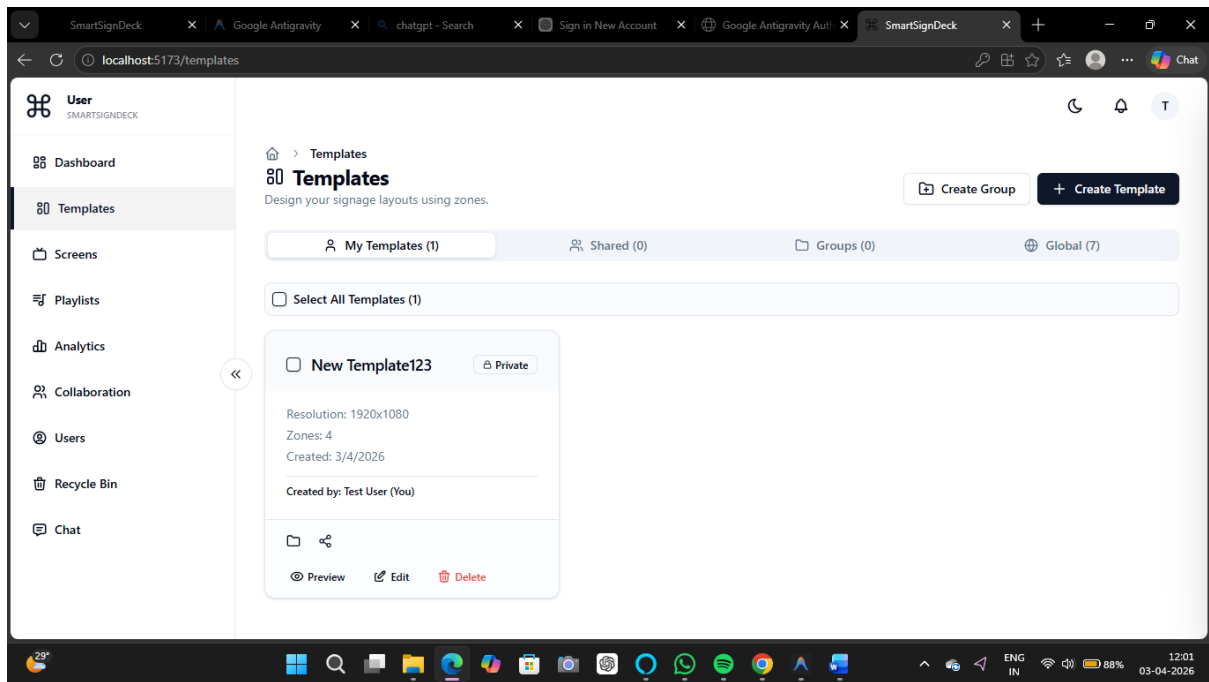
Update Password
Forgot password?

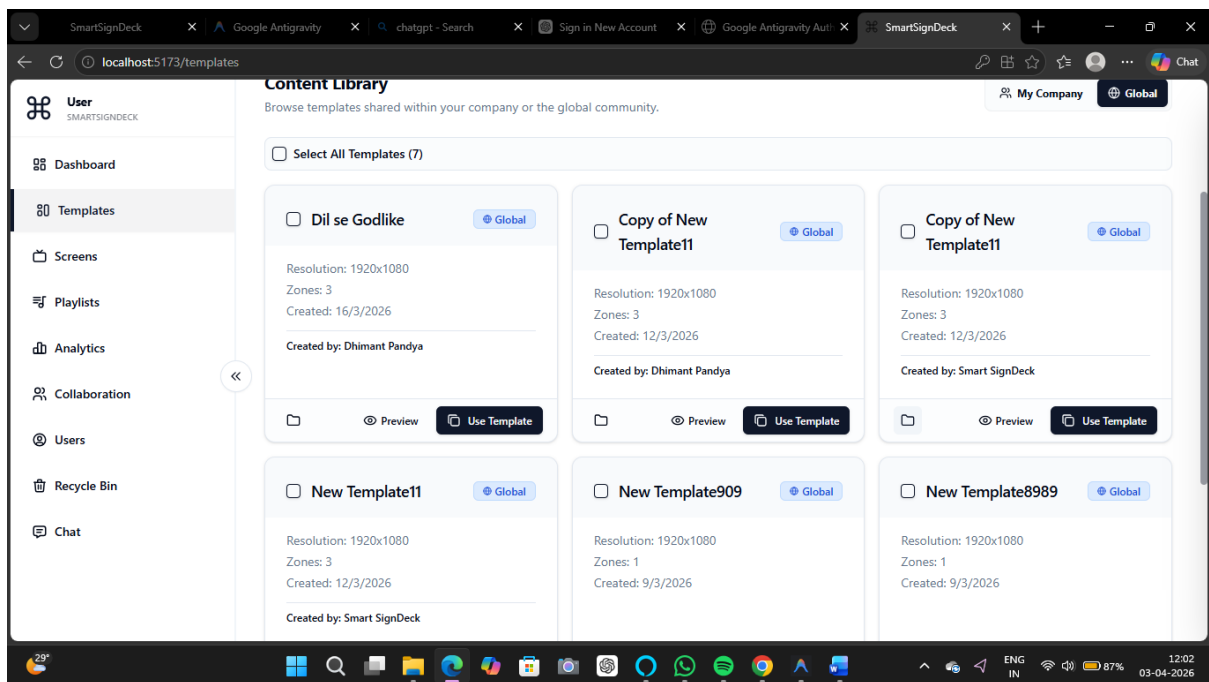
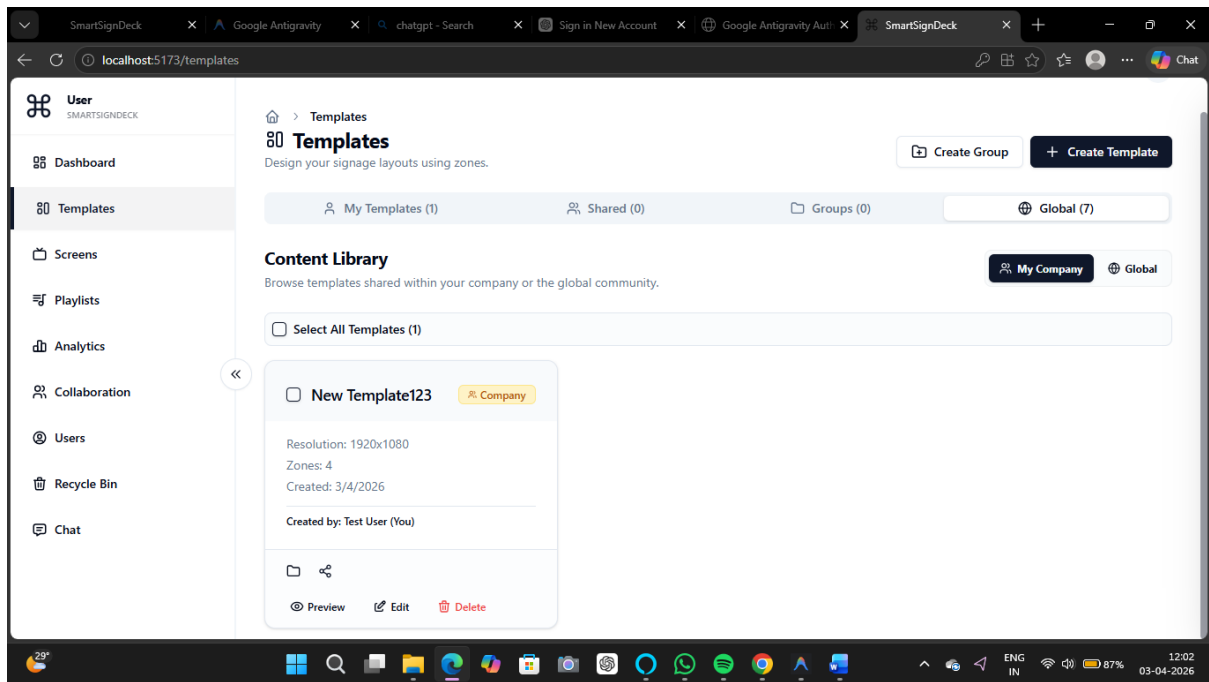
Danger Zone
Delete your account and all associated data.
This action is permanent and cannot be undone.
Delete my account

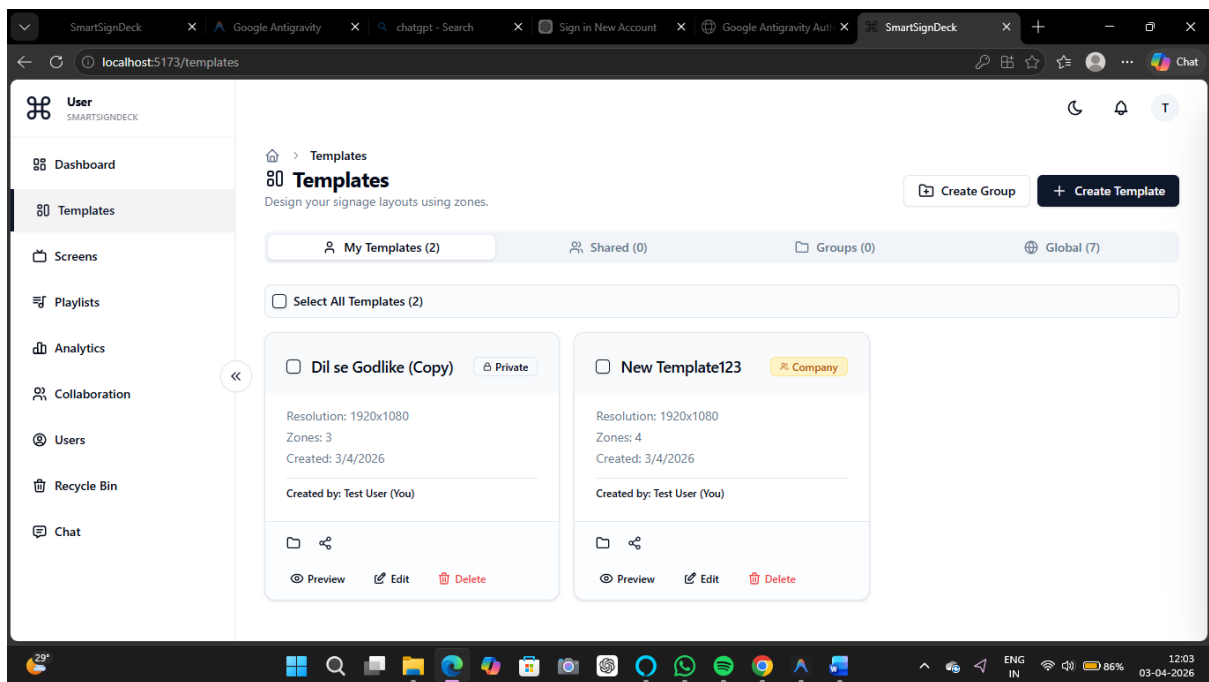
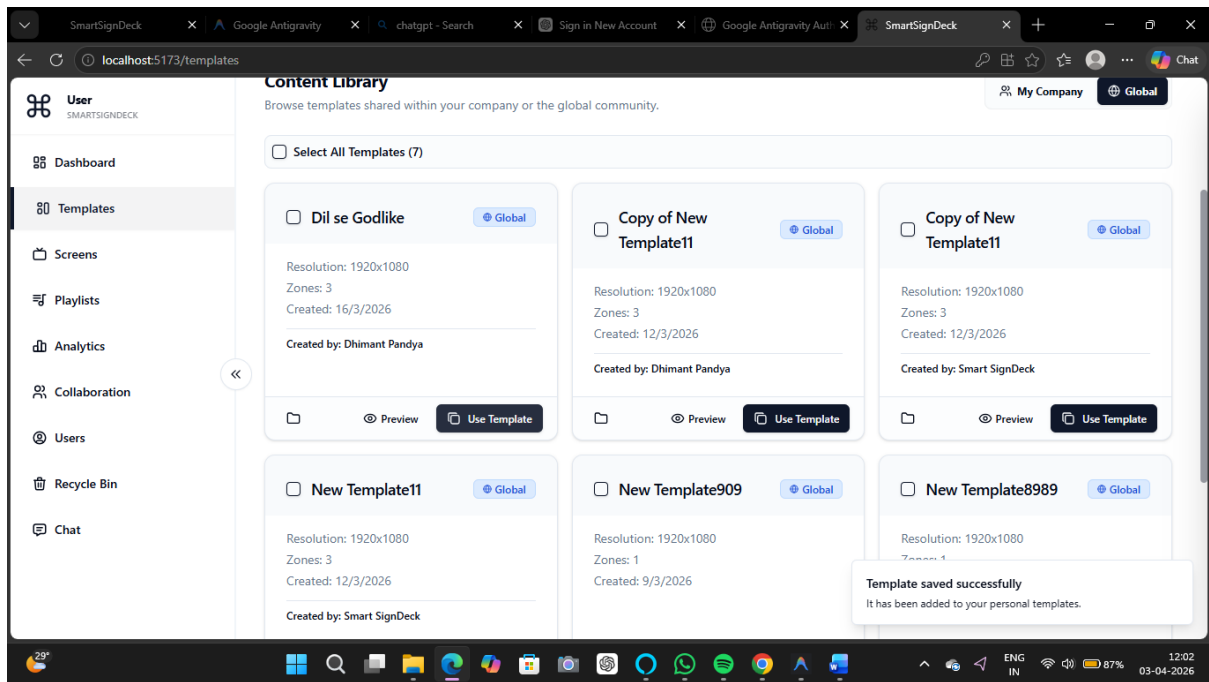
Templates



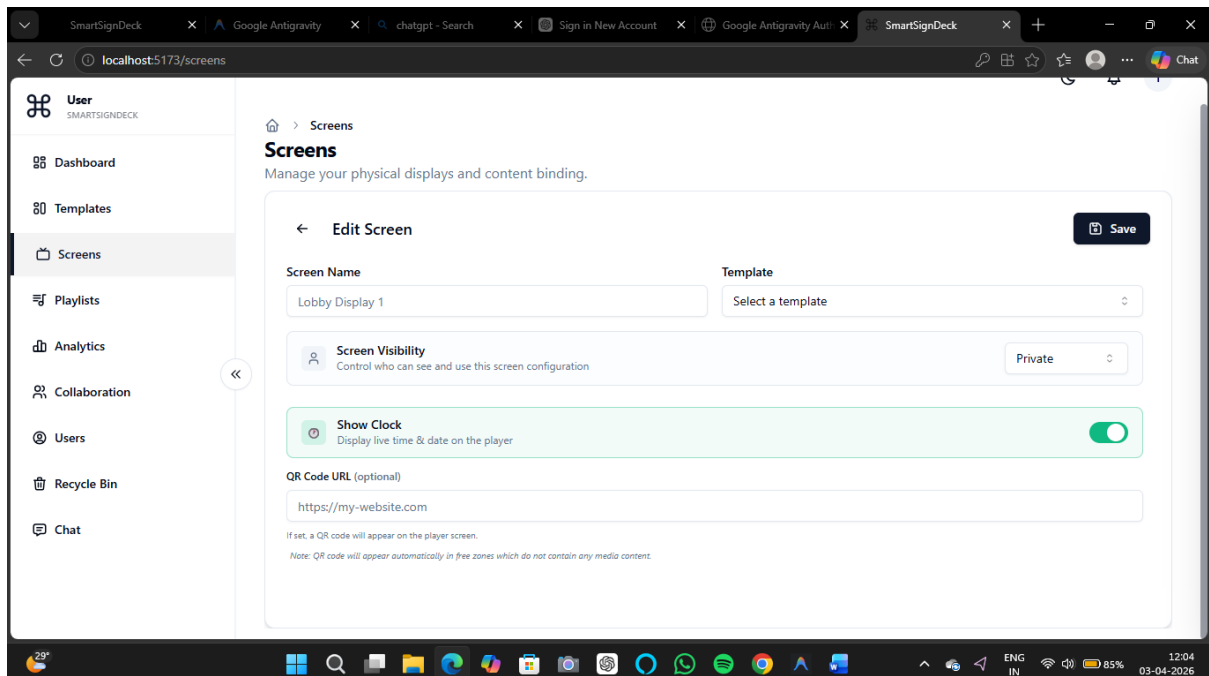
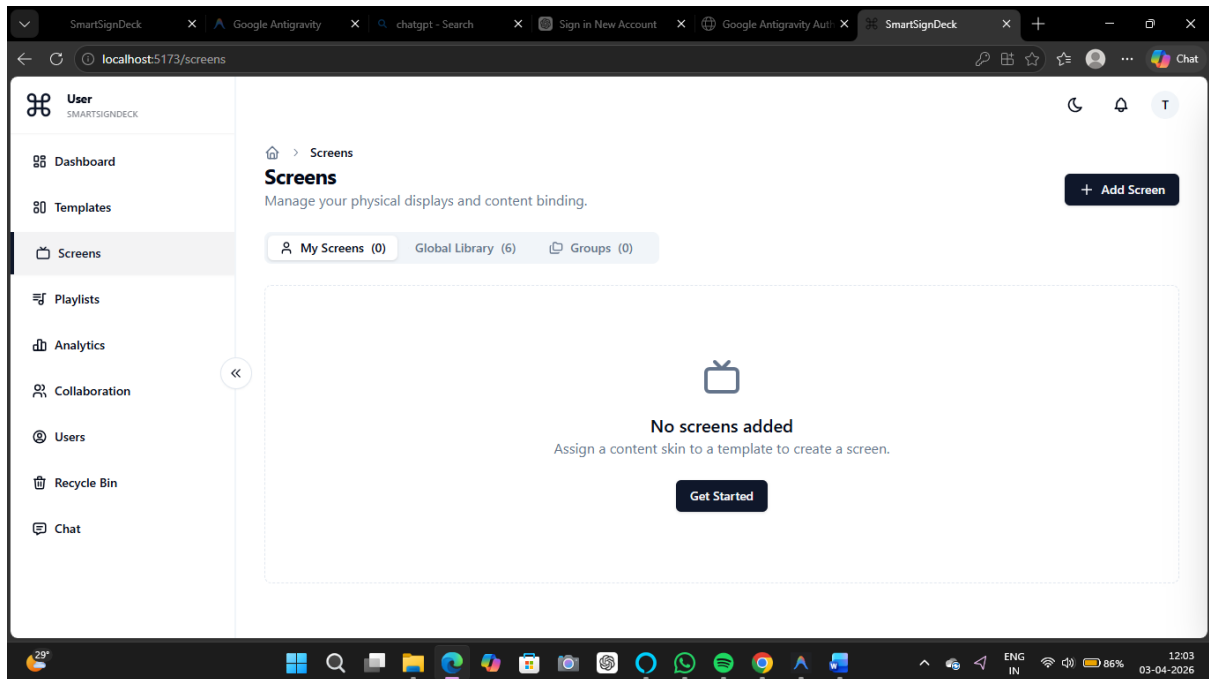


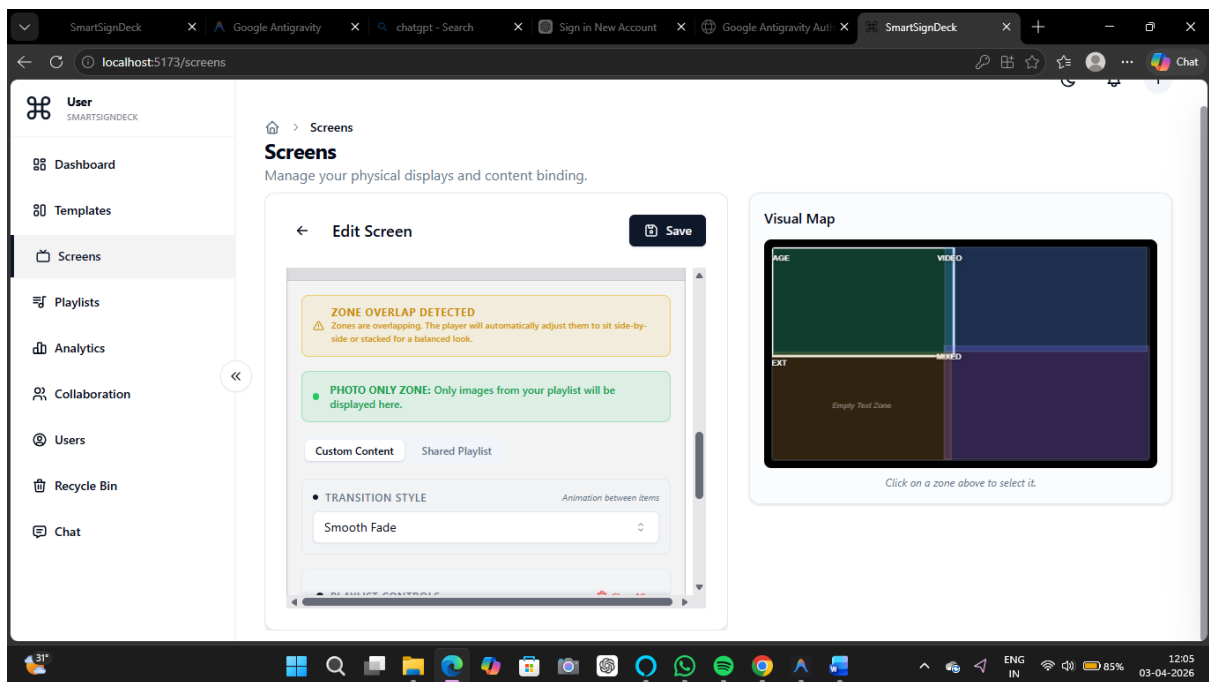
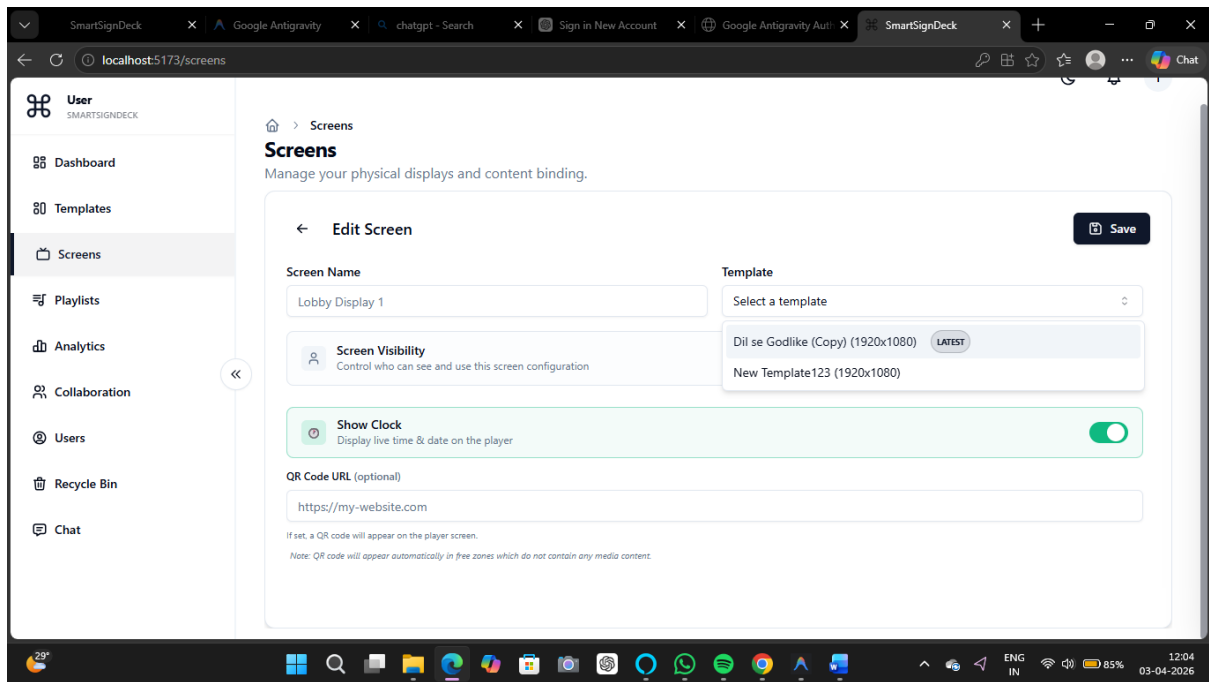


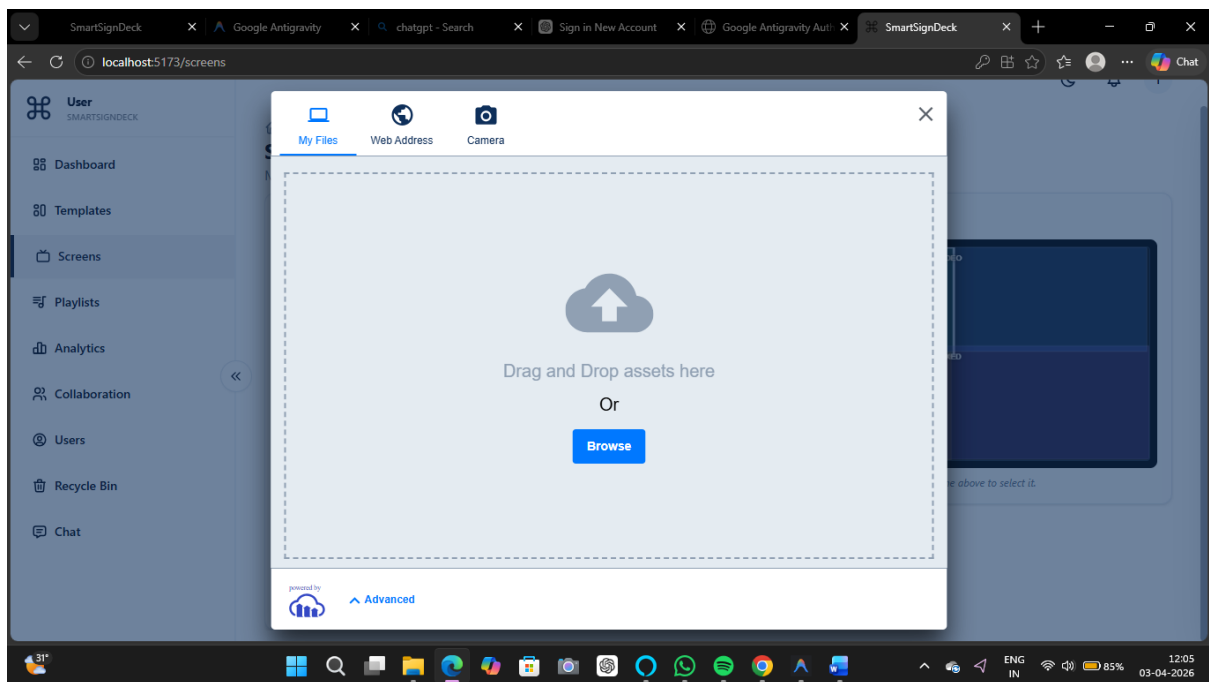
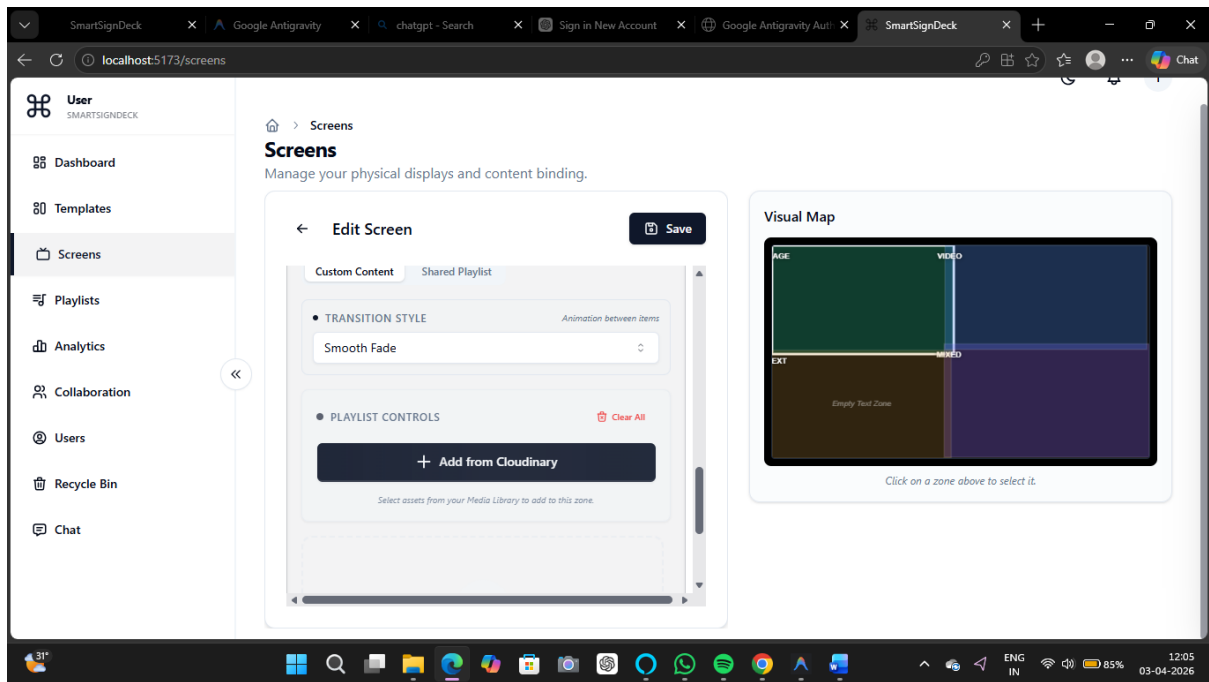


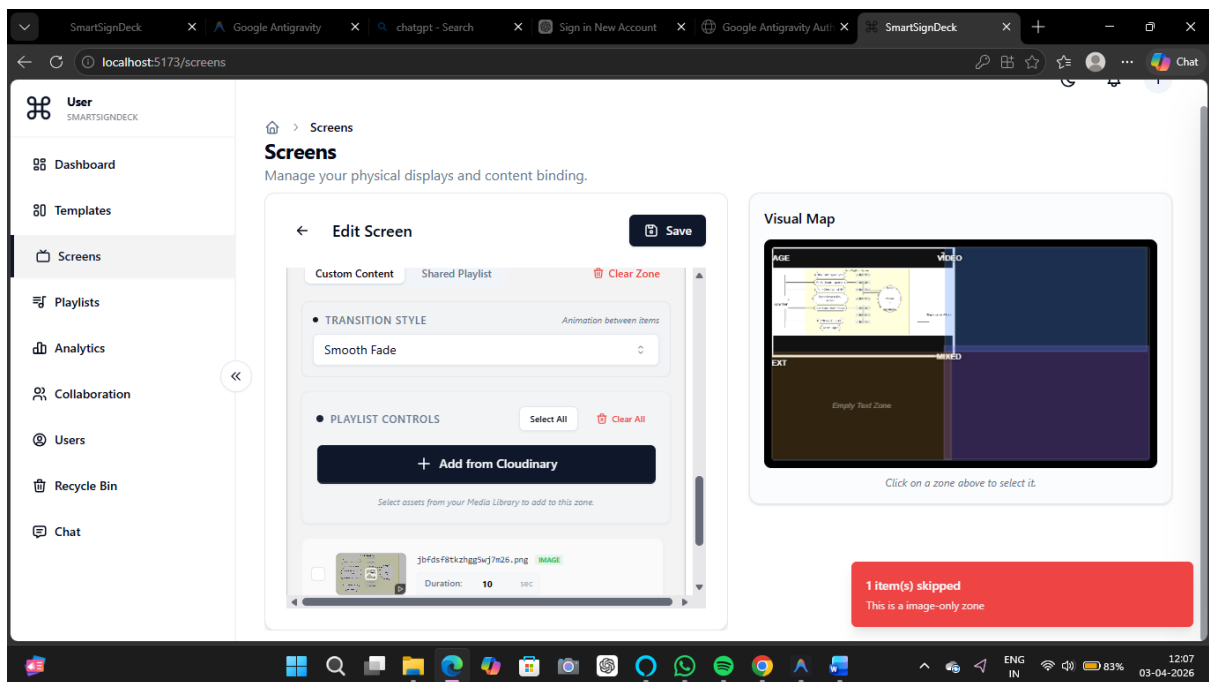
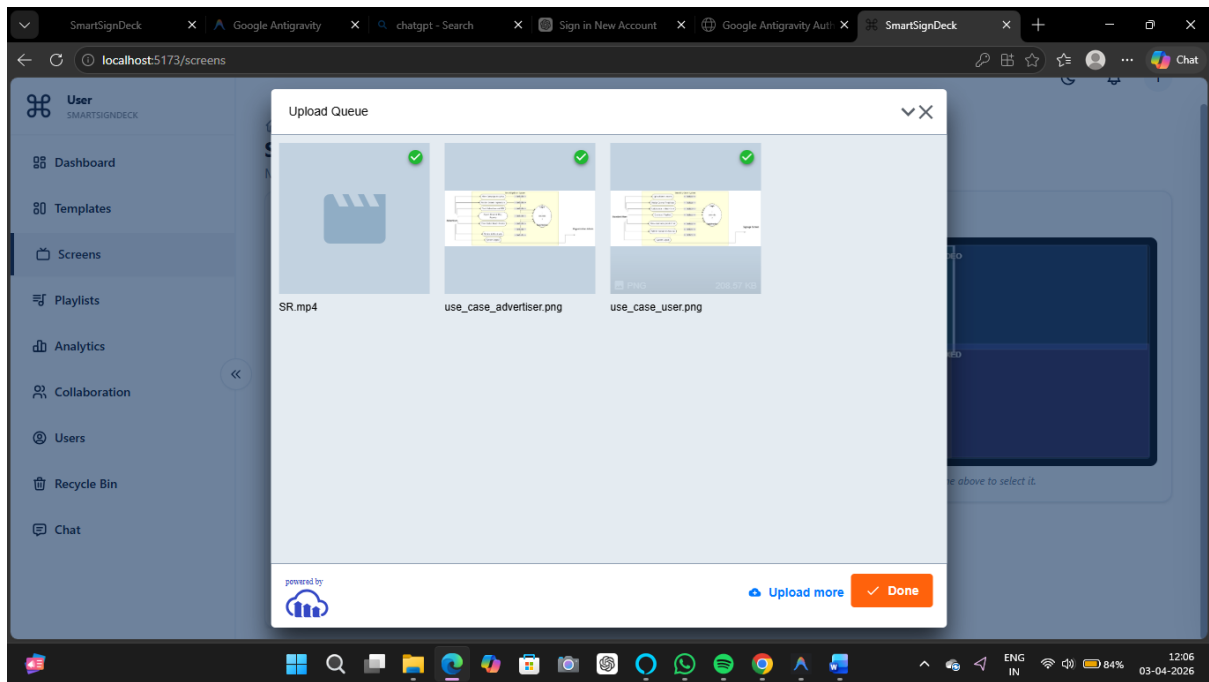


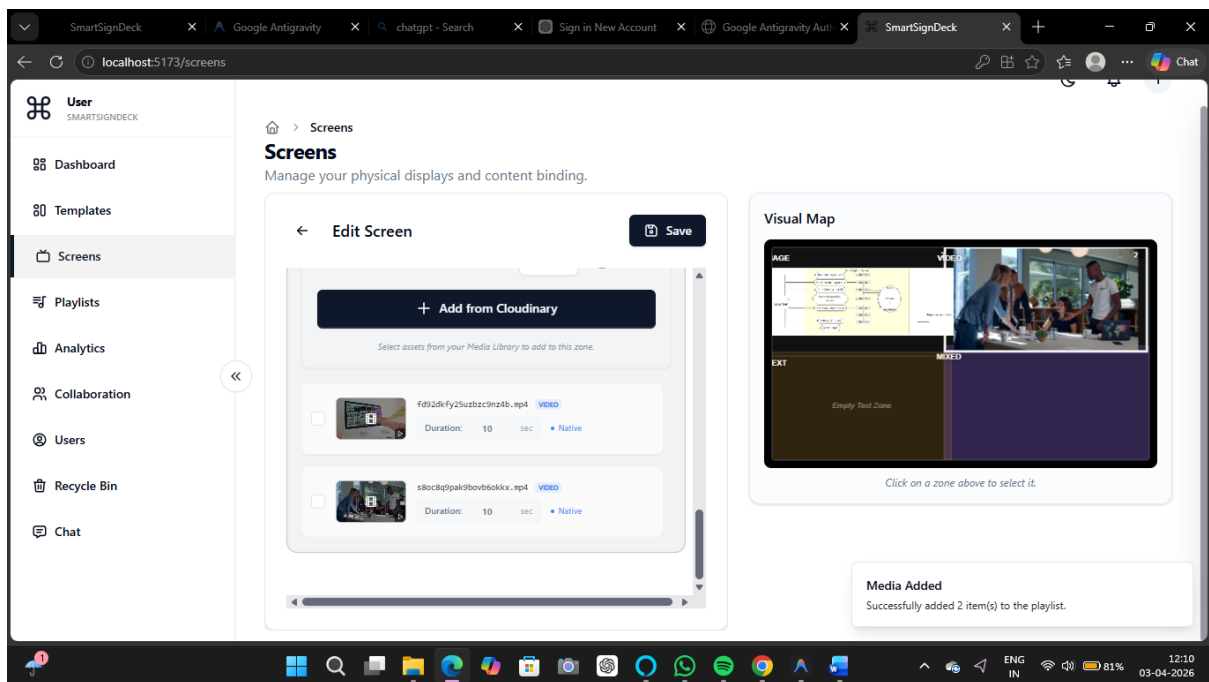
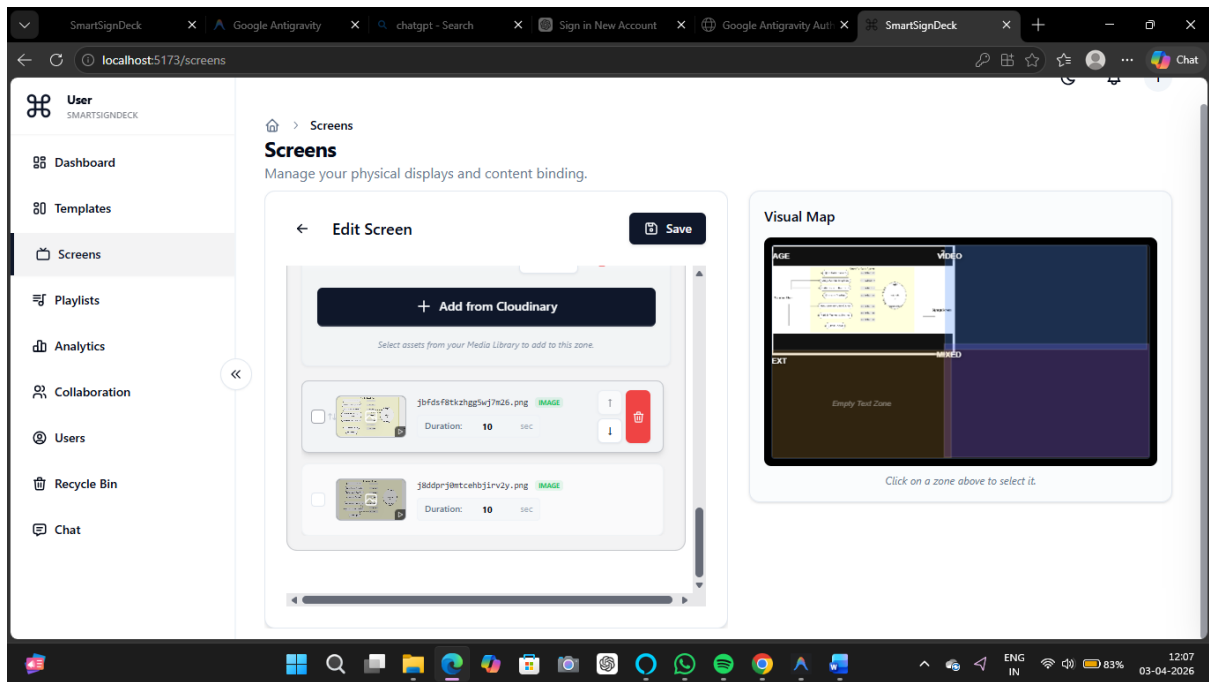
Screen

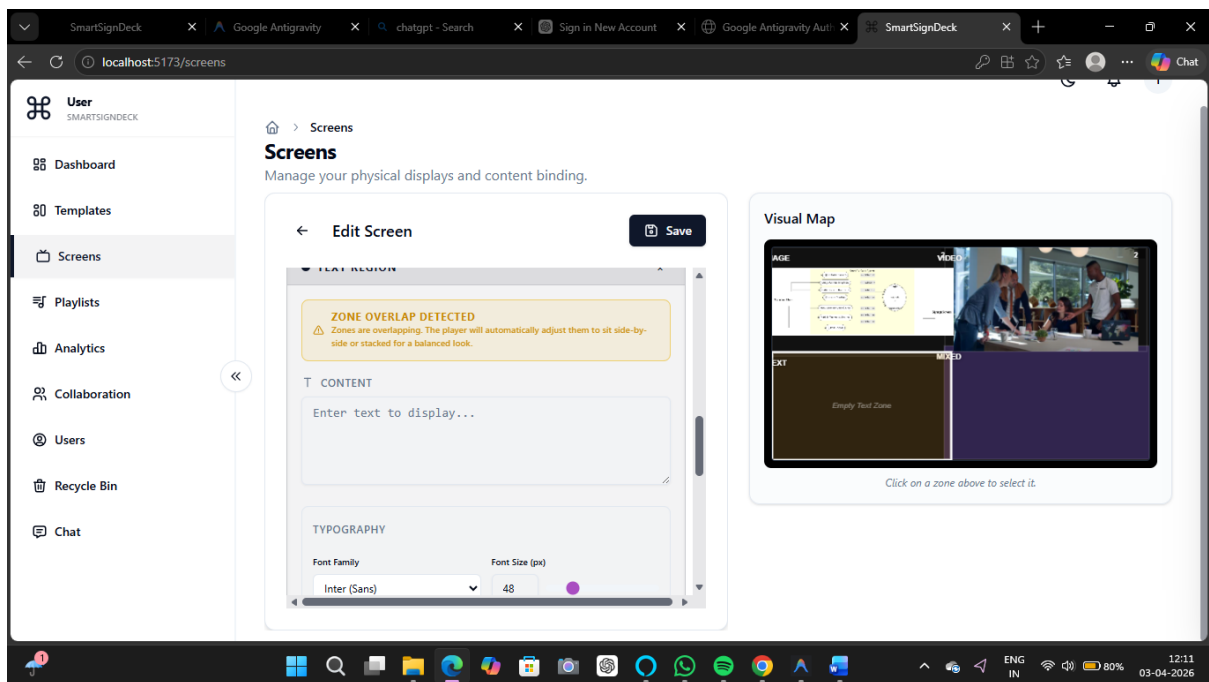
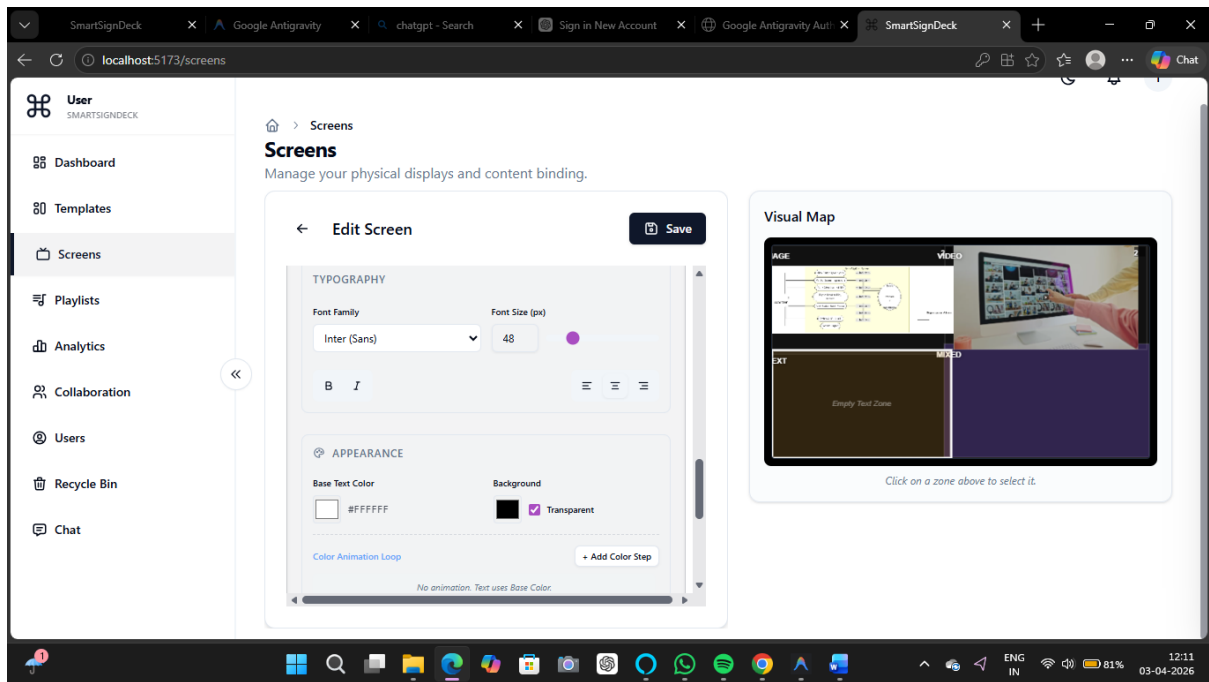


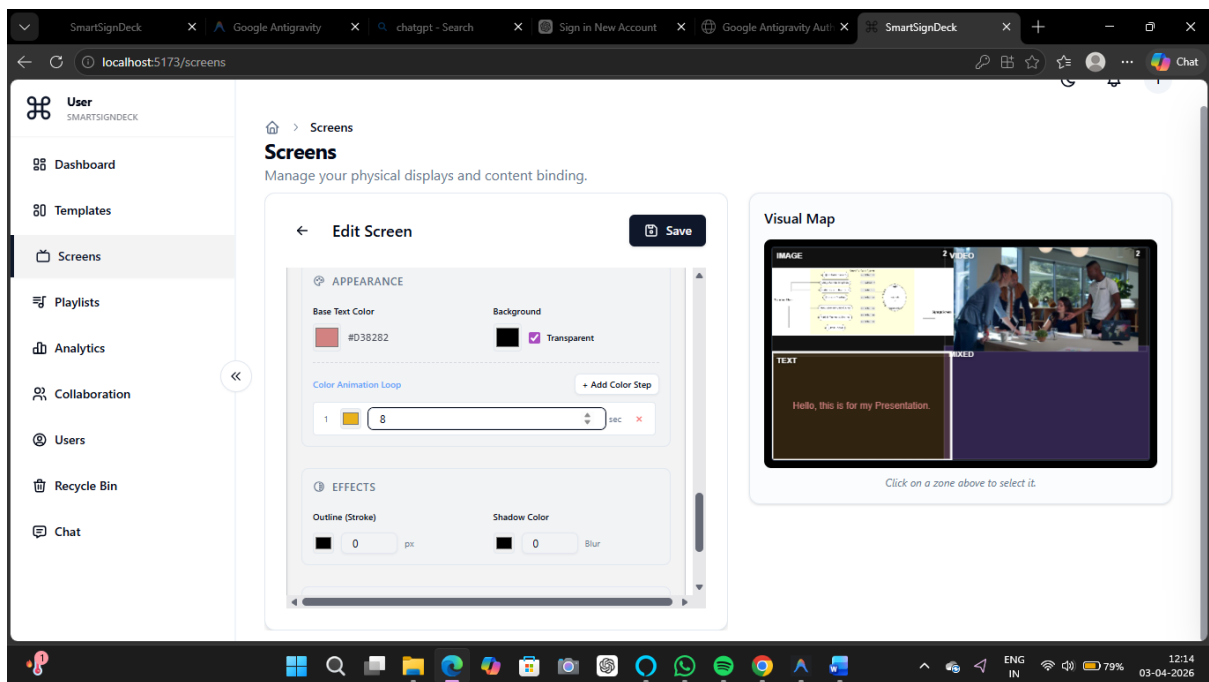
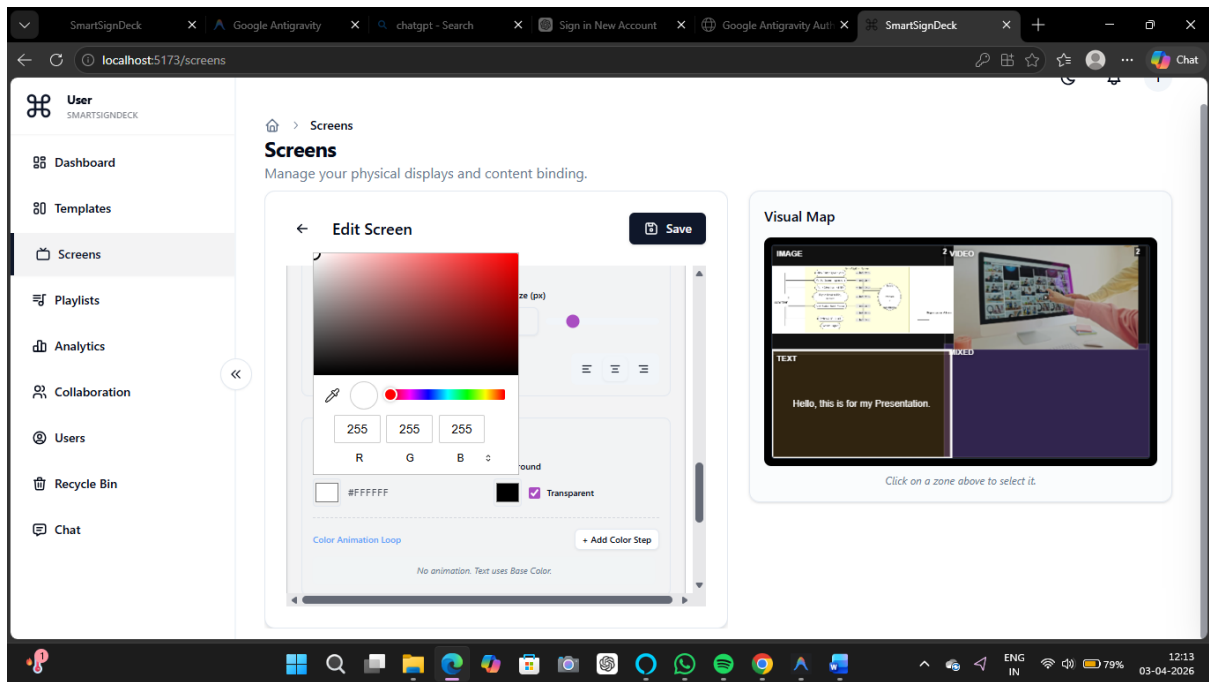


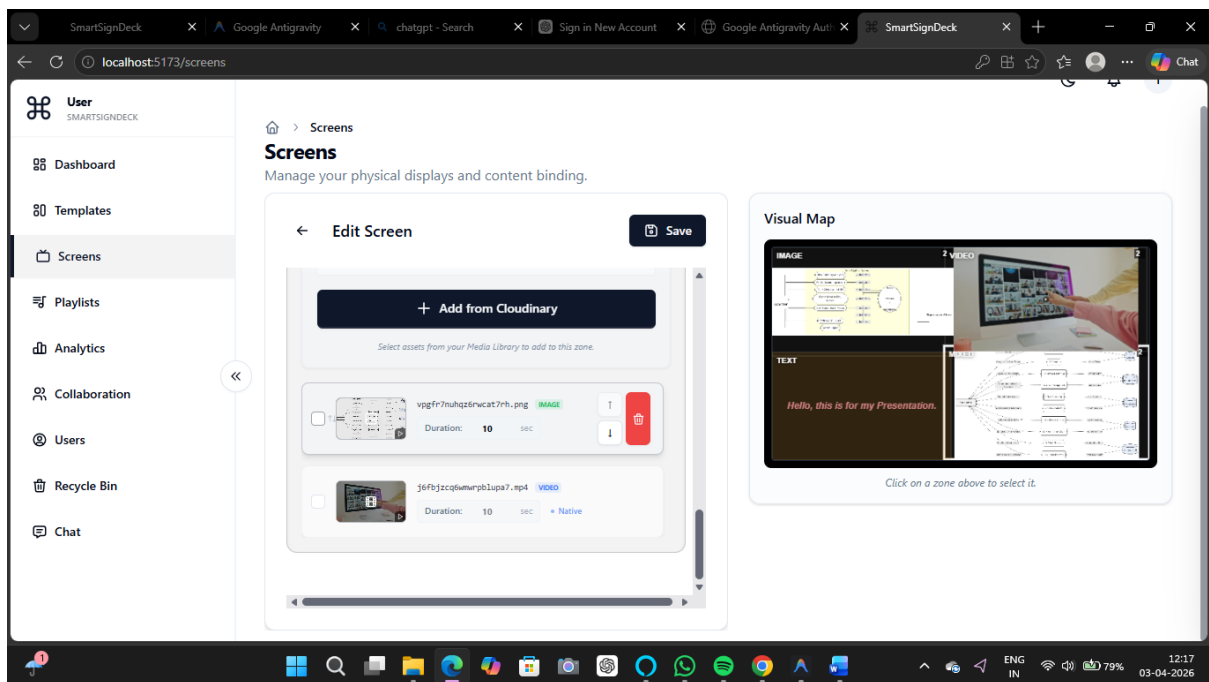
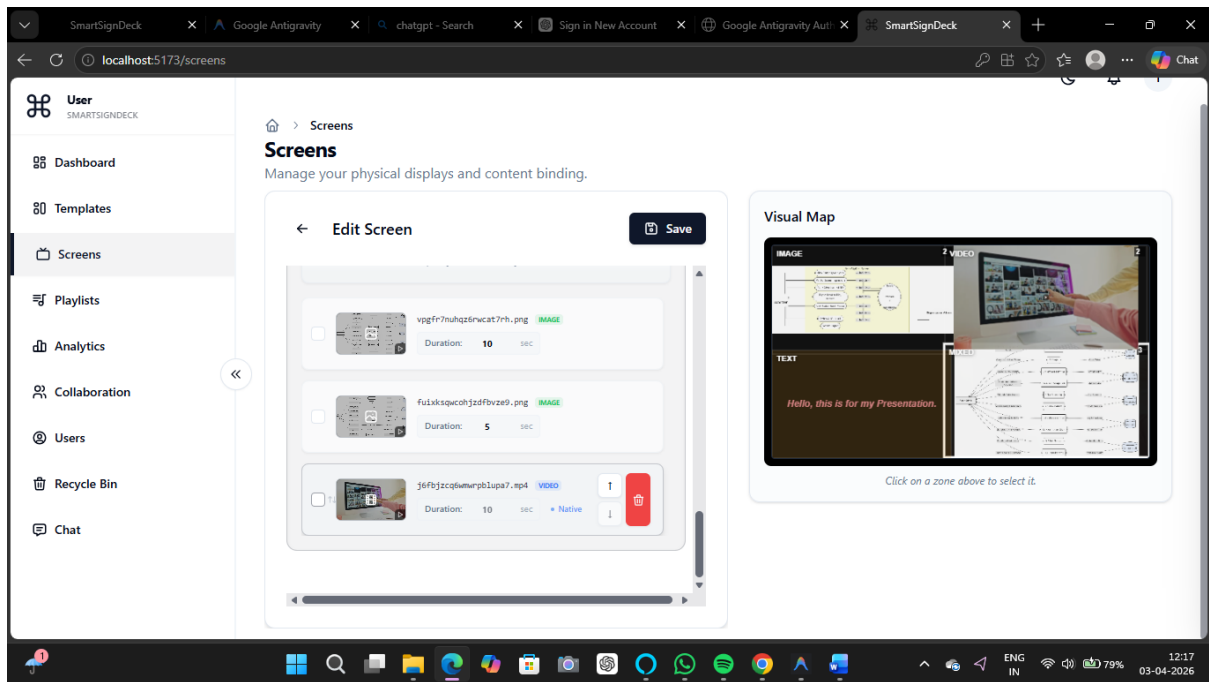


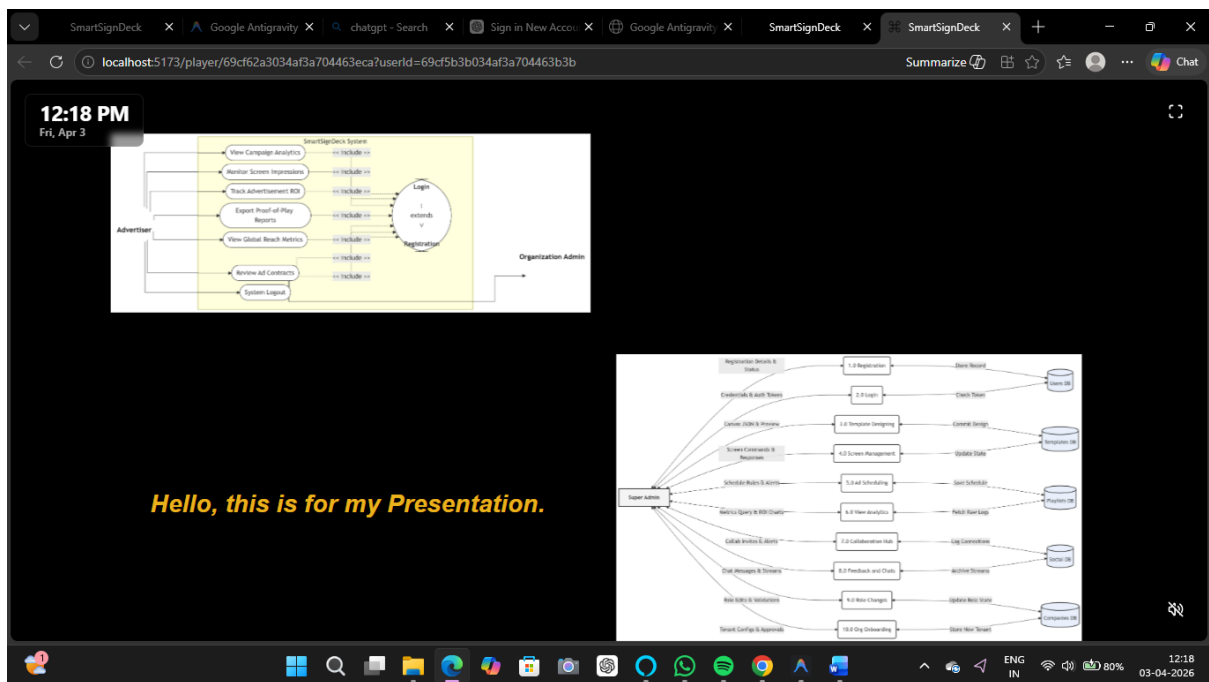
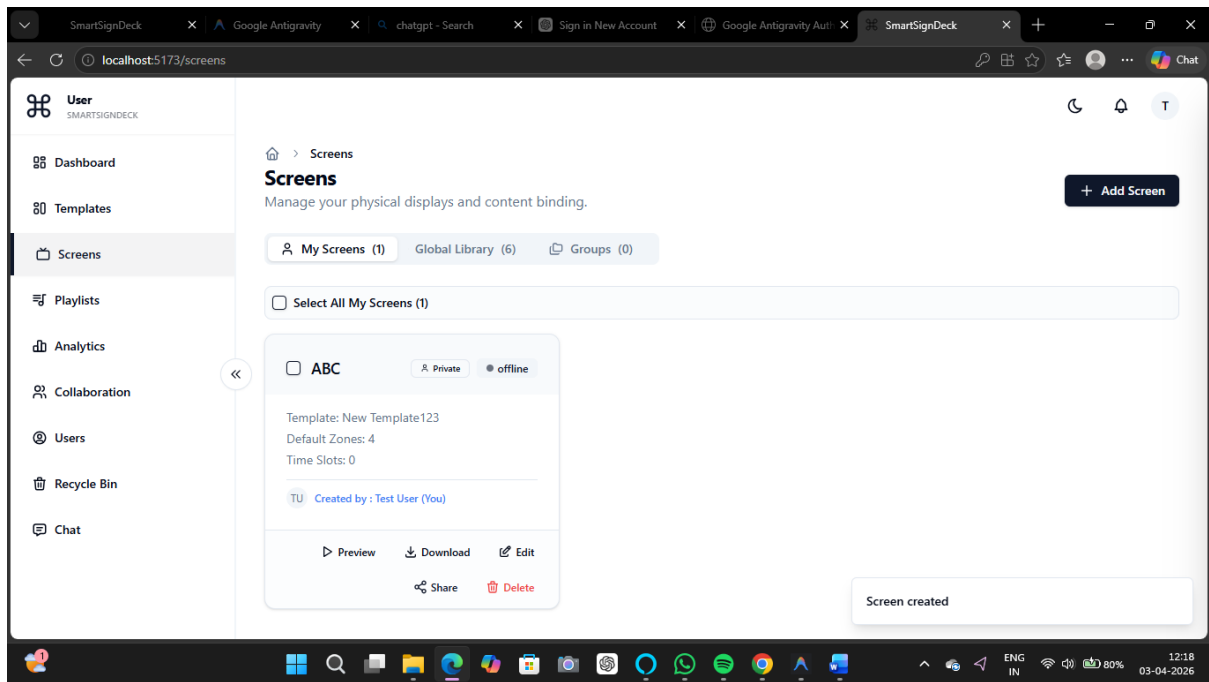


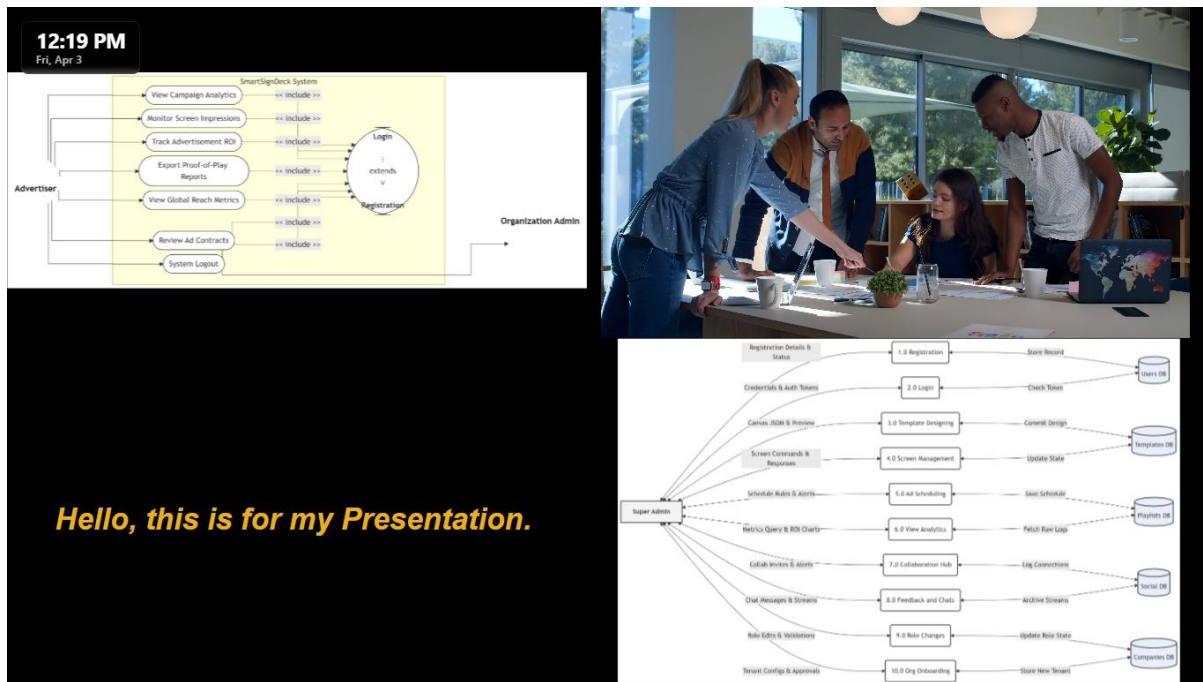




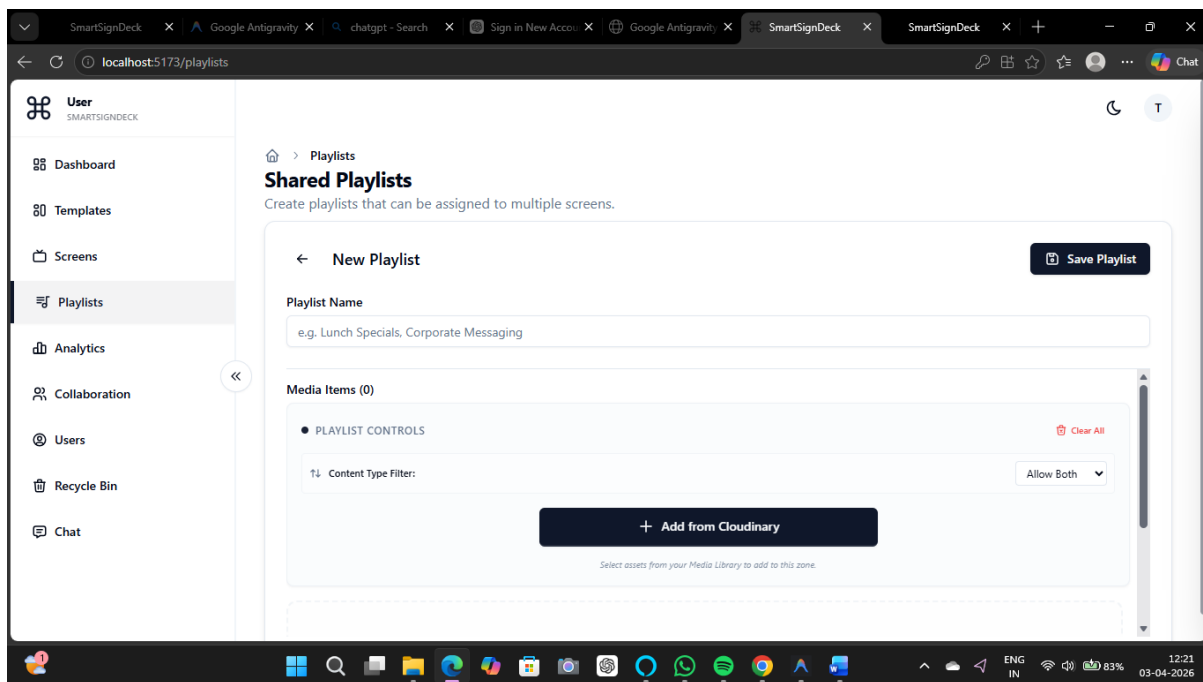


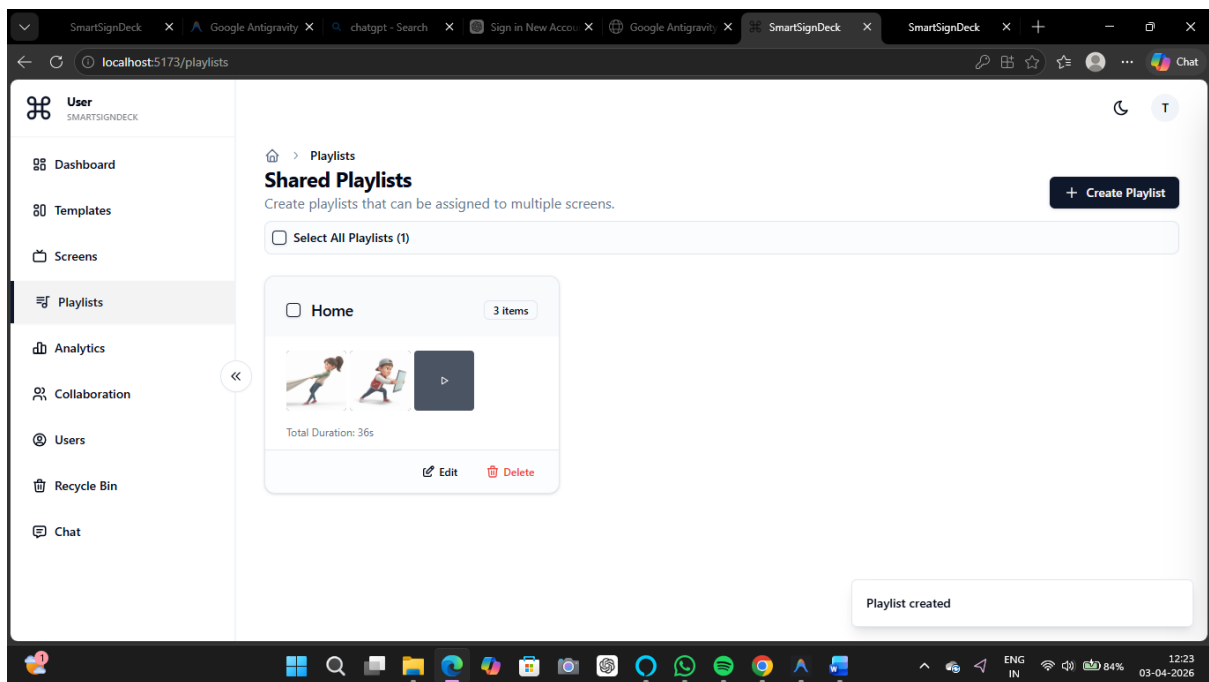
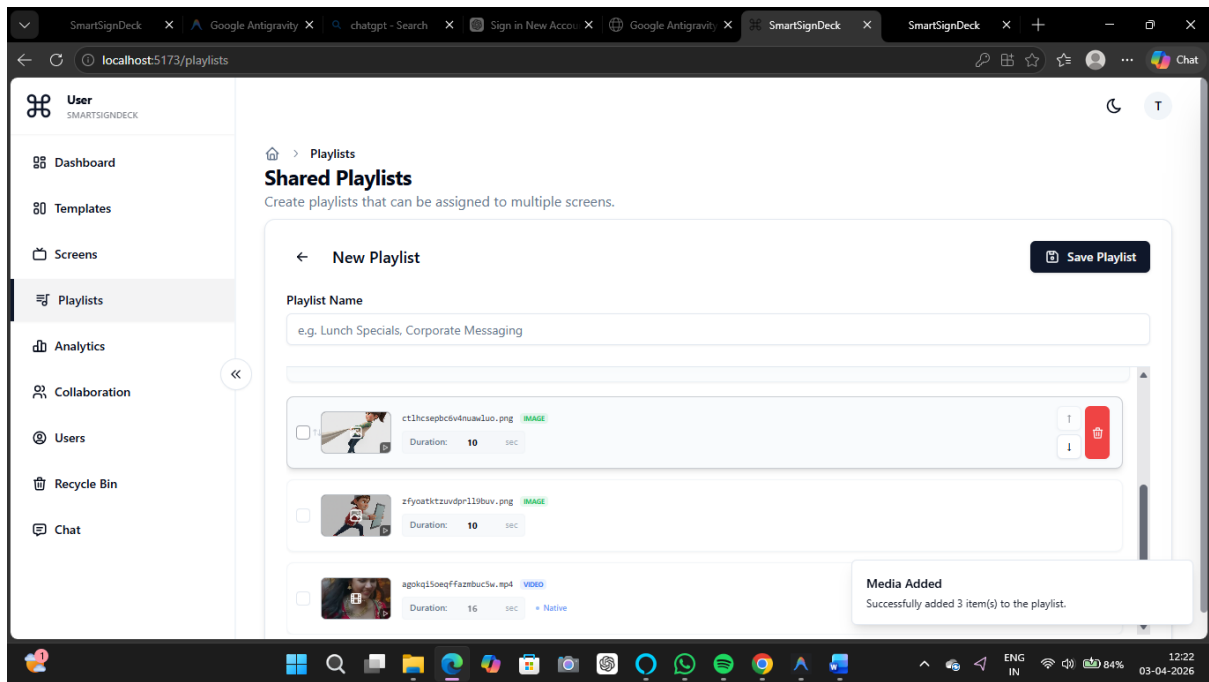


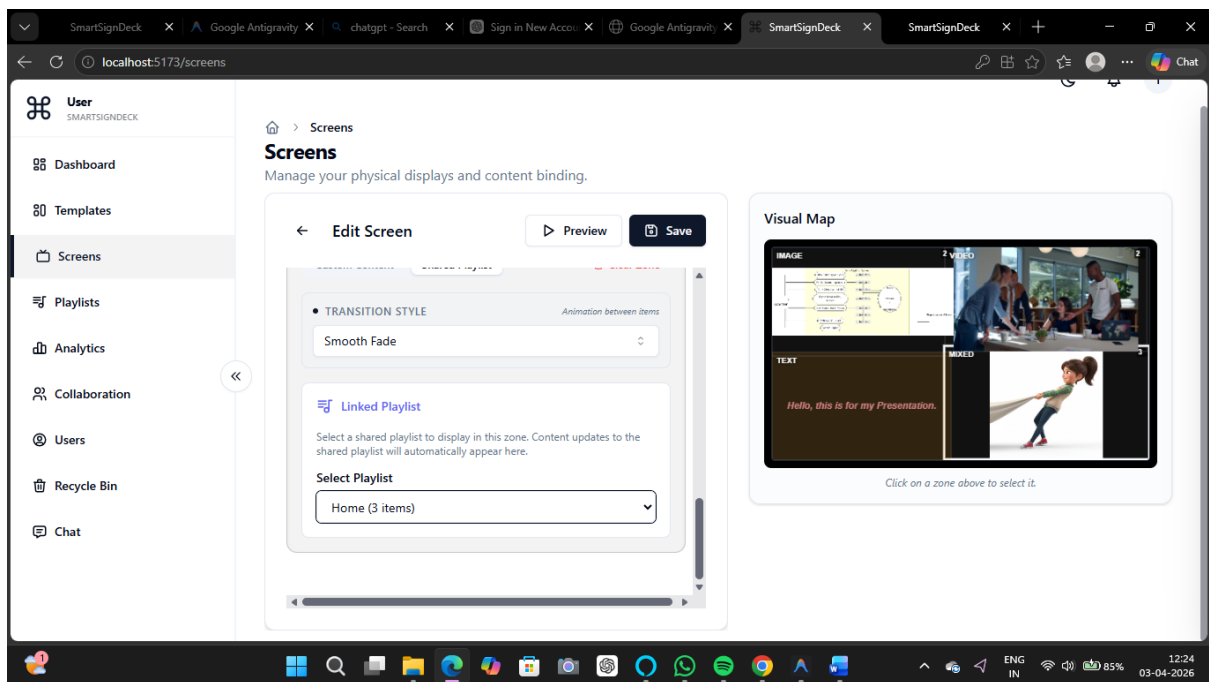
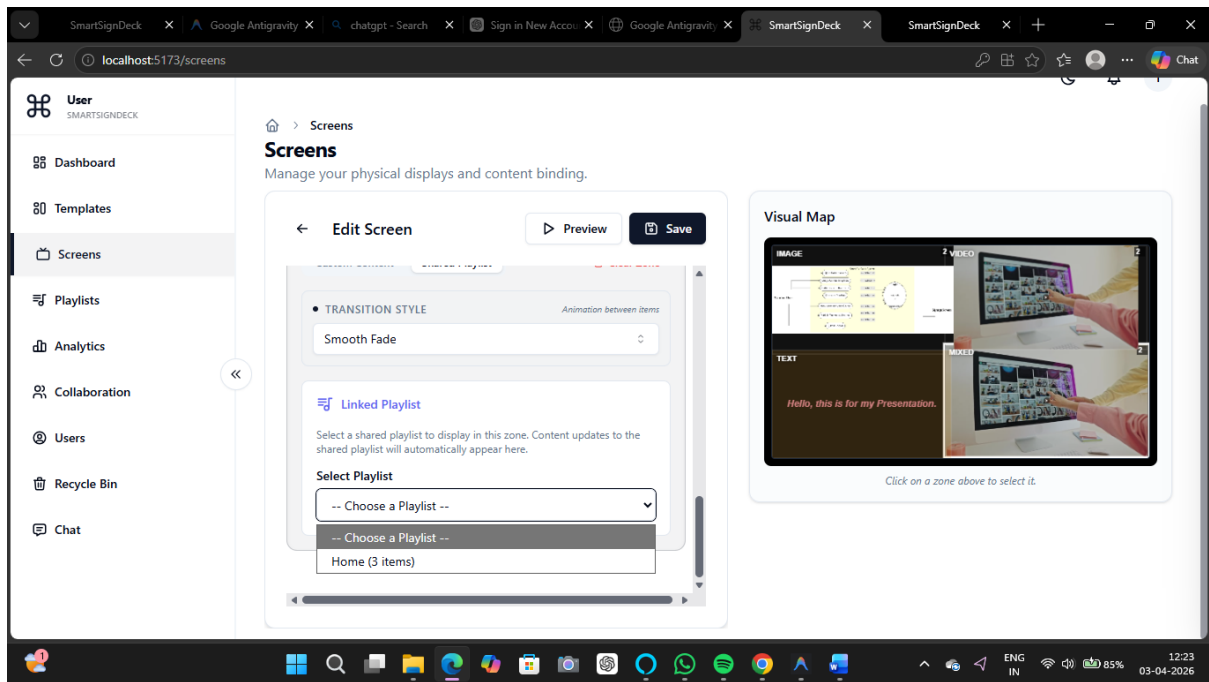


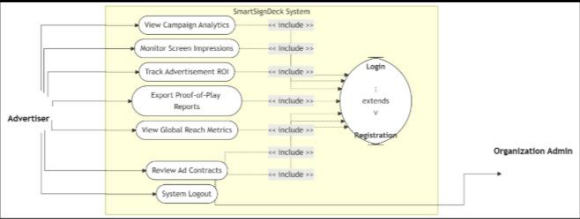


Shared Playlist







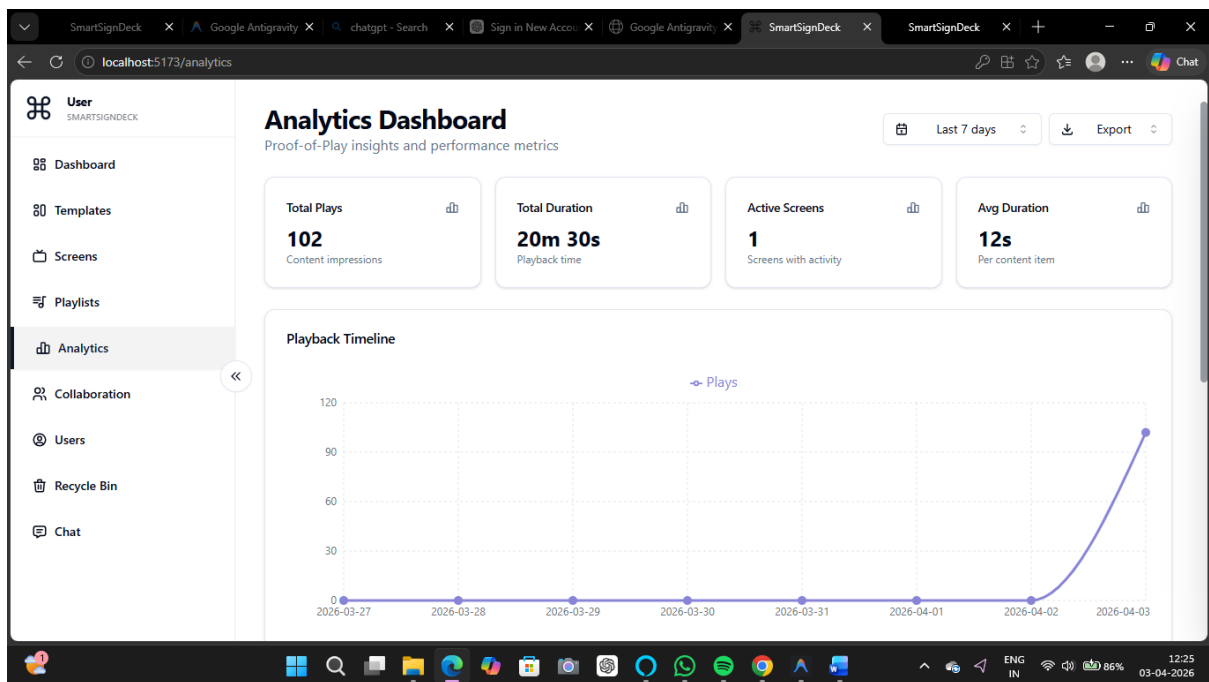


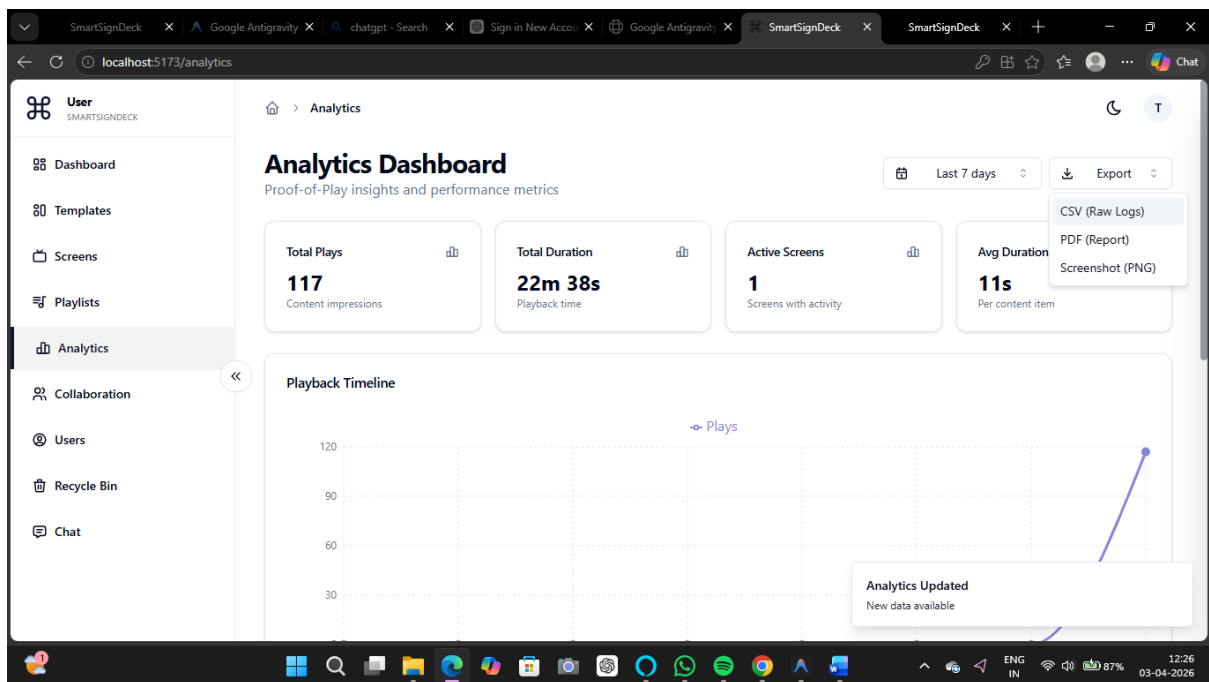
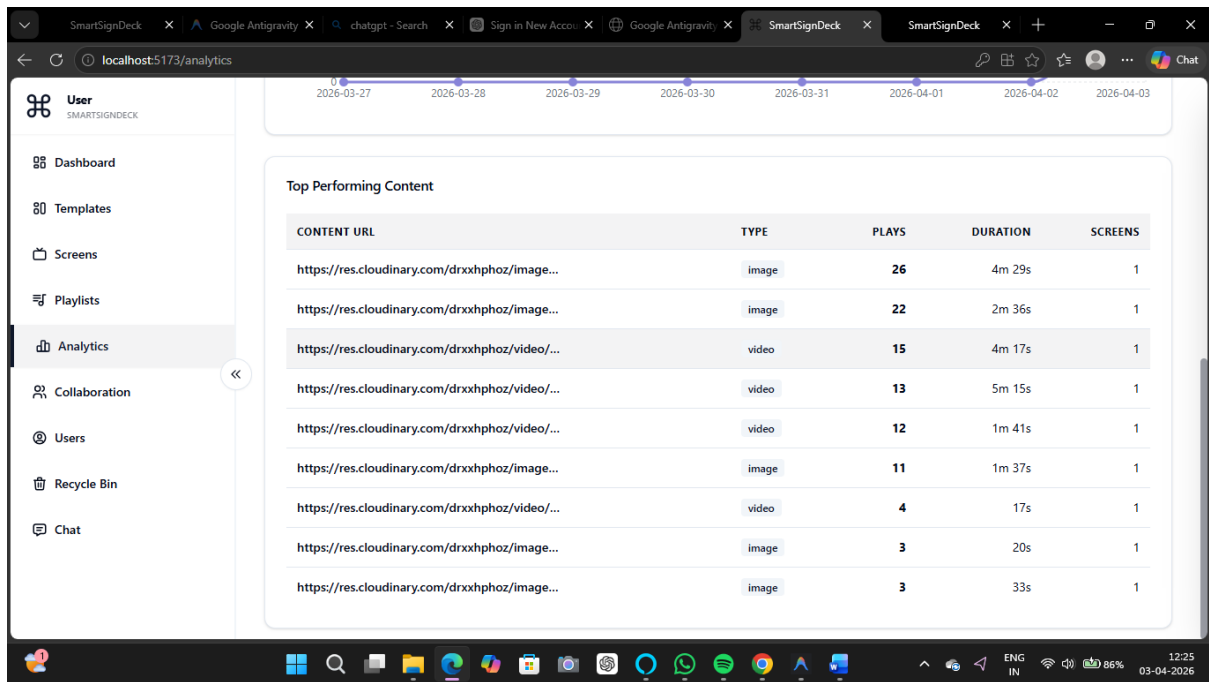
The UML Use Case Diagram for the SmartSignDeck System shows the following components:

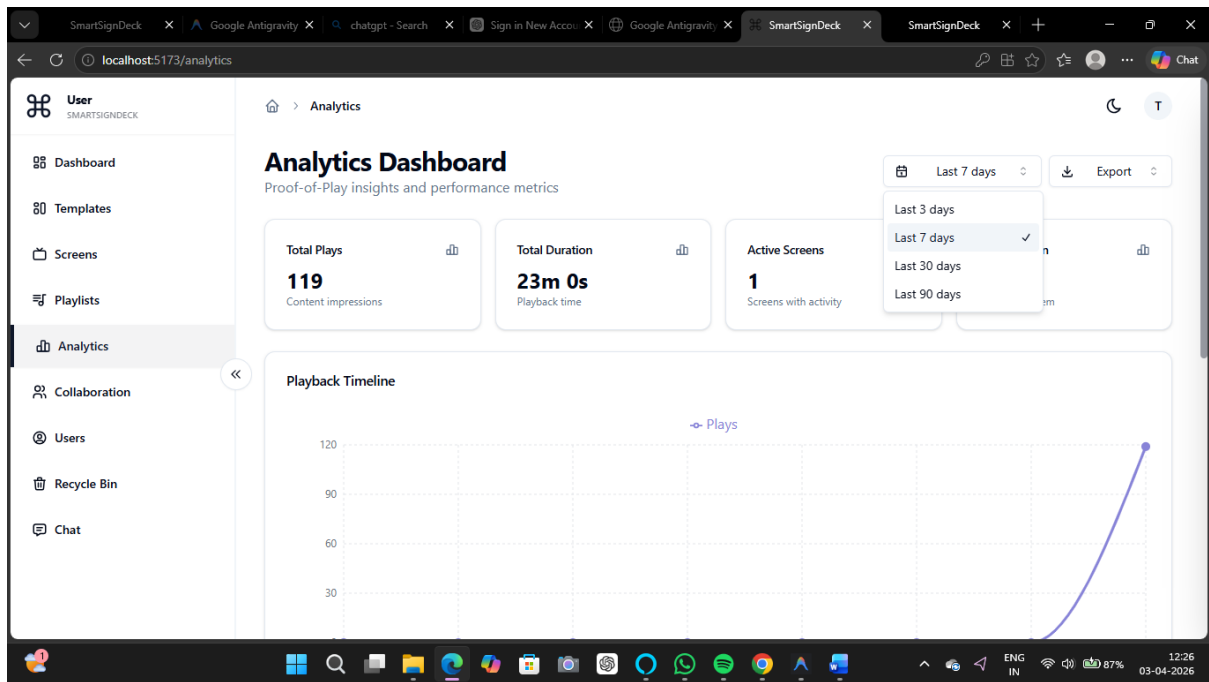
- Actors:** Advertiser, Organization Admin.
- Use Cases:**
 - View Campaign Analytics
 - Monitor Screen Impressions
 - Track Advertisement ROI
 - Export Proof-of-Play Reports
 - View Global Reach Metrics
 - Review Ad Contracts
 - System Logout
 - Login
 - Registration
- Relationships:**
 - Advertiser is associated with View Campaign Analytics, Monitor Screen Impressions, Track Advertisement ROI, Export Proof-of-Play Reports, View Global Reach Metrics, Review Ad Contracts, and System Logout.
 - Organization Admin is associated with System Logout.
 - Registration includes Login.
 - Login includes Registration.
 - View Campaign Analytics includes Monitor Screen Impressions.
 - Monitor Screen Impressions includes Track Advertisement ROI.
 - Track Advertisement ROI includes Export Proof-of-Play Reports.
 - Export Proof-of-Play Reports includes View Global Reach Metrics.
 - View Global Reach Metrics includes Review Ad Contracts.
 - Review Ad Contracts includes System Logout.



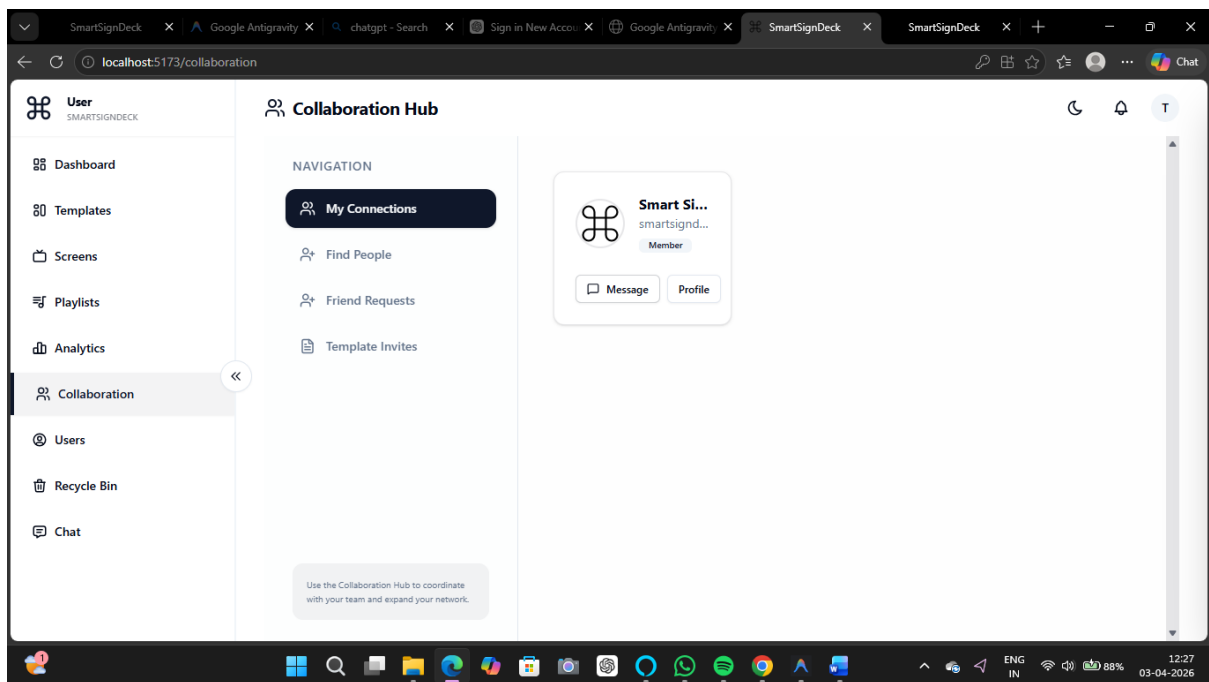
Hello, this is for my Presentation.

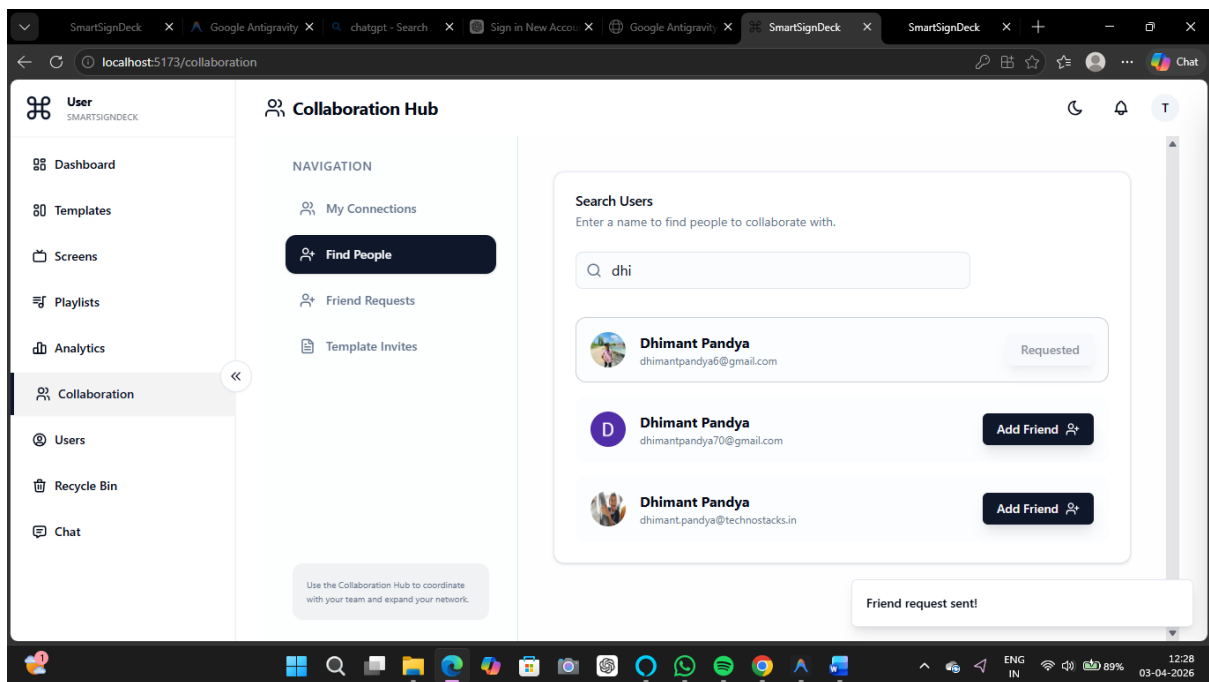
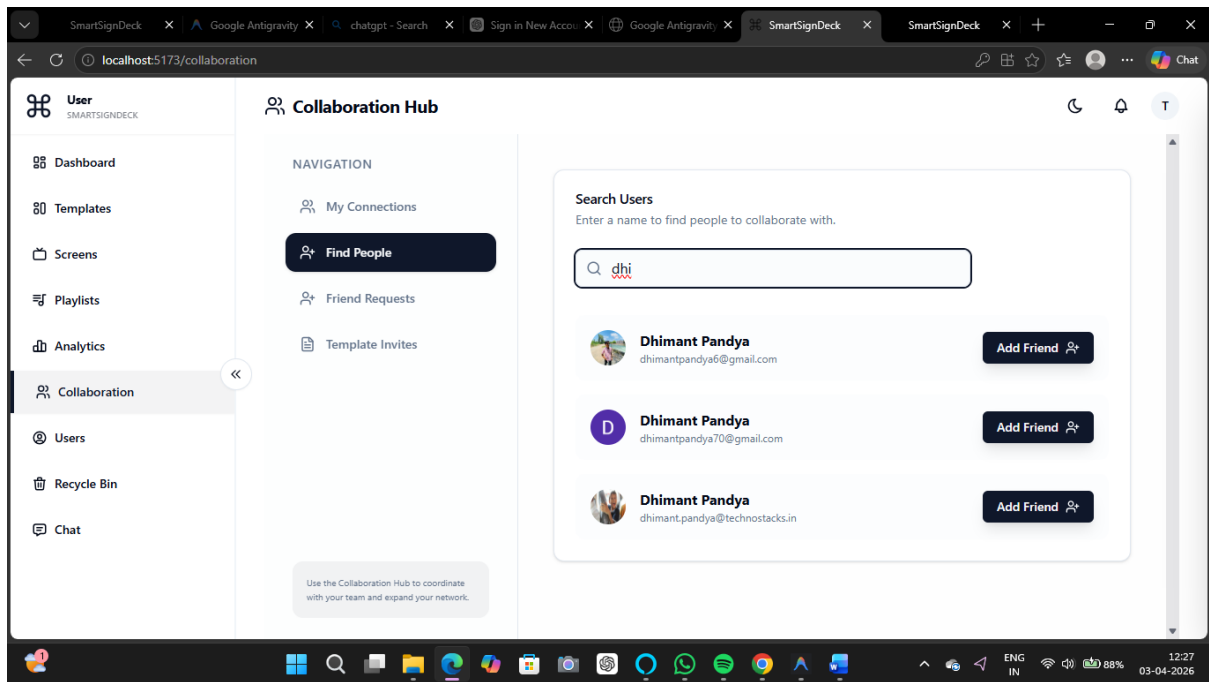


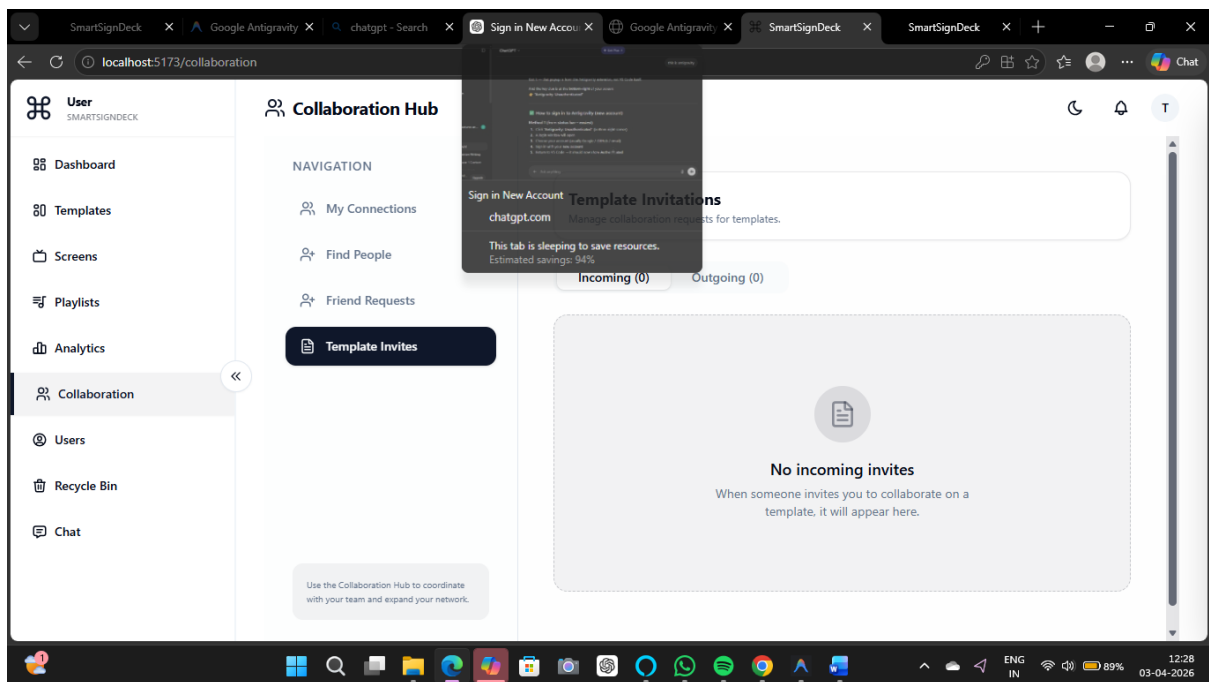
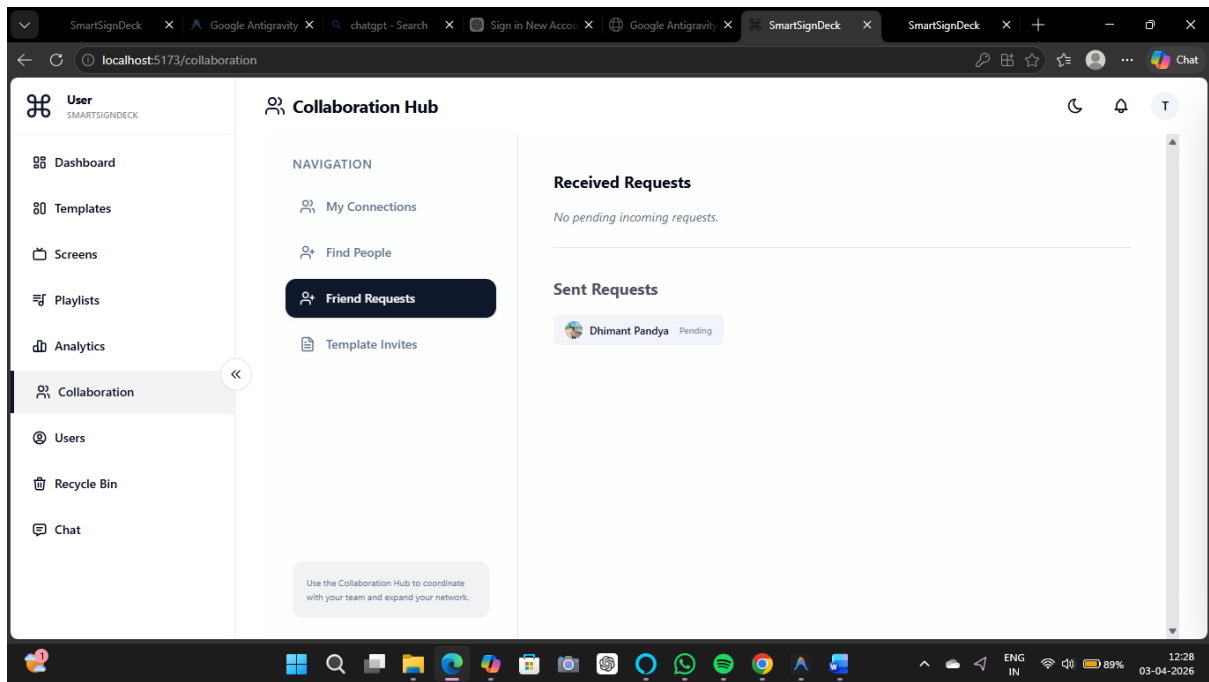




Collabroration Hub







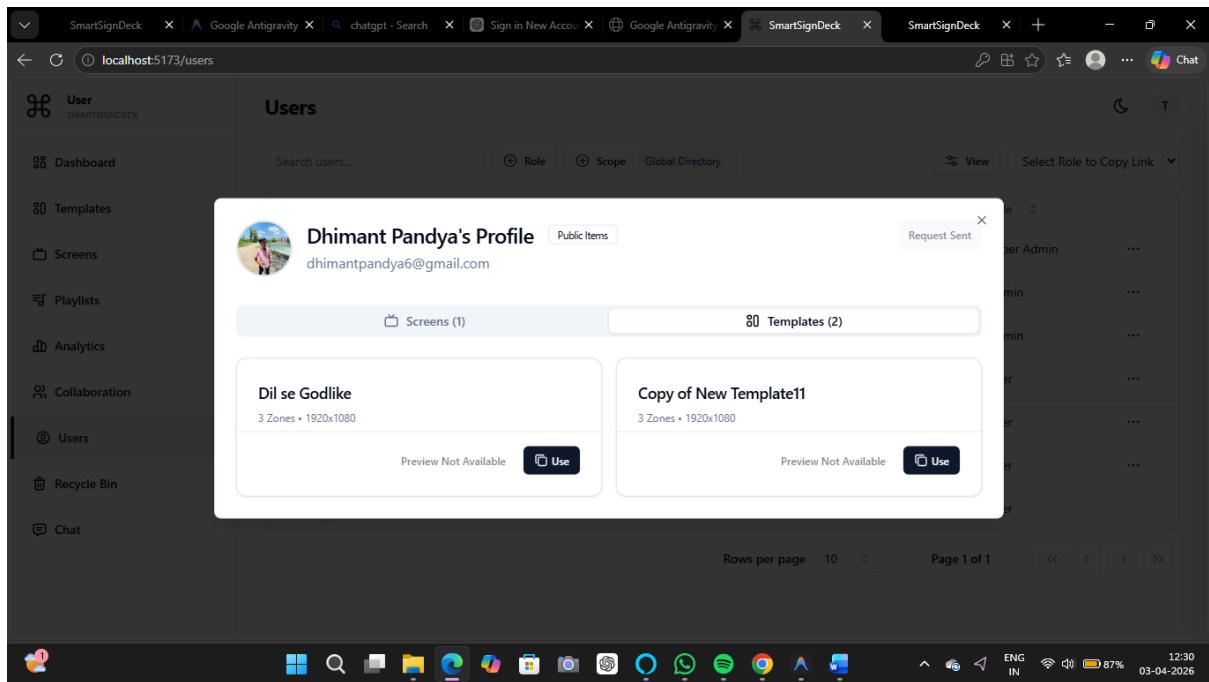
Users

The screenshot shows the SmartSignDeck application interface. On the left is a sidebar with navigation links: Dashboard, Templates, Screens, Playlists, Analytics, Collaboration, Users (selected), Recycle Bin, and Chat. The main content area is titled 'Users' and features a search bar with the placeholder 'Search users...'. Below the search bar are filters for 'Role', 'Scope', and 'Global Directory'. A 'View' button and a 'Select Role to Copy Link' dropdown are also present. The user list is displayed in a table with columns: Id, Name, Email, Company, and Role. The table contains 7 rows of user data. At the bottom right of the table, there are controls for 'Rows per page' (set to 10) and 'Page 1 of 1'.

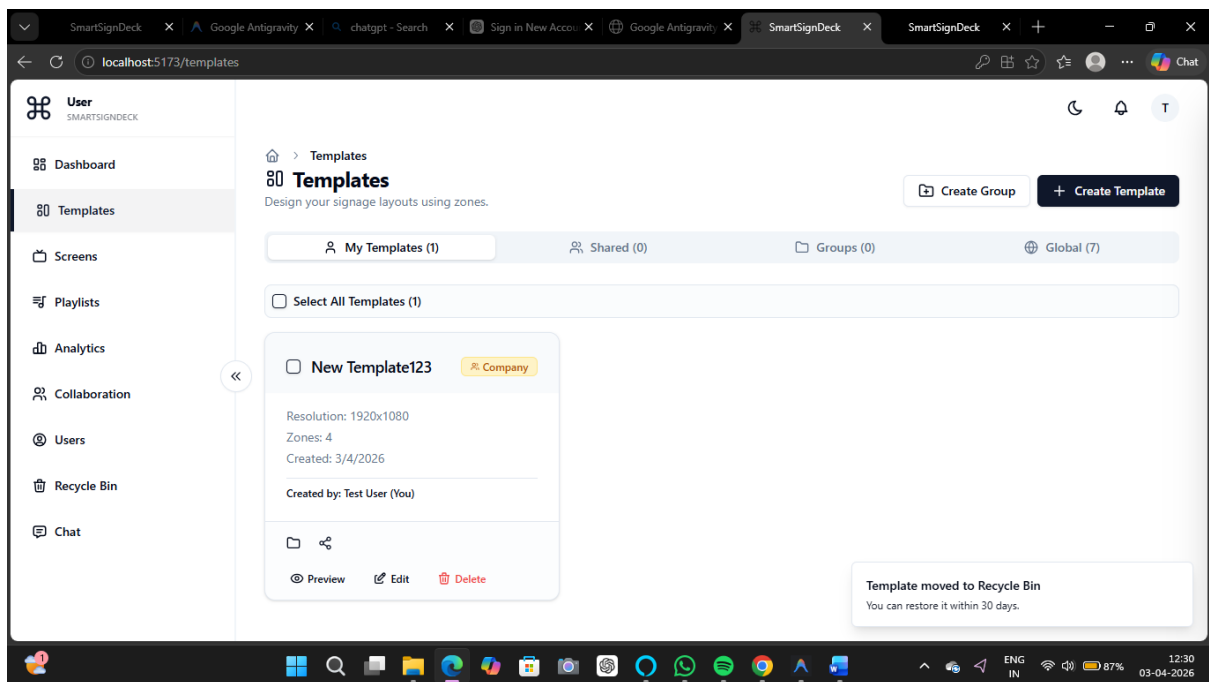
Id	Name	Email	Company	Role
1	Smart SignDeck	smartsigndeckk@gmail.com	smartsigndeck	Super Admin
2	Dhimant Pandya	dhimantpandya6@gmail.com	technostacks	Admin
3	Dhimant Pandya	dhimantpandya70@gmail.com	technostacks	Admin
4	Dhimant Pandya	dhimant.pandya@technostacks.in	smartsigndeck	User
5	Tathya Dave	tathyadave009@gmail.com	tathya dave	User
6	Akshay Pathak	akshay@yopmail.com	name	User
7	Test User	testuser@example.com	abc	User

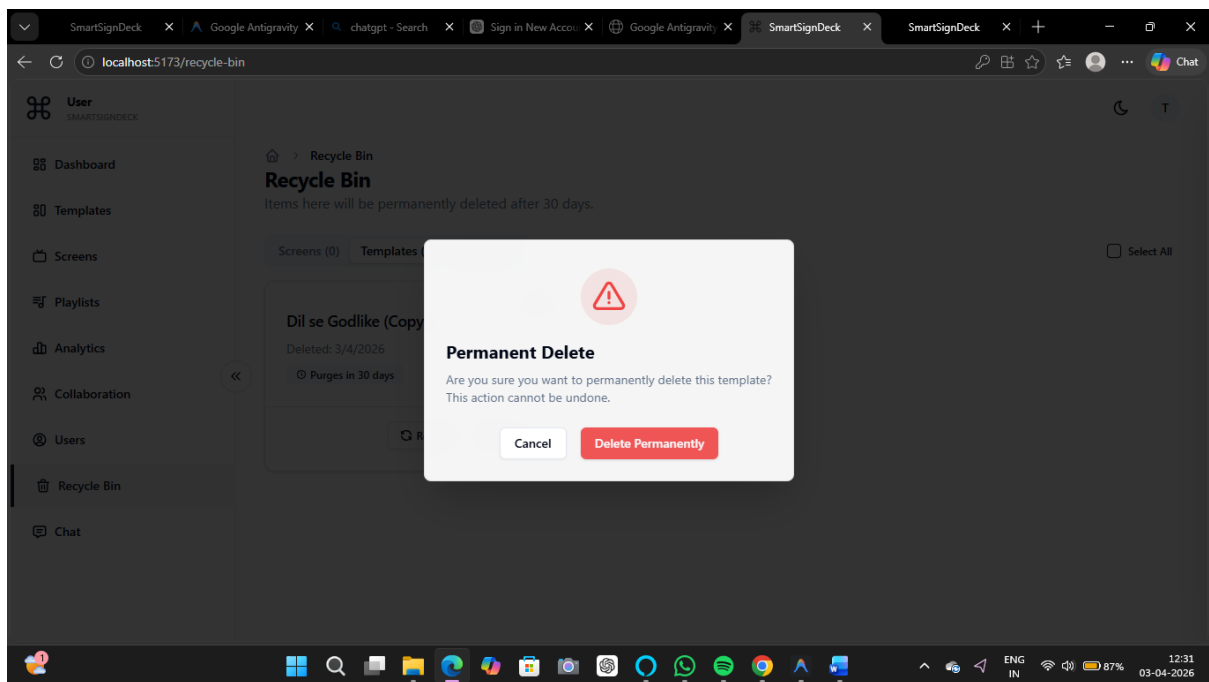
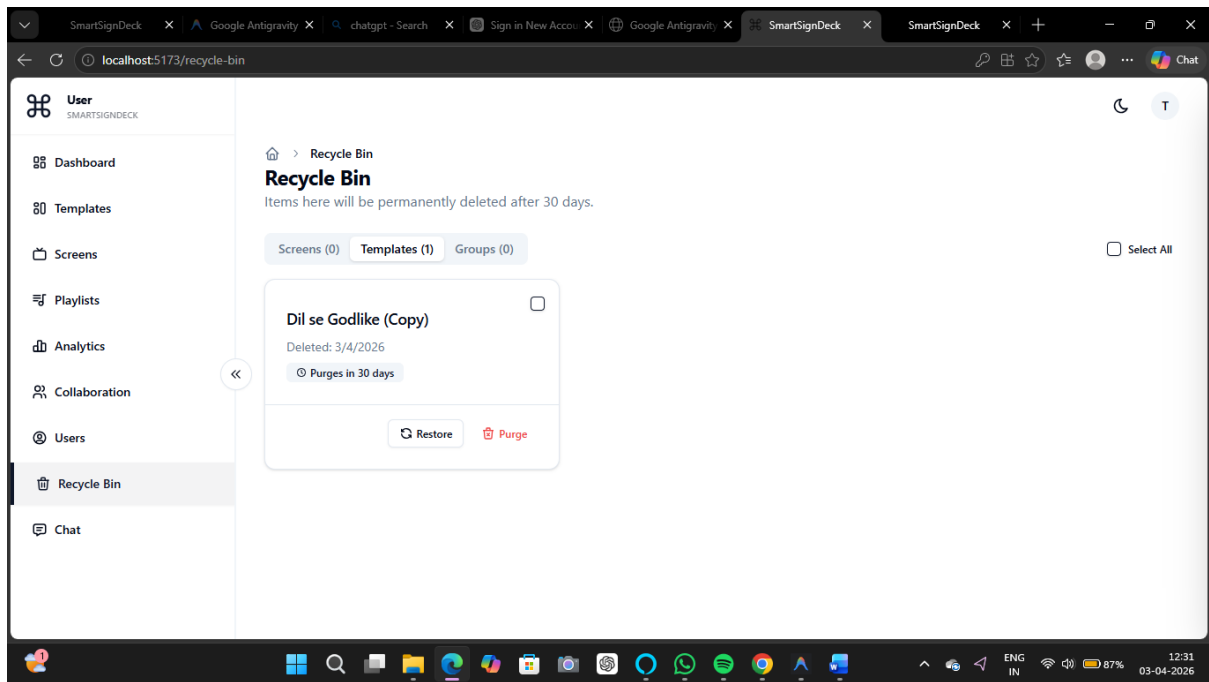
This screenshot shows the same SmartSignDeck 'Users' page, but with a search filter applied. The search bar now contains the text 'dhi'. The table below the search bar is filtered to show only 3 rows of users whose names start with 'dhi'. The table columns remain the same: Id, Name, Email, Company, and Role. The 'Rows per page' and 'Page 1 of 1' controls are still visible at the bottom right.

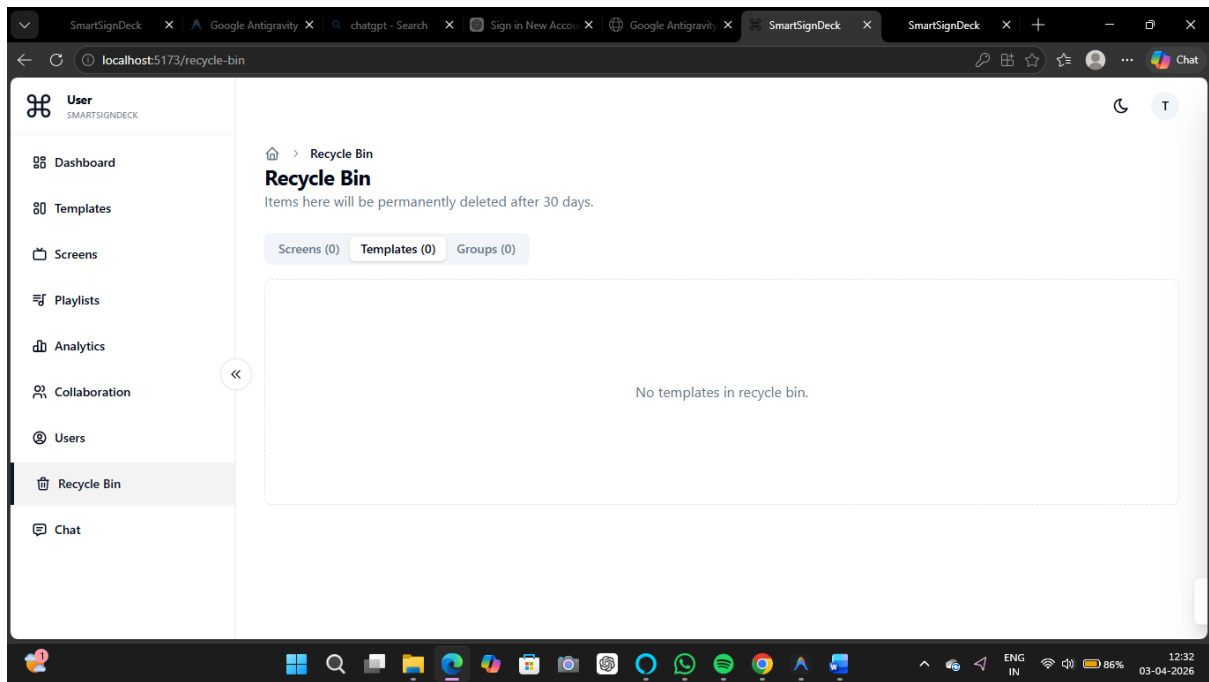
Id	Name	Email	Company	Role
1	Dhimant Pandya	dhimantpandya6@gmail.com	technostacks	Admin
2	Dhimant Pandya	dhimantpandya70@gmail.com	technostacks	Admin
3	Dhimant Pandya	dhimant.pandya@technostacks.in	smartsigndeck	User



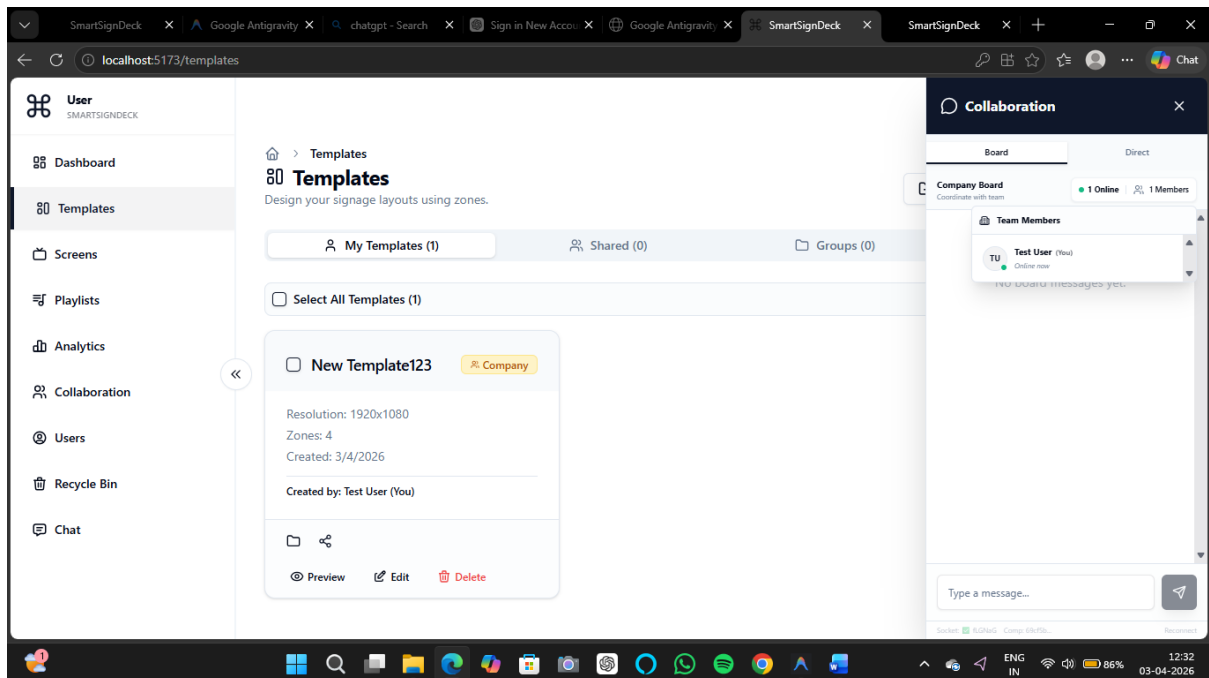
Recycle Bin

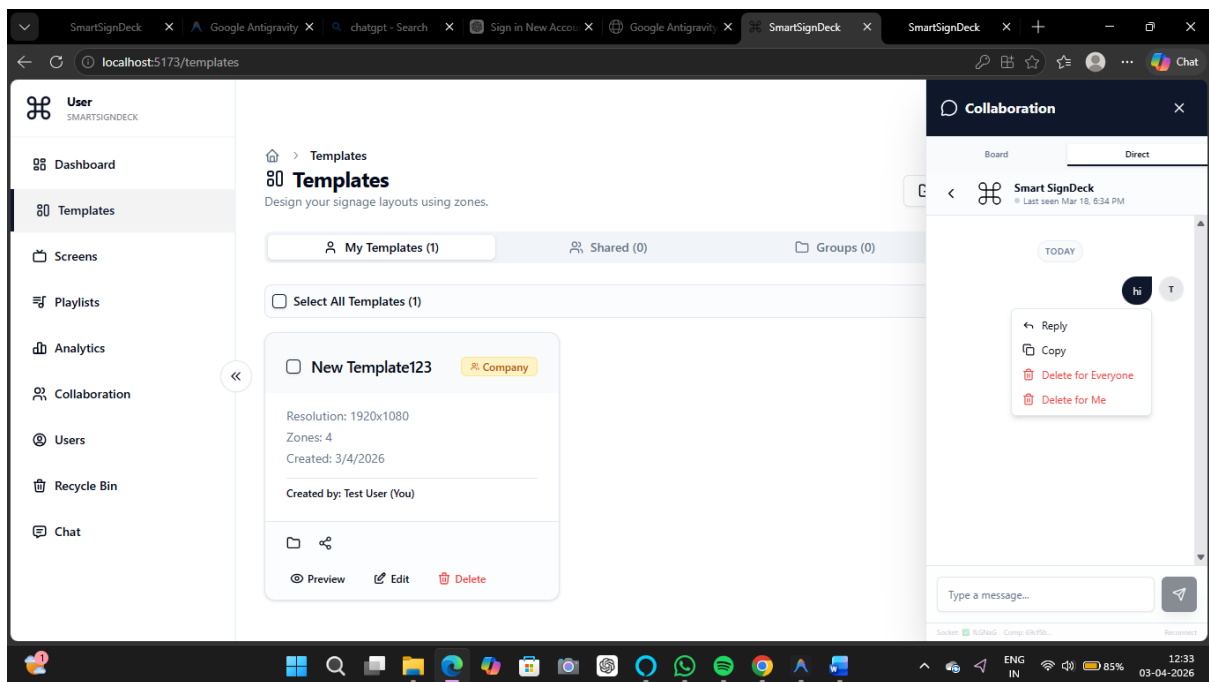
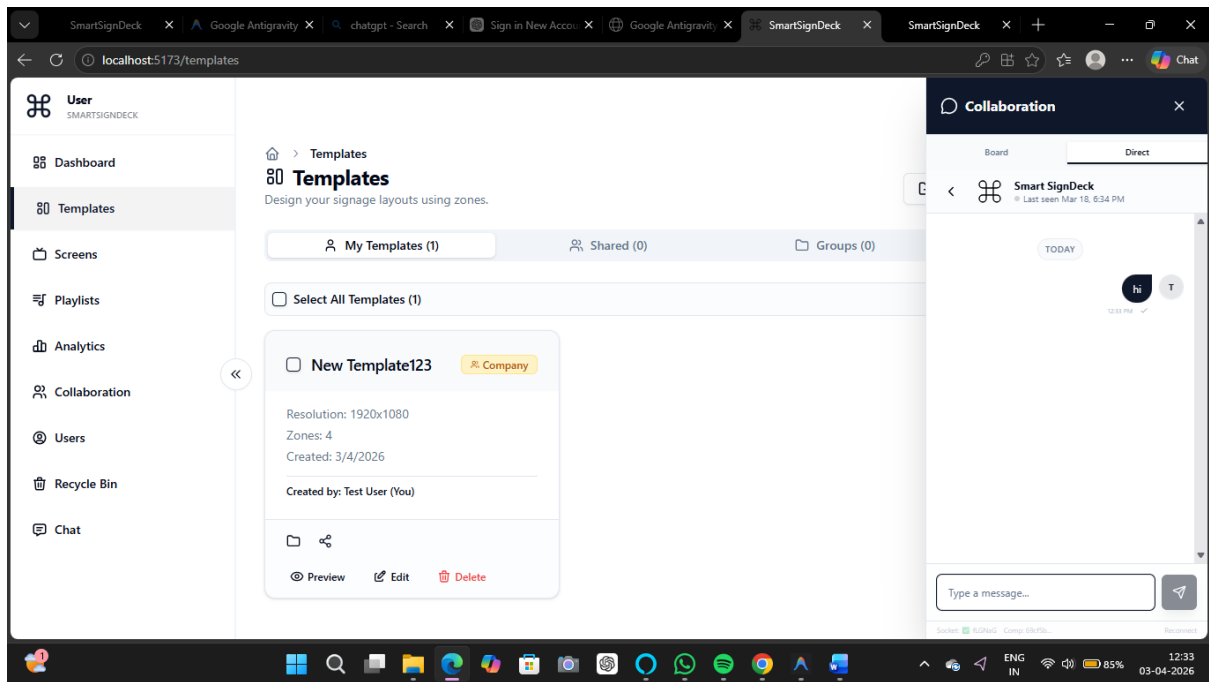




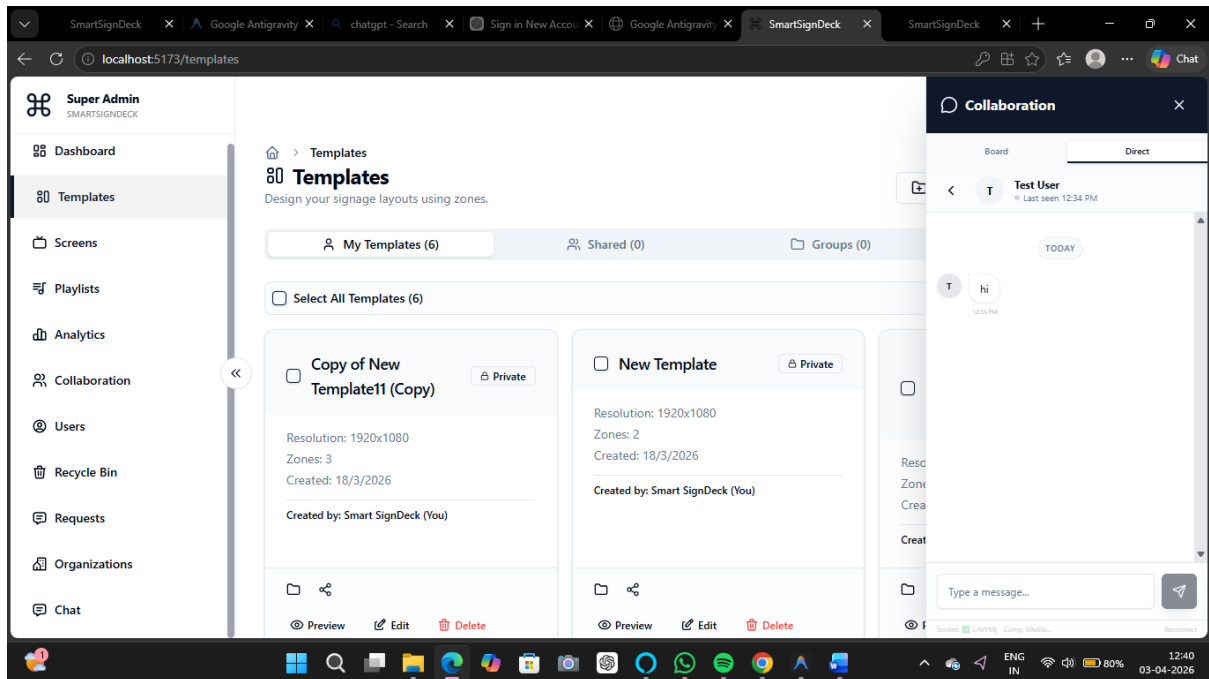


Chat

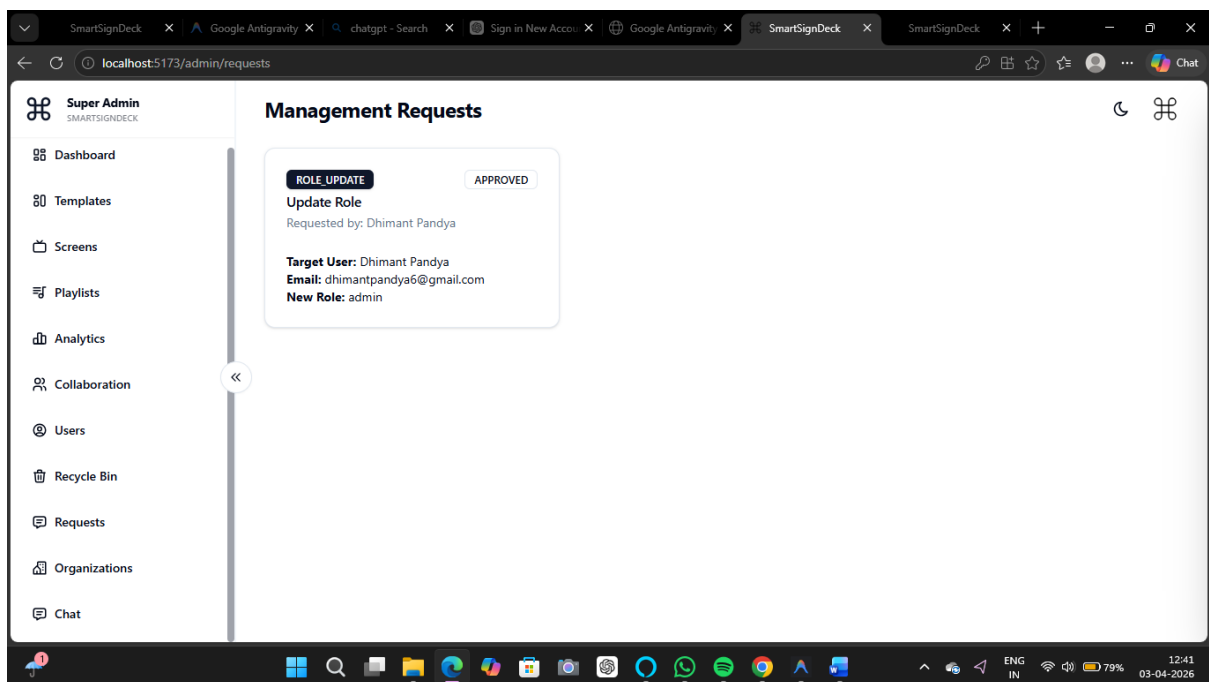




Super Admin



Mgmt Role Request & Super Admin



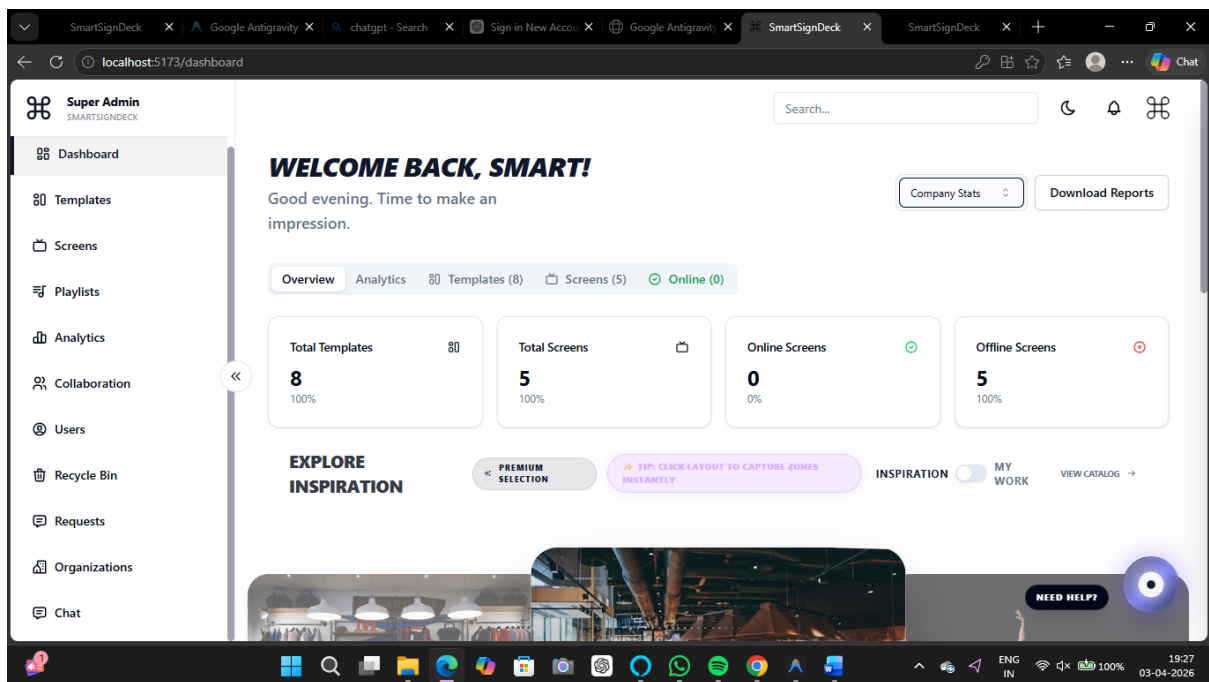
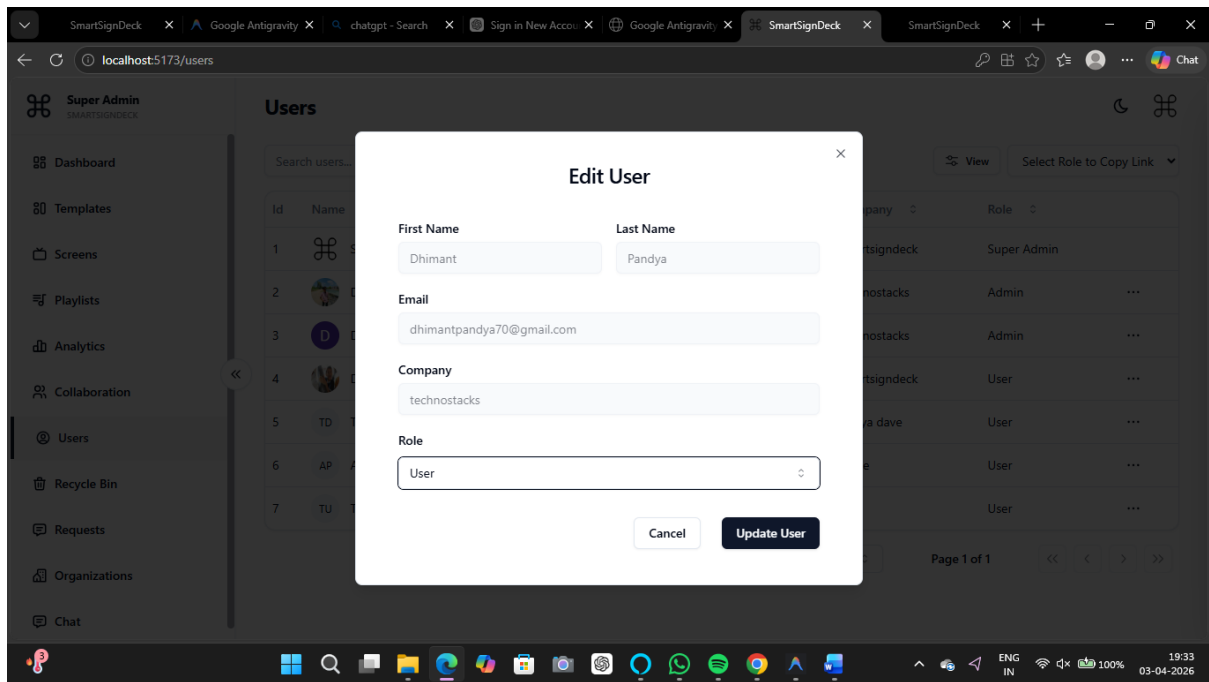
The screenshot displays the 'System Organizations' page in the SmartSignDeck Super Admin interface. The left sidebar contains a navigation menu with options: Dashboard, Templates, Screens, Playlists, Analytics, Collaboration, Users, Recycle Bin, Requests, Organizations, and Chat. The main content area shows a table of organizations with columns: Organization, Status, Records, Visibility, and Actions.

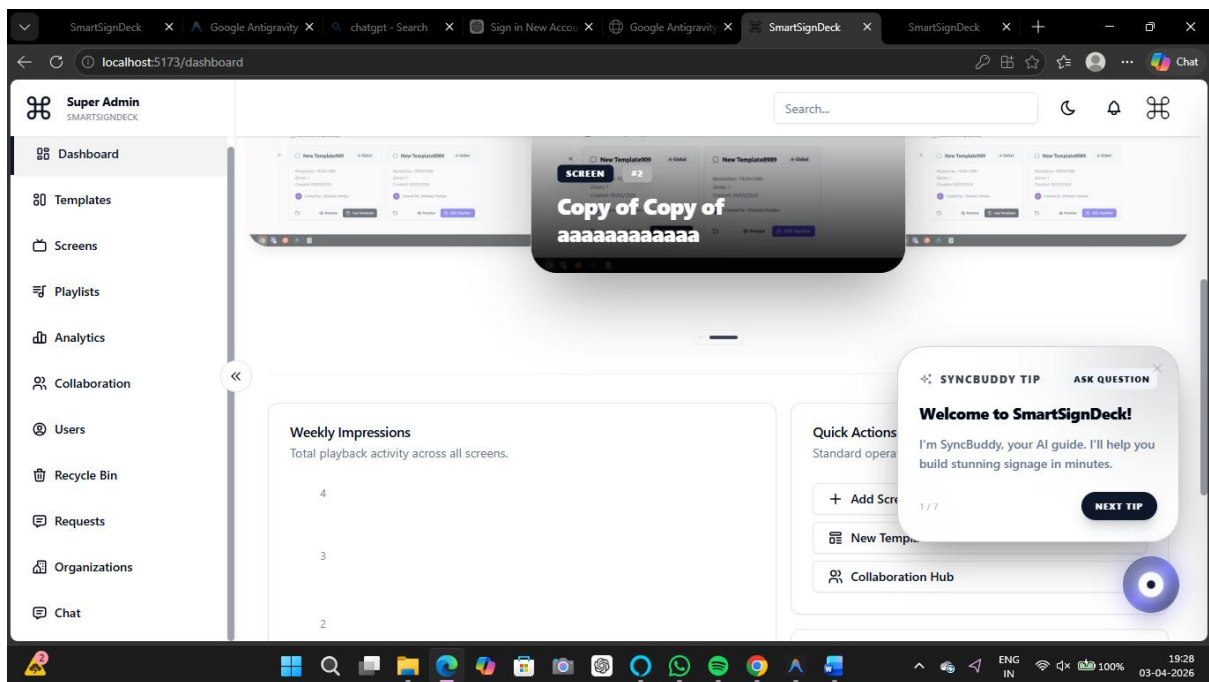
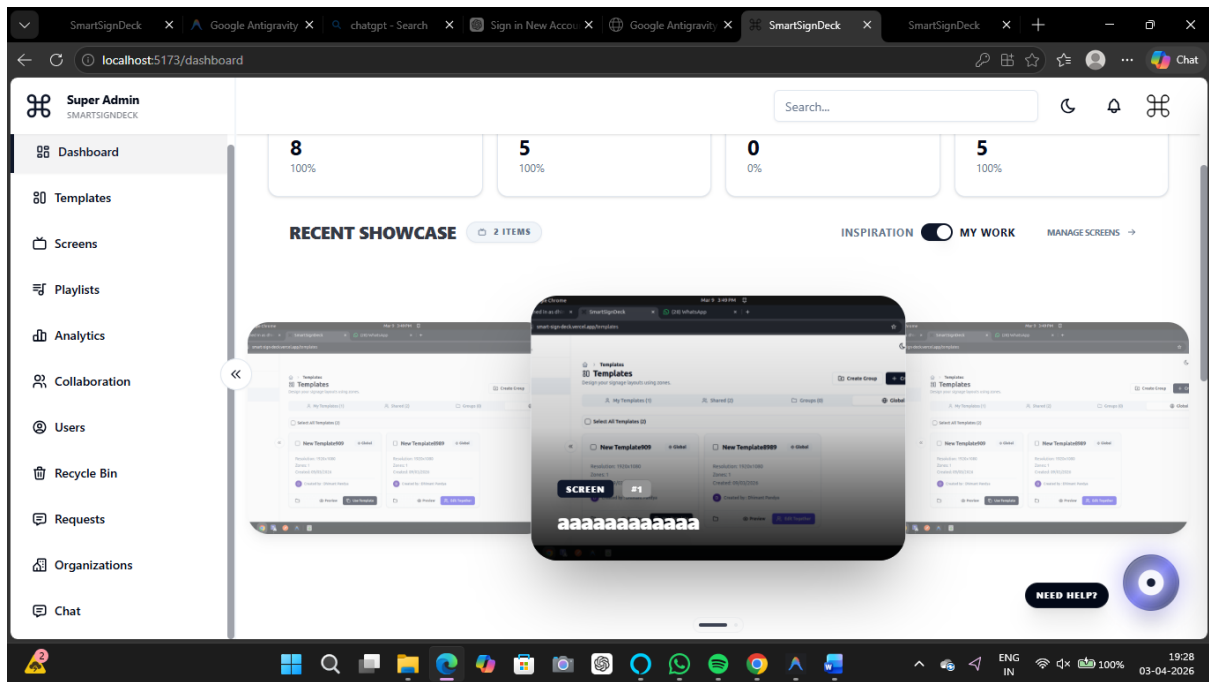
Organization	Status	Records	Visibility	Actions
Abc GROUPED ORGANIZATION	Active	1 Members	Public Connectivity	[Settings] [Delete]
Name GROUPED ORGANIZATION	Active	1 Members	Public Connectivity	[Settings] [Delete]
Smartsigndeck GROUPED ORGANIZATION	Active	3 Members	Public Connectivity	[Settings]
Tathya Dave GROUPED ORGANIZATION	Active	1 Members	Public Connectivity	[Settings] [Delete]
Technostacks GROUPED ORGANIZATION	Active	2 Members	Public Connectivity	[Settings] [Delete]

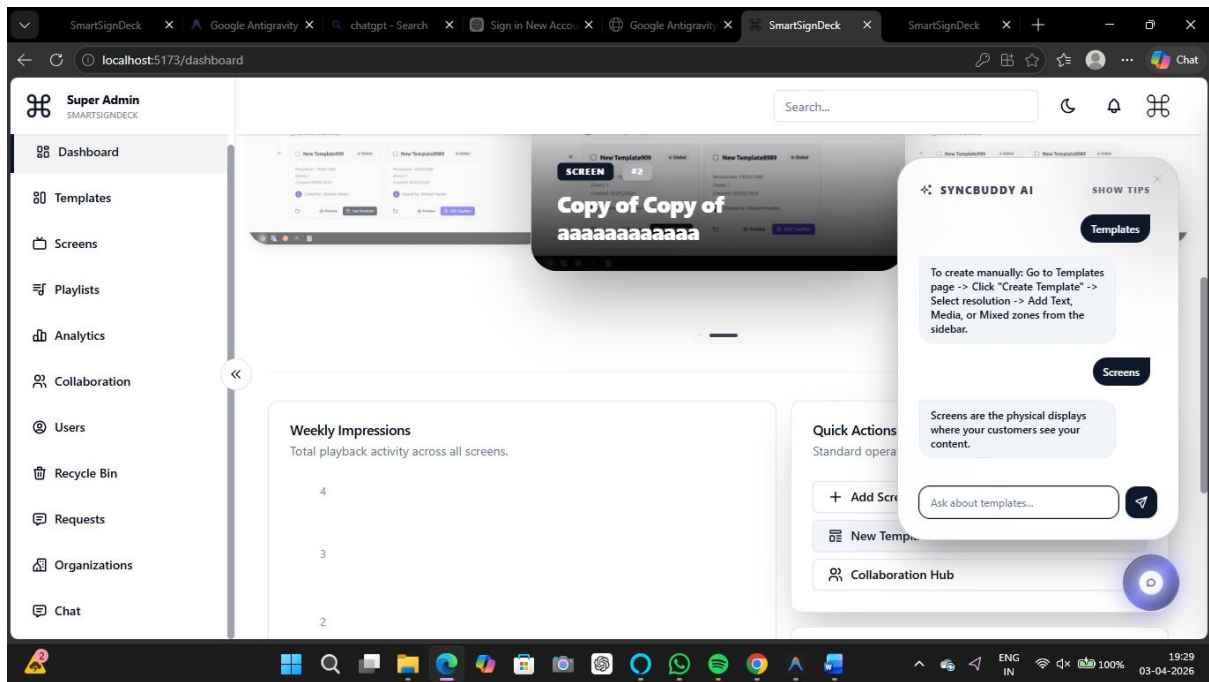
The browser's address bar shows 'localhost:5173/admin/companies'. The system tray at the bottom indicates the time is 12:42 on 03-04-2026, with 78% battery and English as the system language.

This screenshot shows the 'Users' tab for the organization 'abc' in the SmartSignDeck Super Admin interface. The modal window displays the 'Platform Users' section, which lists one user: 'Test User' with the email 'testuser@example.com'. The user has a 'User' role. The background shows the same navigation sidebar as the previous screenshot, but the main content area is dimmed.

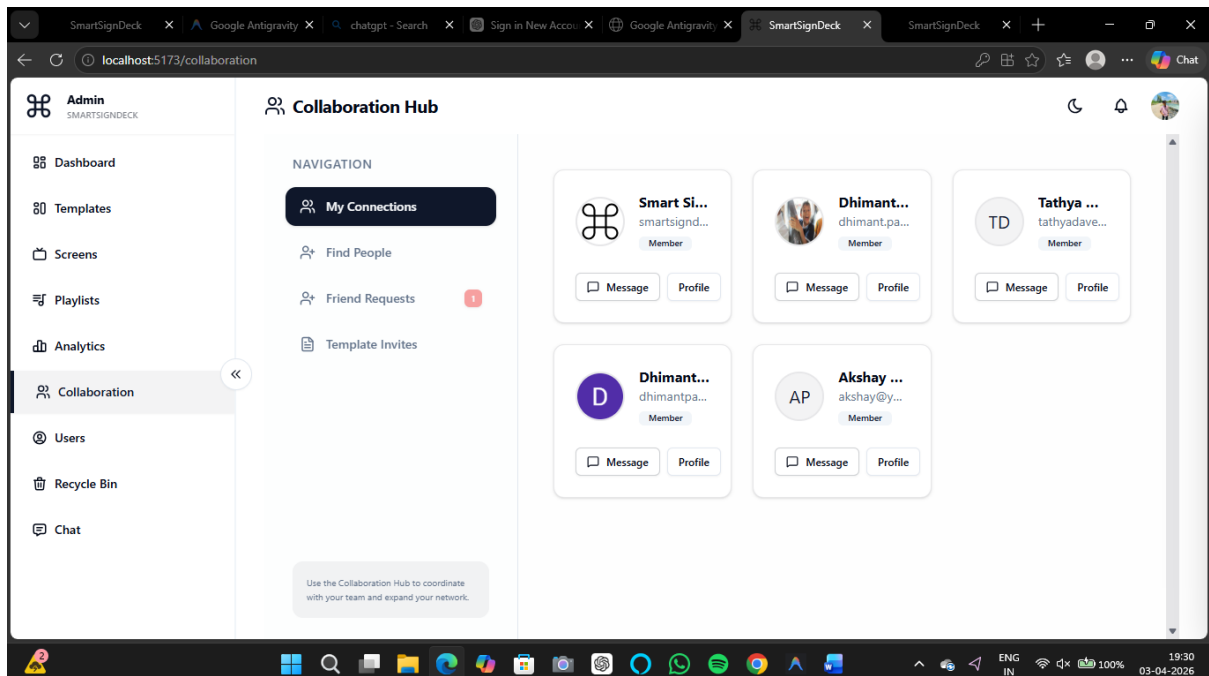
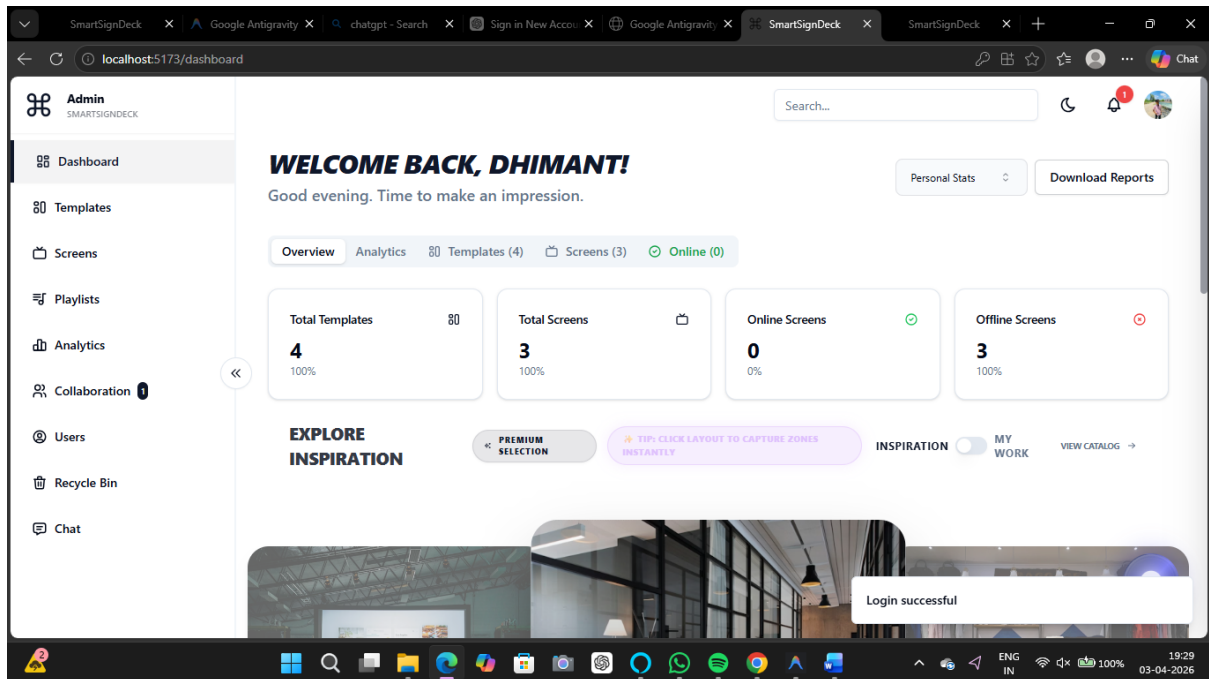
The browser's address bar shows 'localhost:5173/admin/companies'. The system tray at the bottom indicates the time is 19:21 on 03-04-2026, with 100% battery and English as the system language.

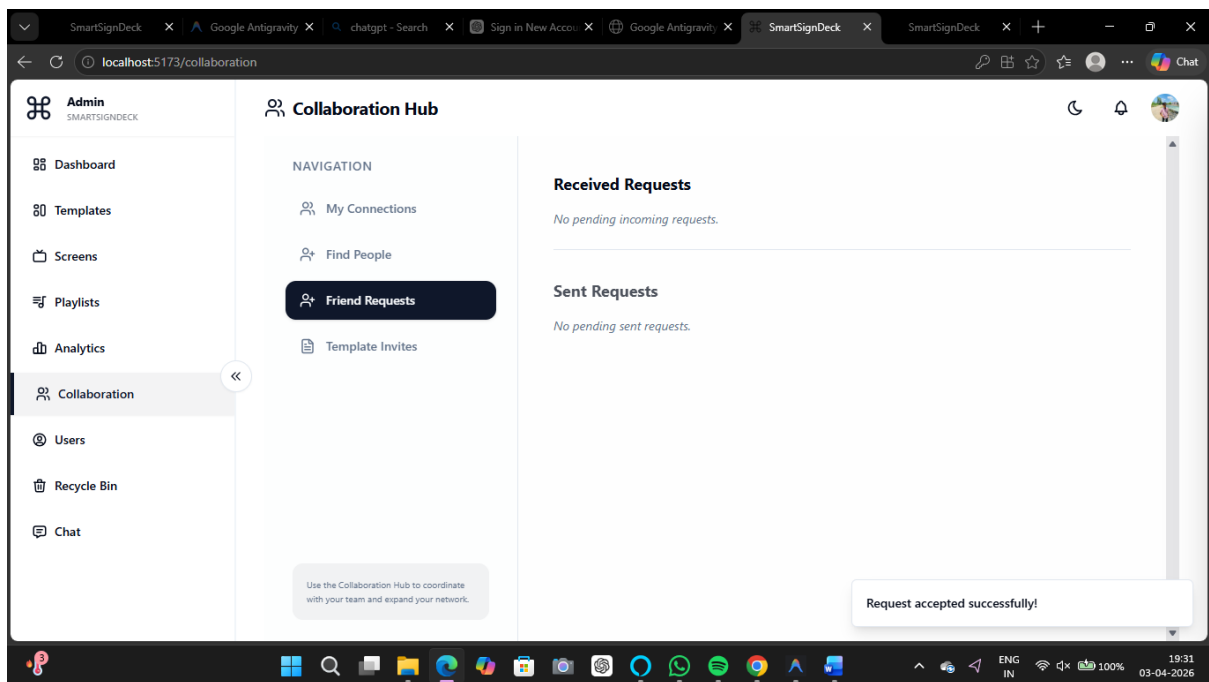
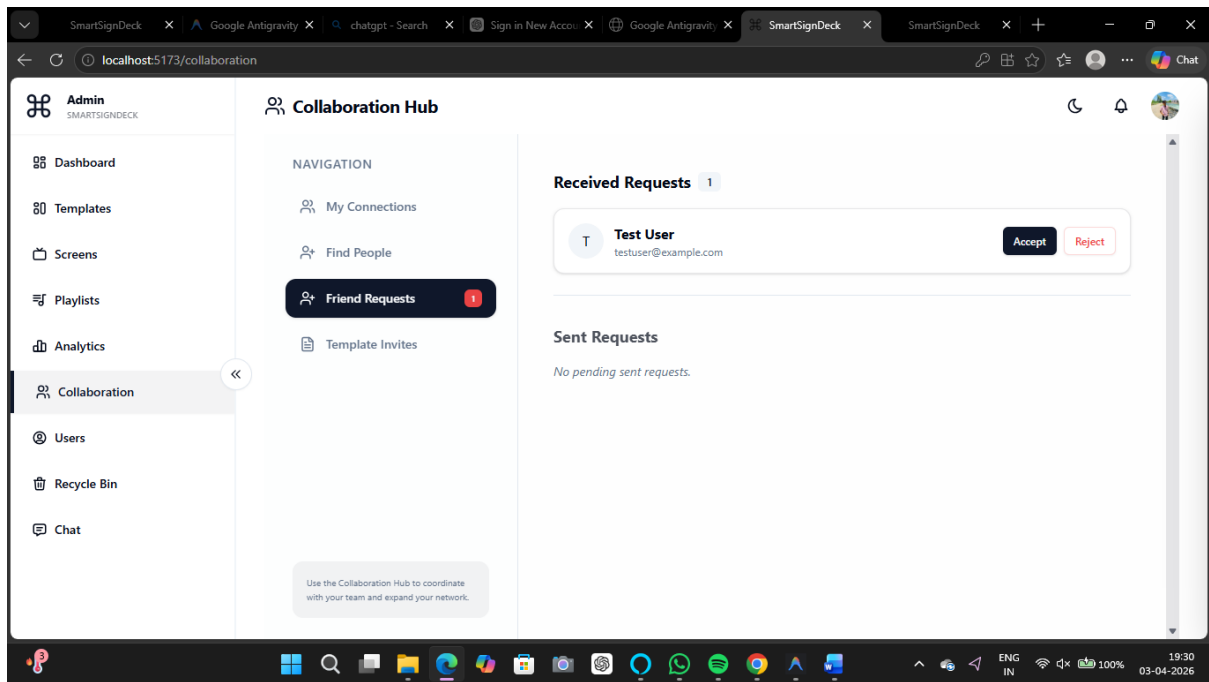


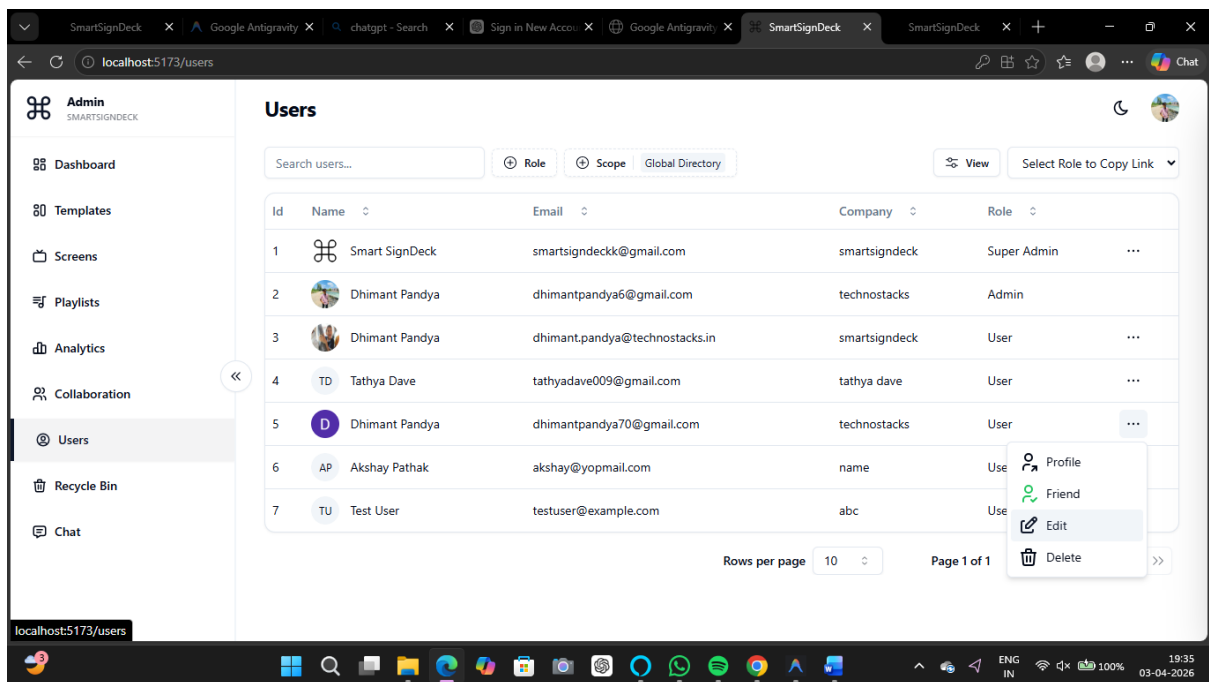
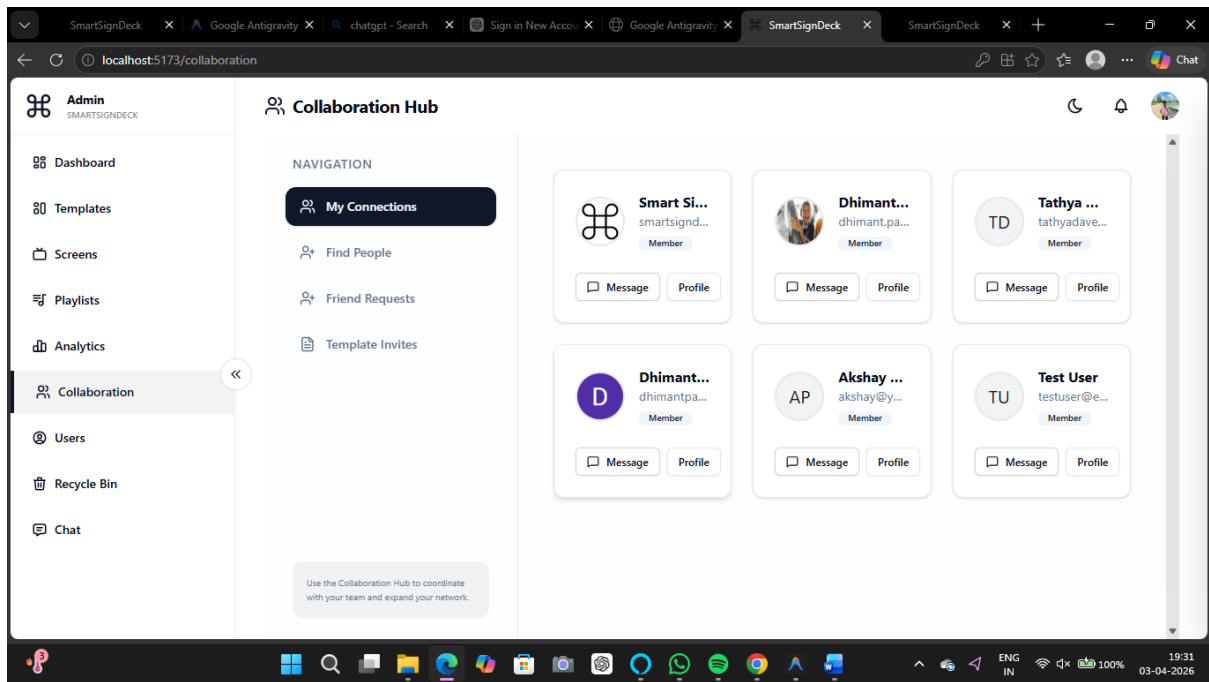


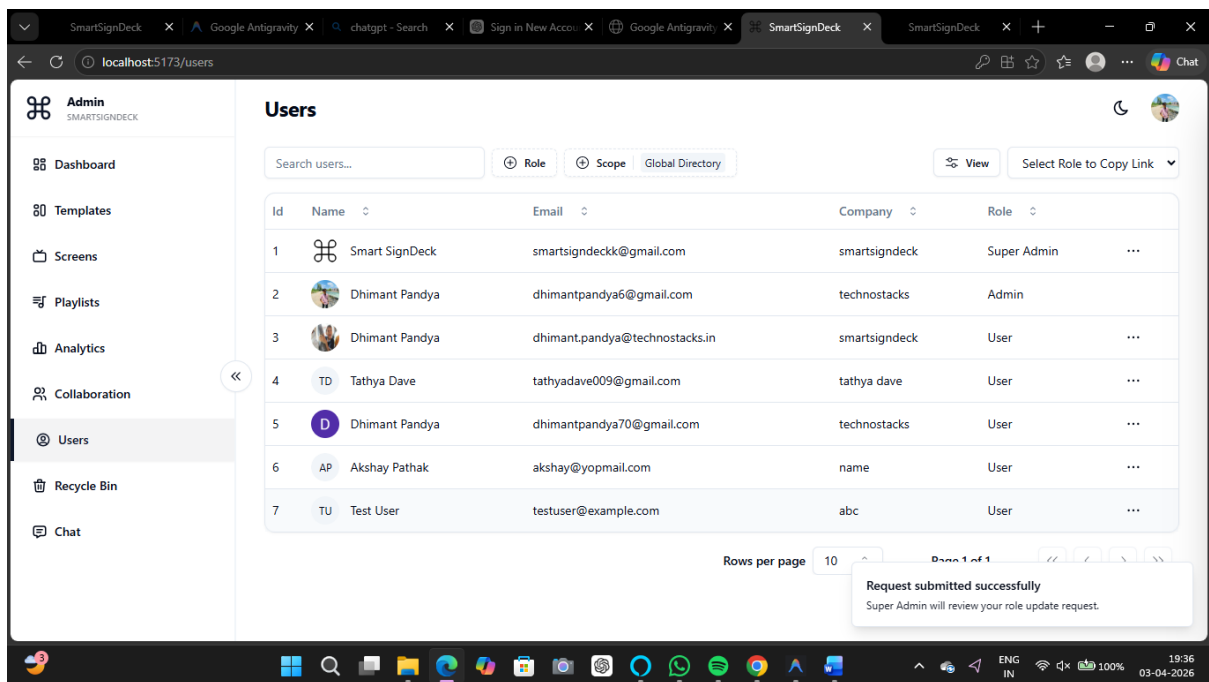
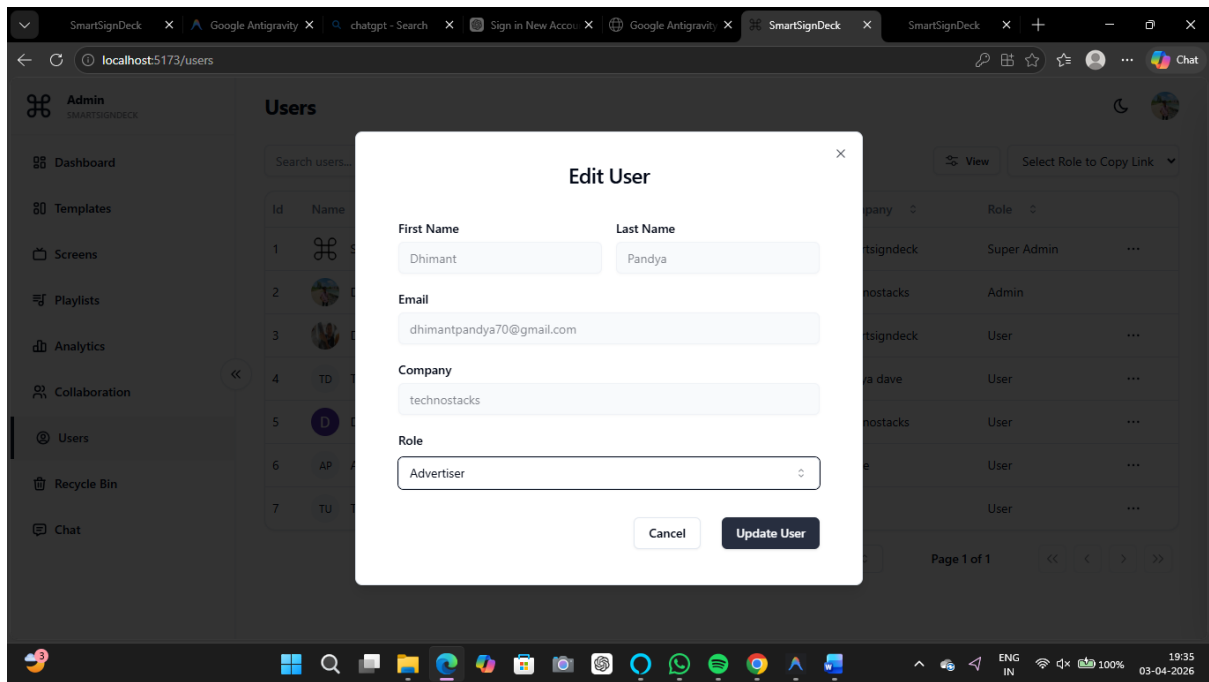


Notification Friend Request and Admin

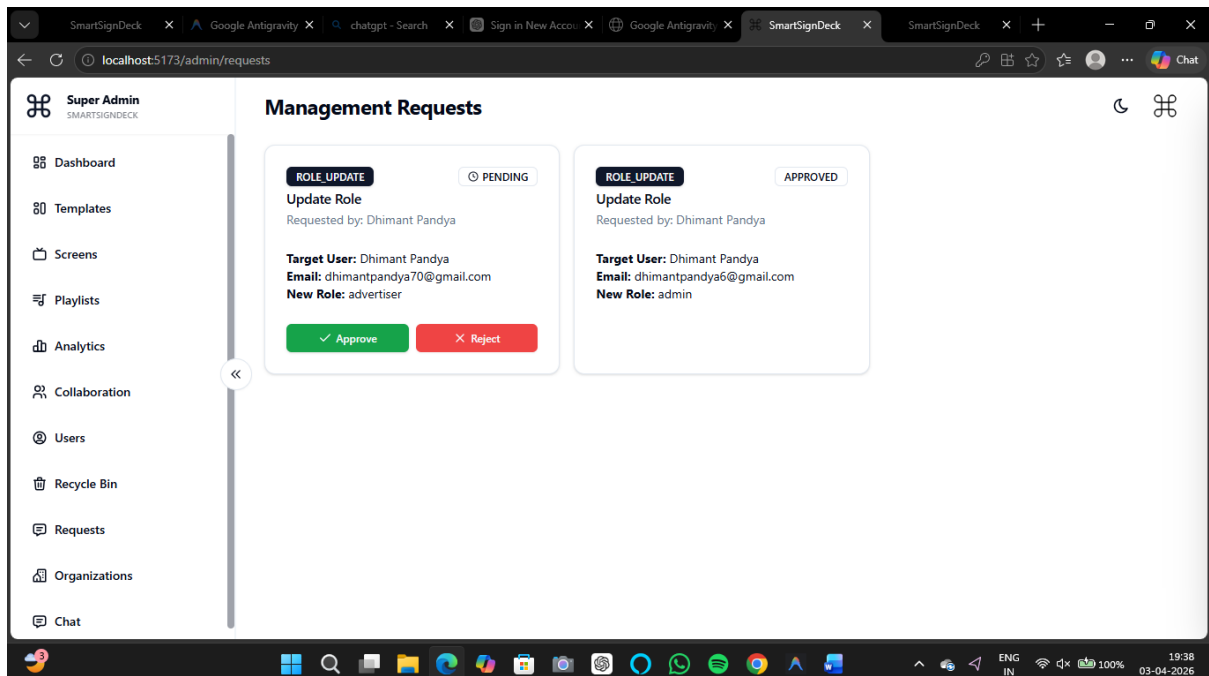
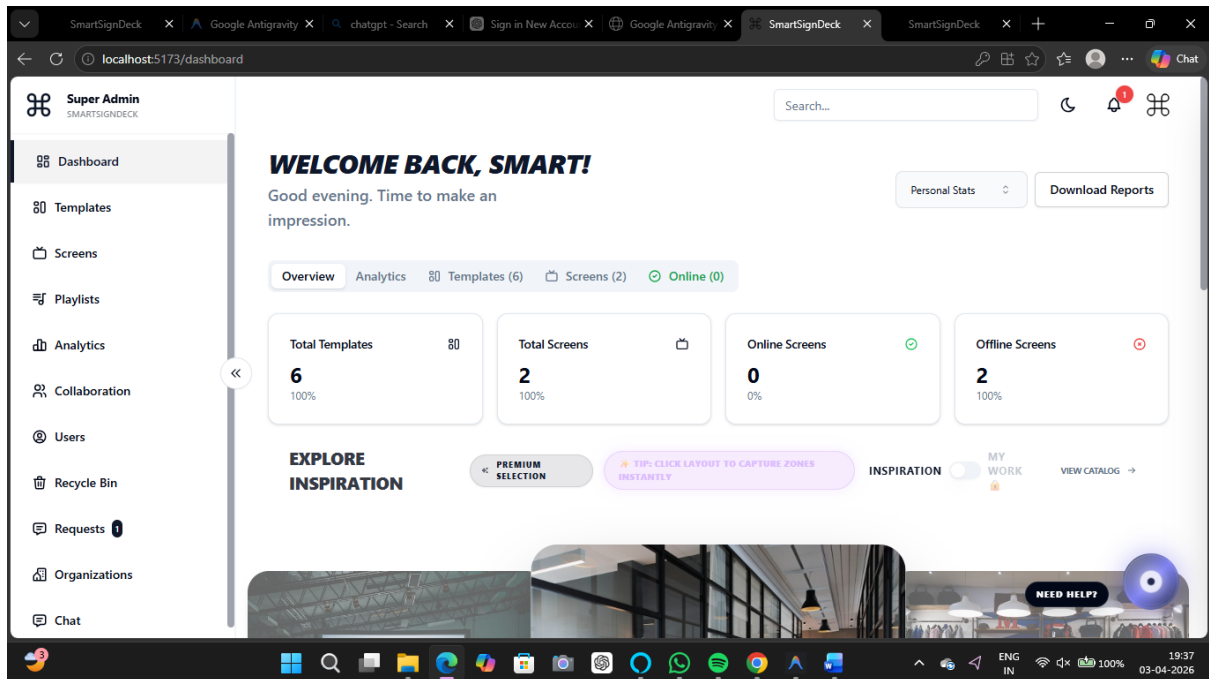


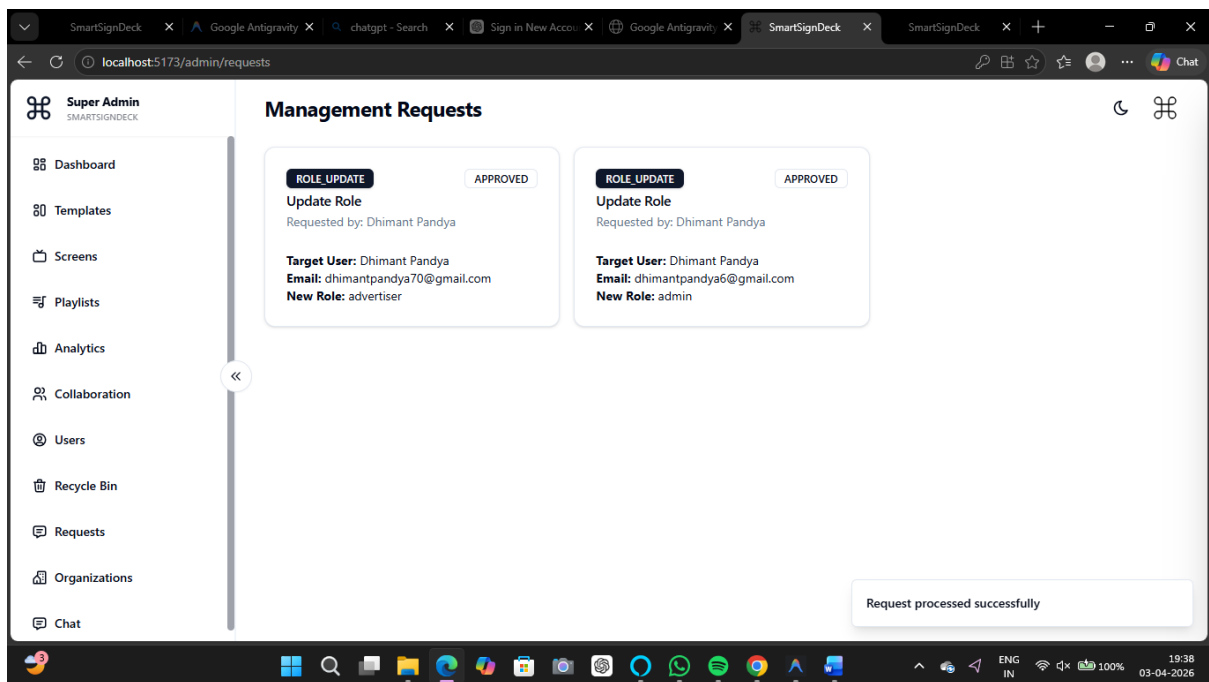
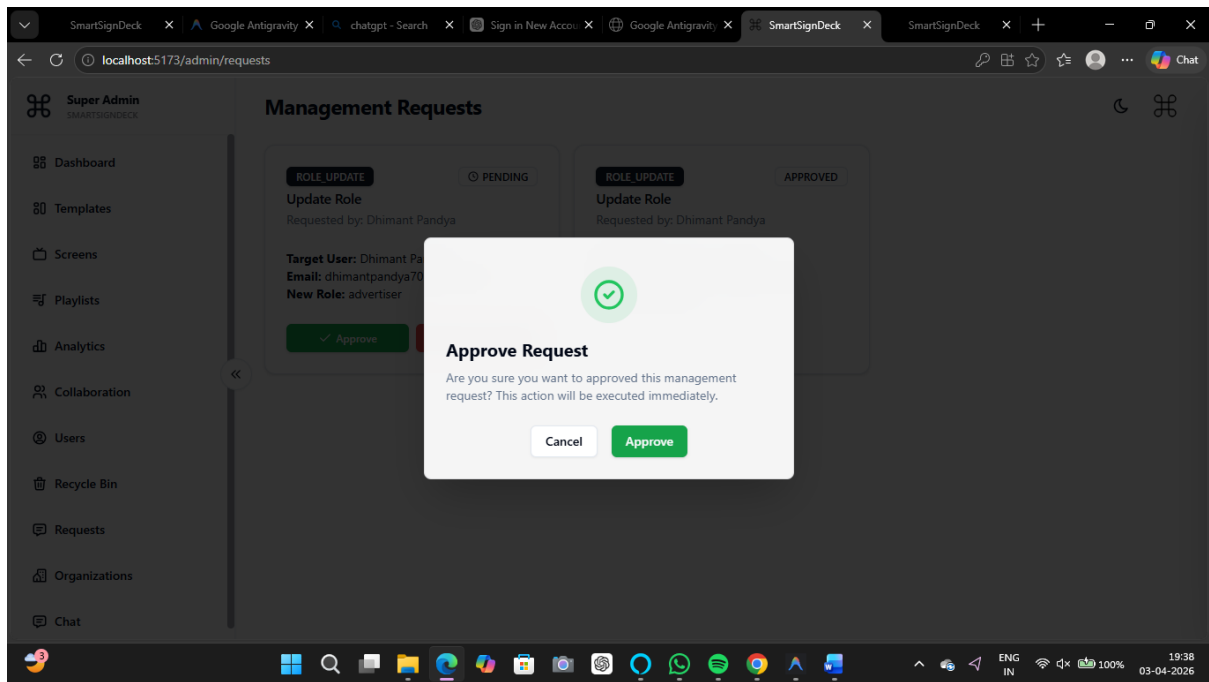






Super Admin Request



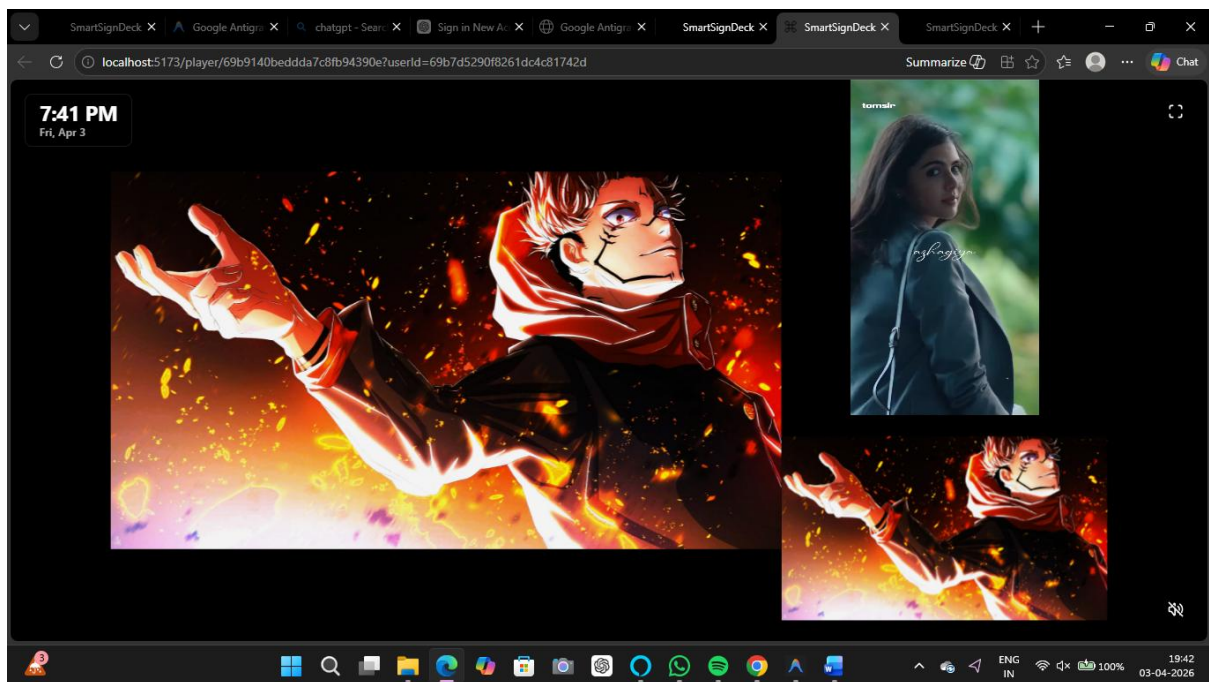
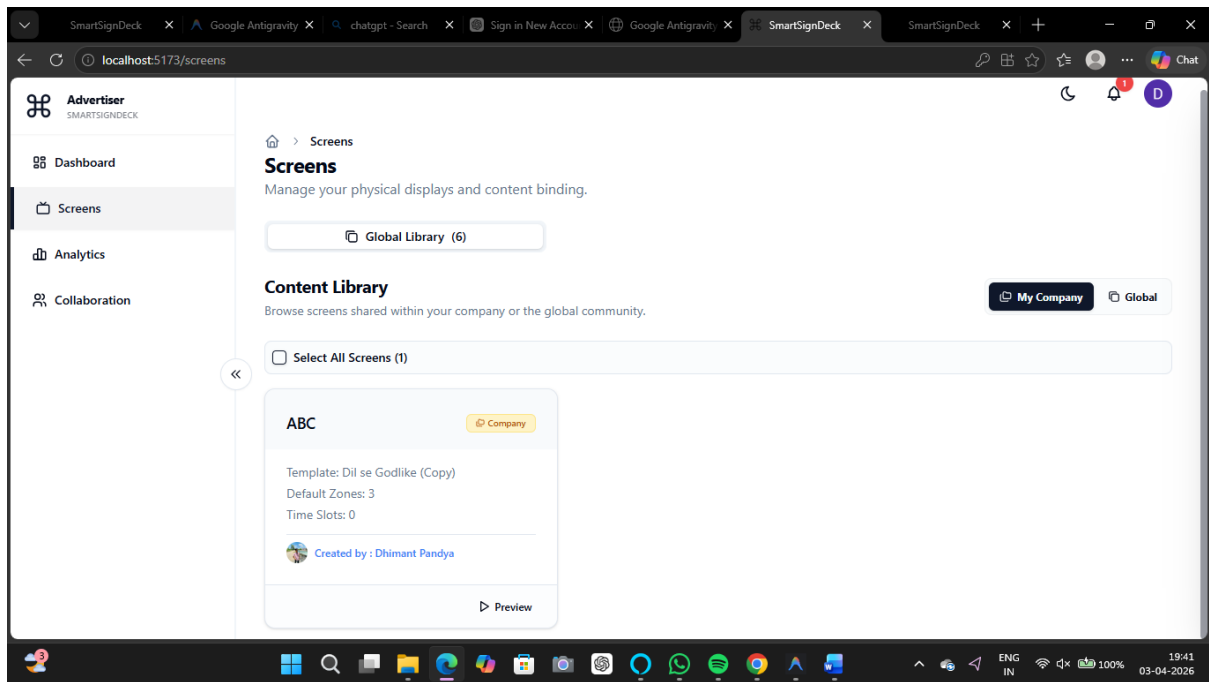


The screenshot shows the 'Super Admin' interface of SmartSignDeck. The left sidebar contains navigation links: Dashboard, Templates, Screens, Playlists, Analytics, Collaboration, Users (selected), Recycle Bin, Requests, Organizations, and Chat. The main content area is titled 'Users' and features a search bar, filters for Role, Scope, and Global Directory, and a 'View' button. A table lists 7 users with columns for Id, Name, Email, Company, and Role. The roles include Super Admin, Admin, User, and Advertiser. At the bottom, there are pagination controls showing 'Page 1 of 1' and 'Rows per page 10'.

Id	Name	Email	Company	Role
1	Smart SignDeck	smartsigndeckk@gmail.com	smartsigndeck	Super Admin
2	Dhimant Pandya	dhimantpandya6@gmail.com	technostacks	Admin
3	Dhimant Pandya	dhimant.pandya@technostacks.in	smartsigndeck	User
4	TD Tathya Dave	tathyadave009@gmail.com	tathya dave	User
5	AP Akshay Pathak	akshay@yopmail.com	name	User
6	TU Test User	testuser@example.com	abc	User
7	Dhimant Pandya	dhimantpandya70@gmail.com	technostacks	Advertiser

Advertiser

The screenshot shows the 'Advertiser' interface of SmartSignDeck. The left sidebar contains navigation links: Dashboard (selected), Screens, Analytics, and Collaboration. The main content area is titled 'WELCOME TO SMARTSIGNDECK, DHIMANT!' and includes a 'Download Reports' button. Below the welcome message, there are three cards showing 'Total Screens' (0), 'Online Screens' (0), and 'Offline Screens' (0). A section titled 'EXPLORE INSPIRATION' features a 'PREMIUM SELECTION' button and a tip: 'TIP: CLICK LAYOUT TO CAPTURE ZONES INSTANTLY'. At the bottom, there is a banner image of a modern office interior and a 'Login successful' notification.



❖ Test Case

1. Registration

Sr.No	Field	Input	Expected Output	Actual Output	Result
1	First Name	Dhimant	Data accepted; OTP sent to email	Data accepted; OTP sent to email	Data added successfully
		D1 (less than 2 chars / numeric)	Error: Min 2 characters required	Error	Data not added
		(empty)	Error: First name is required	Error	Data not added
2	Last Name	Pandya	Data accepted and stored	Data accepted and stored	Data added successfully
		7890 (numeric)	Error: Min 2 characters	Error	Data not added
		(empty)	Error: Last name is required	Error	Data not added
3	Email	testuser@example.com	Data accepted; valid email format	Data accepted	Data added successfully
		testusericloudcom (no @)	Error: Invalid email format	Error	Data not added
		test@.com (invalid domain)	Error: Invalid email	Error	Data not added
		(empty)	Error: Email is required	Error	Data not added
4	Password	Password123!	Accepted (min 8, uppercase, lowercase, digit, special char)	Accepted	Data added successfully

Sr.No	Field	Input	Expected Output	Actual Output	Result
		pass (too short)	Error: Min 8 characters	Error	Data not added
		password123 (no uppercase/special)	Error: Must contain uppercase & special char	Error	Data not added
		PASSWORD123! (no lowercase)	Error: Must contain lowercase letter	Error	Data not added
		(empty)	Error: Password is required	Error	Data not added
5	Confirm Password	Password123! (matches)	Accepted; passwords match	Accepted	Data added successfully
		Password456! (mismatch)	Error: Passwords do not match	Error	Data not added
		(empty)	Error: Confirm password required	Error	Data not added
6	Company Name	Antigravity Corp	Data accepted; workspace created	Data accepted	Data added successfully
		(empty)	Error: Company Name is required	Error	Data not added
7	All Fields Null	All fields empty	Error: All fields are required	Error	Please fill all required fields

2. Login

Sr.No	Field	Input	Expected Output	Actual Output	Result
1	Email	testuser@example.com	Login successful; redirect to dashboard	Login successful	Data verified successfully
		testuserexamplecom (no @)	Error: Invalid email format	Error	Login failed

Sr.No	Field	Input	Expected Output	Actual Output	Result
		(empty)	Error: Email is required	Error	Login failed
2	Password	Password123!	Login successful	Login successful	Data verified successfully
		wrongpassword (incorrect)	Error: Incorrect password	Error	Login failed
		pass (too short)	Error: Invalid credentials	Error	Login failed
		(empty)	Error: Password is required	Error	Login failed
3	All Fields Null	Both fields empty	Error: Fields are required	Error	Please insert all field data

3. OTP Verification (Email Verification)

Sr.No	Field	Input	Expected Output	Actual Output	Result
1	OTP Code	642182 (correct, within 2 min)	OTP verified; account created; redirected to dashboard	OTP verified	Data verified successfully
		000000 (wrong OTP)	Error: Invalid OTP	Error	Verification failed
		12345 (5 digits only)	Error: Invalid OTP format	Error	Verification failed
		(left blank)	Error: OTP is required	Error	Verification failed
2	OTP Expiry	OTP entered after 2 minutes	Error: OTP has expired	Error	OTP expired

Sr.No	Field	Input	Expected Output	Actual Output	Result
3	Resend OTP	Click "Didn't receive code? Resend OTP"	New OTP generated and sent to email	New OTP sent	Resend successful

4. Forgot Password

Sr.No	Field	Input	Expected Output	Actual Output	Result
1	Email	testuser@example.com (registered)	OTP sent to email for password reset	OTP sent	Data sent successfully
		notregistered@example.com	Security: Success shown (user not revealed)	Success message	Generic success
		testuserexamplecom (invalid)	Error: Invalid email format	Error	Data not sent
		(empty)	Error: Email is required	Error	Data not sent
2	Reset OTP	Correct 6-digit OTP within 10 min	Reset token generated; proceed to new password	Token generated	Verified successfully
		Wrong OTP	Error: Invalid OTP	Error	Verification failed
		Expired OTP (after 10 min)	Error: OTP has expired	Error	Verification failed
3	All Fields Null	Empty email	Error: Email is required	Error	Please insert all field data

5. Change Password

Sr.No	Field	Input	Expected Output	Actual Output	Result
1	Current / Old Password	Password123! (correct)	Accepted; proceed to new password	Accepted	Data verified successfully
		WrongPass1! (incorrect)	Error: Incorrect current password	Error	Data not changed
		pass (too short)	Error: Must meet password requirements	Error	Data not changed
		(empty)	Error: Old password is required	Error	Data not changed
2	New Password	NewPassword456!	Accepted (meets all requirements)	Accepted	Data changed successfully
		pass (too short, no upper/special)	Error: Password requirements not met	Error	Data not changed
		Password123! (same as old)	Error: New password must differ from old	Error	Data not changed
		(empty)	Error: New password is required	Error	Data not changed
3	Confirm New Password	NewPassword456! (matches new)	Accepted	Accepted	Data changed successfully
		DifferentPass1! (mismatch)	Error: Passwords do not match	Error	Data not changed
		(empty)	Error: Confirm password required	Error	Data not changed
4	All Fields Null	All empty	Error: All fields required	Error	Please insert all field data

6. Profile Update

Sr.No	Field	Input	Expected Output	Actual Output	Result
1	First Name	Dhimant	Profile updated successfully	Updated	Data updated successfully
		D (less than 2 chars)	Error: Minimum 2 characters	Error	Data not updated
		(empty)	Ignored (optional update)	No change	No update
2	Last Name	Pandya	Profile updated successfully	Updated	Data updated successfully
		1 (too short)	Error: Minimum 2 characters	Error	Data not updated
3	Avatar / Profile Photo	Valid image URL	Avatar updated and displayed	Updated	Data updated successfully
		invalid_url (non-URL format)	Error: Invalid URL format	Error	Data not updated
4	Gender	male / female / other	Profile updated	Updated	Data updated successfully
		xyz (invalid option)	Error: Invalid gender value	Error	Data not updated
5	Date of Birth	2000-01-15 (ISO format)	DOB saved	Updated	Data updated successfully
		15/01/2000 (wrong format)	Error: Invalid date format	Error	Data not updated
6	Language	english / french / german	Language preference saved	Updated	Data updated successfully
		hindi (not in options)	Error: Invalid language	Error	Data not updated

7. Template Management

Sr.No	Field	Input	Expected Output	Actual Output	Result
1	Template Name	Welcome Board	Template created successfully	Created	Data added successfully
		(empty)	Error: Name is required	Error	Data not added
2	Resolution	1920x1080	Template resolution set	Set	Data added successfully
		(empty)	Error: Resolution is required	Error	Data not added
3	Zones	At least 1 zone defined	Template zones saved	Saved	Data added successfully
		(empty zones array)	Error: Zones are required	Error	Data not added
4	Visibility	private / company / public	Visibility set correctly	Set	Data added successfully
		hidden (invalid option)	Error: Invalid visibility value	Error	Data not added
5	Preview URL	Valid image/video URL	Preview saved and shown	Saved	Data added successfully
		not-a-url (invalid format)	Ignored or error	Error	Data not added
6	Preview Type	image or video	Preview type saved	Saved	Data added successfully
		pdf (invalid type)	Error: Must be image or video	Error	Data not added
7	Update Template	Change name to "New Board"	Template updated successfully	Updated	Data updated successfully

Sr.No	Field	Input	Expected Output	Actual Output	Result
		Empty body	Error: At least one field required	Error	Data not updated
8	Delete Template	Valid template ID	Template moved to Recycle Bin	Soft-deleted	Data deleted successfully
		Already deleted template	Error: Template not found	Error	Data not deleted
9	All Fields Null	No name, resolution, zones	Error: Required fields missing	Error	Please fill all required fields

8. Screen Management

Sr.No	Field	Input	Expected Output	Actual Output	Result
1	Screen Name	Lobby Display	Screen created successfully	Created	Data added successfully
		(empty)	Error: Screen name is required	Error	Data not added
2	Location	Main Entrance	Location saved	Saved	Data added successfully
		(empty)	Accepted (optional field)	Accepted	Data added successfully
3	Template ID	Valid template ObjectID	Screen linked to template	Linked	Data added successfully
		invalid_id (not ObjectID)	Error: Invalid template ID	Error	Data not added
		(empty)	Error: Template ID is required	Error	Data not added
4	Schedule Name	Morning Shift	Schedule created	Created	Data added successfully

Sr.No	Field	Input	Expected Output	Actual Output	Result
		(empty)	Error: Schedule name required	Error	Data not added
5	Schedule Start Time	08:00	Start time saved	Saved	Data added successfully
		25:00 (invalid time)	Error: Invalid time format	Error	Data not added
6	Schedule End Time	18:00	End time saved	Saved	Data added successfully
		Before start time	Error: End must be after start	Error	Data not added
7	Days of Week	[1,2,3,4,5] (Mon-Fri)	Days saved correctly	Saved	Data added successfully
		[7,8] (out of 0-6 range)	Error: Invalid day values	Error	Data not added
8	Visibility	private / company / public	Visibility set	Set	Data added successfully
		secret (invalid)	Error: Invalid visibility	Error	Data not added
9	Show Clock	true / false	Clock preference saved	Saved	Data added successfully
10	Delete Screen	Valid screen ID	Screen moved to Recycle Bin	Soft-deleted	Data deleted successfully
11	All Fields Null	No name, templateId	Error: Required fields missing	Error	Please fill all required fields

9. Playlist Management

Sr.No	Field	Input	Expected Output	Actual Output	Result
1	Playlist Name	Morning Content	Playlist created successfully	Created	Data added successfully
		(empty)	Error: Name is required	Error	Data not added
2	Add Media Item	Valid media URL	Item added to playlist	Added	Data added successfully
		invalid-url (non-URI)	Error: Invalid media URL	Error	Data not added
3	Item Duration	10 (seconds)	Duration set correctly	Set	Data added successfully
		0 or negative	Error: Duration must be positive	Error	Data not added
4	Update Playlist	Change name to "Evening Content"	Playlist updated	Updated	Data updated successfully
		Empty update body	Error: At least one field required	Error	Data not updated
5	Delete Playlist	Valid playlist ID	Playlist deleted	Deleted	Data deleted successfully
6	All Fields Null	No name	Error: Required fields missing	Error	Please fill all required fields

10. User Invitation (Invite Team Member)

Sr.No	Field	Input	Expected Output	Actual Output	Result
1	Company ID	Valid company ObjectId	Invite token generated	Generated	Invite link created

Sr.No	Field	Input	Expected Output	Actual Output	Result
		invalid_id	Error: Invalid company ID	Error	Invite not created
2	Role	user / admin	Invite token with correct role	Generated	Invite link created
		super_admin (restricted)	Error: Cannot invite super admin	Error	Invite not created
		advertiser trying to invite	Error: Advertisers cannot generate invite	Error	Invite not created
3	Invite Link Expiry	Used within 7 days	Registration completes via invite	Completed	Registration successful
		Expired invite token	Error: Invalid or expired invitation	Error	Registration failed

11. Collaboration Request

Sr.No	Field	Input	Expected Output	Actual Output	Result
1	Send Request	Valid user ID	Collaboration request sent	Sent	Data added successfully
		Own user ID	Error: Cannot collaborate with self	Error	Data not added
2	Accept Request	Valid request ID	Request accepted; collaboration active	Accepted	Updated successfully
		Already accepted request	Error: Already accepted	Error	Data not updated
3	Reject Request	Valid request ID	Request rejected	Rejected	Updated successfully

12. Analytics

Sr.No	Field	Input	Expected Output	Actual Output	Result
1	Start Date	2026-03-27 (ISO format)	Analytics filtered from start date	Filtered	Data displayed
		2026/03/27 (wrong format)	Error: Invalid date format	Error	Data not displayed
2	End Date	2026-04-03	Analytics filtered to end date	Filtered	Data displayed
		Before start date	Error: End date before start date	Error	Data not displayed
3	Interval	day / week / month	Data grouped by interval	Grouped	Data displayed
		year (not supported)	Error: Invalid interval	Error	Data not displayed
4	Scope	personal / company	Data filtered by scope	Filtered	Data displayed

SCOPES AND LIMITATIONS

1 . Scopes

- **Super Admin** can manage all users, companies, templates, screens, playlists, notifications, and analytics across the entire platform.
- **Admin** can manage their own company's users, templates, screens, playlists, and view company-wide analytics. Can generate invite links for user and admin roles.
- **User** can create, edit, and manage their own templates, screens, and playlists. Can send collaboration requests and invite other users to their company.

- **Advertiser** can view templates and analytics but cannot invite other users or generate invite links.
- Any user can design digital signage templates using the drag-and-drop editor with zones for text, images, videos, clock, and QR codes.
- The system supports real-time screen content delivery via Socket.IO without a page refresh.
- Supports Google (Firebase) OAuth for authentication alongside standard email/password registration.
- Email-based OTP verification (expiry 2 minutes for registration; 10 minutes for password reset) ensures account security.
- Content scheduling allows screens to display different content on specific days and time ranges.
- The Recycle Bin feature provides soft-delete for safe recovery of deleted templates and screens.
- Analytics module provides timeline-based insights on template usage, screen activity, and online status.
- Multi-company architecture: each company has its own workspace, users, templates, and screens.

2) Limitations

- **Guest users** cannot create templates, manage screens, or access the dashboard without registering and verifying their email.
- **OTP is valid for only 2 minutes** during registration. Expired OTPs require the user to click "Resend OTP".
- **Advertisers** are restricted from generating invite links or inviting any user.
- **Google OAuth users** cannot log in using email/password and vice versa — auth providers are locked per account.

- **File uploads** (images/videos) require cloud storage (Cloudinary/AWS S3). Large file uploads may be slow depending on internet speed.
- The platform does **not support offline mode** — an active internet connection is required for all operations.
- **Screen Player** requires a network connection to fetch real-time content updates from the server.
- **No mobile app** is currently available; the system is web-based only.
- **Template Editor** is browser-based (Fabric.js canvas) — complex designs may be slow on low-end devices.
- **Rate limiting** is enforced on login and OTP endpoints to prevent brute-force attacks; excessive attempts are blocked temporarily.
- **Password policy** is strict: minimum 8 characters, must contain uppercase, lowercase, digit, and special character — simple passwords are rejected.
- **Cross-company access** is restricted: users can only see their own company's data unless they are Super Admin.

FUTURE EXPANSION

- **Mobile Application:** A dedicated iOS and Android app for SmartSignDeck to allow users to manage screens, templates, and analytics on the go.
- **AI-Powered Template Suggestions:** Integrate AI to automatically suggest template designs and content based on the company's industry and usage patterns.
- **Advanced Scheduling:** Support for recurring events, holiday-based schedules, and weather-triggered content changes on displays.
- **Payment & Subscription Integration:** Implement tiered subscription plans (Free, Pro, Enterprise) with billing via Stripe or PayPal for SaaS monetization.
- **Offline Player Mode:** Allow screens to cache content locally so they continue displaying even without an internet connection, syncing when reconnected.
- **Video Conferencing Integration:** Embed live video meeting links (Zoom/Google Meet) directly into digital signage templates for meeting room displays.
- **Custom Fonts & Brand Kits:** Allow companies to upload their brand fonts, colors, and logos and apply them system-wide across all templates.
- **Multi-Language Support:** Extend the platform to support multiple languages in the UI (Hindi, Gujarati, Spanish, French, etc.) for global usage.
- **Third-Party Integrations:** API integrations with external content sources such as Google Slides, RSS Feeds, Live Stock Tickers, Social Media Walls, and news feeds for dynamic content.
- **Analytics Export & Reporting:** Allow users to export analytics data as PDF / Excel reports for business presentations.

- **Push Notification Support:** Send push notifications to registered mobile/desktop devices for critical alerts and screen status changes.
- **Role-Based Content Approval Workflow:** Introduce a content approval flow where admin must approve templates/content before it goes live on public screens.

REFERENCES AND BIBLIOGRAPHY

➤ References:-

- <https://www.mongodb.com/docs> — MongoDB Atlas Database Documentation
- <https://expressjs.com> — Express.js Backend Framework Documentation
- <https://react.dev> — React.js Frontend Library Documentation
- <https://www.typescriptlang.org/docs> — TypeScript Language Documentation
- <https://socket.io/docs> — Socket.IO Real-time Communication Documentation
- <https://www.npmjs.com> — NPM Package Registry (dependencies reference)
- <https://firebase.google.com/docs> — Firebase Authentication (Google OAuth) Documentation
- <https://cloudinary.com/documentation> — Cloudinary Media Upload Documentation
- <https://www.radiustheme.com/signage> — Digital Signage industry reference
- <https://www.w3schools.com> — CSS, HTML, JavaScript learning referenc

➤ Bibliography:-

- ✓ From tutorial point we refer information about magento, CSS and bootstrap
- ✓ From nykaa we get the idea about different category, product and offers and theme
- ✓ From Amazon we get the idea of different filter.
- ✓ From w3school we learn coding in bootstrap and CSS.
- ✓ From Google we get idea about the website theme. Idea about feasibility study, data dictionary, ERdiagram, data flow diagrams.
- ✓ From YouTube we get idea about the coding and error solving at the time of preparation of system